Day 001 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Add Decimals with Place Value

Standards: 6.NS.B.3

Learning target: I can add decimals by aligning equal place values.

1. 48.9

Align decimal points; 40.5 + 8.4 = 48.9.

6.NS.B.3 | Constructed response

2. 86.7

Align decimal points; 84.7 + 2 = 86.7.

6.NS.B.3 | Constructed response

3. 61.3

Align decimal points; 59.4 + 1.9 = 61.3.

6.NS.B.3 | Constructed response

4. 4.17

Align decimal points; 1.67 + 2.5 = 4.17.

6.NS.B.3 | Constructed response

5. 14.24

Align decimal points; 6.34 + 7.9 = 14.24.

6.NS.B.3 | Constructed response

6. 11.63

Align decimal points; 4.93 + 6.7 = 11.63.

6.NS.B.3 | Constructed response

7. 6.48

Align decimal points; 5.68 + 0.8 = 6.48.

6.NS.B.3 | Constructed response

8. B - 423/50

Align decimal points; 7.99 + 0.47 = 8.46.

6.NS.B.3 | Multiple choice

9. A - 72/25

Align decimal points; 2.67 + 0.21 = 2.88.

6.NS.B.3 | Multiple choice

10. 5.11 km

Align decimal points; 4.4 + 0.71 = 5.11.

6.NS.B.3 | Constructed response

Day 002 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Subtract Decimals and Check

Standards: 6.NS.B.3

Learning target: I can subtract decimals and explain regrouping.

1. 38.9

Align decimal points; 46.2 - 7.3 = 38.9.

6.NS.B.3 | Constructed response

2. 81

Align decimal points; 82.5 - 1.5 = 81.

6.NS.B.3 | Constructed response

3. 29.4

Align decimal points; 36.6 - 7.2 = 29.4.

6.NS.B.3 | Constructed response

4. 3.7

Align decimal points; 8.2 - 4.5 = 3.7.

6.NS.B.3 | Constructed response

5. 2.04

Align decimal points; 8.04 - 6 = 2.04.

6.NS.B.3 | Constructed response

6. 5.07

Align decimal points; 6.47 - 1.4 = 5.07.

6.NS.B.3 | Constructed response

7. 5.1

Align decimal points; 5.45 - 0.35 = 5.1.

6.NS.B.3 | Constructed response

8. D - 41/5

Align decimal points; 8.54 - 0.34 = 8.2.

6.NS.B.3 | Multiple choice

9. B - 657/100

Align decimal points; 6.71 - 0.14 = 6.57.

6.NS.B.3 | Multiple choice

10. 4.9 L

Align decimal points; 5.59 - 0.69 = 4.9.

6.NS.B.3 | Constructed response

Day 003 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Multiply Decimals and Estimate

Standards: 6.NS.B.3

Learning target: I can multiply decimals and use an estimate to place the decimal point.

1. 34.56

Compute the whole-number product and restore decimal places: 19.2 × 1.8 = 34.56.

6.NS.B.3 | Constructed response

2. 107.64

Compute the whole-number product and restore decimal places: 46.8 × 2.3 = 107.64.

6.NS.B.3 | Constructed response

3. 82.51

Compute the whole-number product and restore decimal places: 22.3 × 3.7 = 82.51.

6.NS.B.3 | Constructed response

4. 53.058

Compute the whole-number product and restore decimal places: 7.17 × 7.4 = 53.058.

6.NS.B.3 | Constructed response

5. 47.889

Compute the whole-number product and restore decimal places: 9.39 × 5.1 = 47.889.

6.NS.B.3 | Constructed response

6. 11.842

Compute the whole-number product and restore decimal places: 3.82 × 3.1 = 11.842.

6.NS.B.3 | Constructed response

7. 1.86

Compute the whole-number product and restore decimal places: 3 × 0.62 = 1.86.

6.NS.B.3 | Constructed response

8. B - 692/125

Compute the whole-number product and restore decimal places: 6.92 × 0.8 = 5.536.

6.NS.B.3 | Multiple choice

9. C - 2314/625

Compute the whole-number product and restore decimal places: 7.12 × 0.52 = 3.7024.

6.NS.B.3 | Multiple choice

10. 3.1108 dollars

Compute the whole-number product and restore decimal places: 7.07 × 0.44 = 3.1108.

6.NS.B.3 | Constructed response

Day 004 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Divide Whole Numbers

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

1. 67

53 ÷ 8: write 6, subtract 48, remainder 5; bring down the next digit: 56 ÷ 8: write 7, subtract 56, remainder 0. Check: 8 × 67 = 536.

6.NS.B.2 | Constructed response

2. 64

25 ÷ 4: write 6, subtract 24, remainder 1; bring down the next digit: 16 ÷ 4: write 4, subtract 16, remainder 0. Check: 4 × 64 = 256.

6.NS.B.2 | Constructed response

3. 74

59 ÷ 8: write 7, subtract 56, remainder 3; bring down the next digit: 32 ÷ 8: write 4, subtract 32, remainder 0. Check: 8 × 74 = 592.

6.NS.B.2 | Constructed response

4. 326

61 ÷ 19: write 3, subtract 57, remainder 4; bring down the next digit: 49 ÷ 19: write 2, subtract 38, remainder 11; bring down the next digit: 114 ÷ 19: write 6, subtract 114, remainder 0. Check: 19 × 326 = 6194.

6.NS.B.2 | Constructed response

5. 357

78 ÷ 22: write 3, subtract 66, remainder 12; bring down the next digit: 125 ÷ 22: write 5, subtract 110, remainder 15; bring down the next digit: 154 ÷ 22: write 7, subtract 154, remainder 0. Check: 22 × 357 = 7854.

6.NS.B.2 | Constructed response

6. 334

40 ÷ 12: write 3, subtract 36, remainder 4; bring down the next digit: 40 ÷ 12: write 3, subtract 36, remainder 4; bring down the next digit: 48 ÷ 12: write 4, subtract 48, remainder 0. Check: 12 × 334 = 4008.

6.NS.B.2 | Constructed response

7. 250 R 5

42 ÷ 17: write 2, subtract 34, remainder 8; bring down the next digit: 85 ÷ 17: write 5, subtract 85, remainder 0; bring down the next digit: 5 ÷ 17: write 0, subtract 0, remainder 5. Check: 17 × 250 + 5 = 4255.

6.NS.B.2 | Constructed response

8. 407 R 21

150 ÷ 37: write 4, subtract 148, remainder 2; bring down the next digit: 28 ÷ 37: write 0, subtract 0, remainder 28; bring down the next digit: 280 ÷ 37: write 7, subtract 259, remainder 21. Check: 37 × 407 + 21 = 15080.

6.NS.B.2 | Model and constructed response

9. 152 R 12

29 ÷ 19: write 1, subtract 19, remainder 10; bring down the next digit: 100 ÷ 19: write 5, subtract 95, remainder 5; bring down the next digit: 50 ÷ 19: write 2, subtract 38, remainder 12. Check: 19 × 152 + 12 = 2900.

6.NS.B.2 | Model and constructed response

10. 394 full packets; 10 cards remain

5132 = 13 × 394 + 10.

6.NS.B.2 | Constructed response

Day 005 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Estimate and Check Whole-Number Division

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

1. 81

72 ÷ 9: write 8, subtract 72, remainder 0; bring down the next digit: 9 ÷ 9: write 1, subtract 9, remainder 0. Check: 9 × 81 = 729.

6.NS.B.2 | Constructed response

2. 63

25 ÷ 4: write 6, subtract 24, remainder 1; bring down the next digit: 12 ÷ 4: write 3, subtract 12, remainder 0. Check: 4 × 63 = 252.

6.NS.B.2 | Constructed response

3. 86

43 ÷ 5: write 8, subtract 40, remainder 3; bring down the next digit: 30 ÷ 5: write 6, subtract 30, remainder 0. Check: 5 × 86 = 430.

6.NS.B.2 | Constructed response

4. 121

22 ÷ 19: write 1, subtract 19, remainder 3; bring down the next digit: 39 ÷ 19: write 2, subtract 38, remainder 1; bring down the next digit: 19 ÷ 19: write 1, subtract 19, remainder 0. Check: 19 × 121 = 2299.

6.NS.B.2 | Constructed response

5. 194

27 ÷ 14: write 1, subtract 14, remainder 13; bring down the next digit: 131 ÷ 14: write 9, subtract 126, remainder 5; bring down the next digit: 56 ÷ 14: write 4, subtract 56, remainder 0. Check: 14 × 194 = 2716.

6.NS.B.2 | Constructed response

6. 156

59 ÷ 38: write 1, subtract 38, remainder 21; bring down the next digit: 212 ÷ 38: write 5, subtract 190, remainder 22; bring down the next digit: 228 ÷ 38: write 6, subtract 228, remainder 0. Check: 38 × 156 = 5928.

6.NS.B.2 | Constructed response

7. 319 R 2

67 ÷ 21: write 3, subtract 63, remainder 4; bring down the next digit: 40 ÷ 21: write 1, subtract 21, remainder 19; bring down the next digit: 191 ÷ 21: write 9, subtract 189, remainder 2. Check: 21 × 319 + 2 = 6701.

6.NS.B.2 | Constructed response

8. 157 R 13

23 ÷ 15: write 1, subtract 15, remainder 8; bring down the next digit: 86 ÷ 15: write 5, subtract 75, remainder 11; bring down the next digit: 118 ÷ 15: write 7, subtract 105, remainder 13. Check: 15 × 157 + 13 = 2368.

6.NS.B.2 | Model and constructed response

9. 305 R 27

113 ÷ 37: write 3, subtract 111, remainder 2; bring down the next digit: 21 ÷ 37: write 0, subtract 0, remainder 21; bring down the next digit: 212 ÷ 37: write 5, subtract 185, remainder 27. Check: 37 × 305 + 27 = 11312.

6.NS.B.2 | Model and constructed response

10. 370 full packets; 12 cards remain

4822 = 13 × 370 + 12.

6.NS.B.2 | Constructed response

Day 006 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Write Zero Digits in Quotients

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

1. 109

9 ÷ 9: write 1, subtract 9, remainder 0; bring down the next digit: 8 ÷ 9: write 0, subtract 0, remainder 8; bring down the next digit: 81 ÷ 9: write 9, subtract 81, remainder 0. Check: 9 × 109 = 981.

6.NS.B.2 | Constructed response

2. 307

15 ÷ 5: write 3, subtract 15, remainder 0; bring down the next digit: 3 ÷ 5: write 0, subtract 0, remainder 3; bring down the next digit: 35 ÷ 5: write 7, subtract 35, remainder 0. Check: 5 × 307 = 1535.

6.NS.B.2 | Constructed response

3. 203

6 ÷ 3: write 2, subtract 6, remainder 0; bring down the next digit: 0 ÷ 3: write 0, subtract 0, remainder 0; bring down the next digit: 9 ÷ 3: write 3, subtract 9, remainder 0. Check: 3 × 203 = 609.

6.NS.B.2 | Constructed response

4. 402

84 ÷ 21: write 4, subtract 84, remainder 0; bring down the next digit: 4 ÷ 21: write 0, subtract 0, remainder 4; bring down the next digit: 42 ÷ 21: write 2, subtract 42, remainder 0. Check: 21 × 402 = 8442.

6.NS.B.2 | Constructed response

5. 105

33 ÷ 32: write 1, subtract 32, remainder 1; bring down the next digit: 16 ÷ 32: write 0, subtract 0, remainder 16; bring down the next digit: 160 ÷ 32: write 5, subtract 160, remainder 0. Check: 32 × 105 = 3360.

6.NS.B.2 | Constructed response

6. 307

89 ÷ 29: write 3, subtract 87, remainder 2; bring down the next digit: 20 ÷ 29: write 0, subtract 0, remainder 20; bring down the next digit: 203 ÷ 29: write 7, subtract 203, remainder 0. Check: 29 × 307 = 8903.

6.NS.B.2 | Constructed response

7. 102 R 3

36 ÷ 36: write 1, subtract 36, remainder 0; bring down the next digit: 7 ÷ 36: write 0, subtract 0, remainder 7; bring down the next digit: 75 ÷ 36: write 2, subtract 72, remainder 3. Check: 36 × 102 + 3 = 3675.

6.NS.B.2 | Constructed response

8. 206 R 13

68 ÷ 33: write 2, subtract 66, remainder 2; bring down the next digit: 21 ÷ 33: write 0, subtract 0, remainder 21; bring down the next digit: 211 ÷ 33: write 6, subtract 198, remainder 13. Check: 33 × 206 + 13 = 6811.

6.NS.B.2 | Model and constructed response

9. 204 R 12

28 ÷ 14: write 2, subtract 28, remainder 0; bring down the next digit: 6 ÷ 14: write 0, subtract 0, remainder 6; bring down the next digit: 68 ÷ 14: write 4, subtract 56, remainder 12. Check: 14 × 204 + 12 = 2868.

6.NS.B.2 | Model and constructed response

10. 203 full packets; 11 cards remain

2447 = 12 × 203 + 11.

6.NS.B.2 | Constructed response

Day 007 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Divide by a Decimal

Standards: 6.NS.B.3

Learning target: I can divide by a decimal using an equivalent whole-number divisor.

1. 79

Multiply both numbers by 10: 553 ÷ 7 = 79. Check: 0.7 × 79 = 55.3.

6.NS.B.3 | Constructed response

2. 77

Multiply both numbers by 10: 154 ÷ 2 = 77. Check: 0.2 × 77 = 15.4.

6.NS.B.3 | Constructed response

3. 43

Multiply both numbers by 10: 215 ÷ 5 = 43. Check: 0.5 × 43 = 21.5.

6.NS.B.3 | Constructed response

4. 4.3

Multiply both numbers by 10: 64.5 ÷ 15 = 4.3. Check: 1.5 × 4.3 = 6.45.

6.NS.B.3 | Constructed response

5. 5.1

Multiply both numbers by 10: 204 ÷ 40 = 5.1. Check: 4 × 5.1 = 20.4.

6.NS.B.3 | Constructed response

6. 7.2

Multiply both numbers by 10: 86.4 ÷ 12 = 7.2. Check: 1.2 × 7.2 = 8.64.

6.NS.B.3 | Constructed response

7. 5.4

Multiply both numbers by 10: 21.6 ÷ 4 = 5.4. Check: 0.4 × 5.4 = 2.16.

6.NS.B.3 | Constructed response

8. C - 13/5

Multiply both numbers by 100: 62.4 ÷ 24 = 2.6. Check: 0.24 × 2.6 = 0.624.

6.NS.B.3 | Multiple choice

9. D - 27/5

Multiply both numbers by 100: 345.6 ÷ 64 = 5.4. Check: 0.64 × 5.4 = 3.456.

6.NS.B.3 | Multiple choice

10. 5.2 bags

Multiply both numbers by 100: 166.4 ÷ 32 = 5.2. Check: 0.32 × 5.2 = 1.664.

6.NS.B.3 | Constructed response

Day 008 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Explain Equivalent Decimal Division

Standards: 6.NS.B.3

Learning target: I can divide by a decimal using an equivalent whole-number divisor.

1. 32

Multiply both numbers by 10: 128 ÷ 4 = 32. Check: 0.4 × 32 = 12.8.

6.NS.B.3 | Constructed response

2. 44

Multiply both numbers by 10: 352 ÷ 8 = 44. Check: 0.8 × 44 = 35.2.

6.NS.B.3 | Constructed response

3. 75

Multiply both numbers by 10: 600 ÷ 8 = 75. Check: 0.8 × 75 = 60.

6.NS.B.3 | Constructed response

4. 7.1

Multiply both numbers by 10: 85.2 ÷ 12 = 7.1. Check: 1.2 × 7.1 = 8.52.

6.NS.B.3 | Constructed response

5. 4.5

Multiply both numbers by 10: 288 ÷ 64 = 4.5. Check: 6.4 × 4.5 = 28.8.

6.NS.B.3 | Constructed response

6. 5

Multiply both numbers by 10: 160 ÷ 32 = 5. Check: 3.2 × 5 = 16.

6.NS.B.3 | Constructed response

7. 7.3

Multiply both numbers by 10: 29.2 ÷ 4 = 7.3. Check: 0.4 × 7.3 = 2.92.

6.NS.B.3 | Constructed response

8. A - 31/5

Multiply both numbers by 100: 74.4 ÷ 12 = 6.2. Check: 0.12 × 6.2 = 0.744.

6.NS.B.3 | Multiple choice

9. A - 6/5

Multiply both numbers by 10: 4.8 ÷ 4 = 1.2. Check: 0.4 × 1.2 = 0.48.

6.NS.B.3 | Multiple choice

10. 7.9 bags

Multiply both numbers by 10: 31.6 ÷ 4 = 7.9. Check: 0.4 × 7.9 = 3.16.

6.NS.B.3 | Constructed response

Day 009 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Apply Decimal Operations

Standards: 6.NS.B.3

Learning target: I can apply Decimal Operations, and explain and check my reasoning.

1. 54.1

Align decimal points; 47.3 + 6.8 = 54.1.

6.NS.B.3 | Constructed response

2. 33.9

Align decimal points; 36.3 - 2.4 = 33.9.

6.NS.B.3 | Constructed response

3. 586.95

Compute the whole-number product and restore decimal places: 90.3 × 6.5 = 586.95.

6.NS.B.3 | Constructed response

4. 4.8

Multiply both numbers by 10: 120 ÷ 25 = 4.8. Check: 2.5 × 4.8 = 12.

6.NS.B.3 | Constructed response

5. 11.75

Align decimal points; 5.05 + 6.7 = 11.75.

6.NS.B.3 | Constructed response

6. 1.67

Align decimal points; 5.77 - 4.1 = 1.67.

6.NS.B.3 | Constructed response

7. 0.553

Compute the whole-number product and restore decimal places: 1.58 × 0.35 = 0.553.

6.NS.B.3 | Constructed response

8. D - 6/5

Multiply both numbers by 100: 30 ÷ 25 = 1.2. Check: 0.25 × 1.2 = 0.3.

6.NS.B.3 | Multiple choice

9. C - 111/20

Align decimal points; 5.31 + 0.24 = 5.55.

6.NS.B.3 | Multiple choice

10. 0.59 L

Align decimal points; 1.26 - 0.67 = 0.59.

6.NS.B.3 | Constructed response

Day 010 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Model Multi-Step Decimal Decisions

Standards: 6.NS.B.3

Learning target: I can model Multi-Step Decimal Decisions, and explain and check my reasoning.

1. 39.7

Align decimal points; 36.5 + 3.2 = 39.7.

6.NS.B.3 | Constructed response

2. 44.6

Align decimal points; 53.1 - 8.5 = 44.6.

6.NS.B.3 | Constructed response

3. 301.43

Compute the whole-number product and restore decimal places: 70.1 × 4.3 = 301.43.

6.NS.B.3 | Constructed response

4. 7.9

Multiply both numbers by 10: 118.5 ÷ 15 = 7.9. Check: 1.5 × 7.9 = 11.85.

6.NS.B.3 | Constructed response

5. 11.89

Align decimal points; 6.99 + 4.9 = 11.89.

6.NS.B.3 | Constructed response

6. 0.35

Align decimal points; 3.65 - 3.3 = 0.35.

6.NS.B.3 | Constructed response

7. 1.3005

Compute the whole-number product and restore decimal places: 2.55 × 0.51 = 1.3005.

6.NS.B.3 | Constructed response

8. B - 21/5

Multiply both numbers by 10: 16.8 ÷ 4 = 4.2. Check: 0.4 × 4.2 = 1.68.

6.NS.B.3 | Multiple choice

9. C - 203/25

Align decimal points; 7.27 + 0.85 = 8.12.

6.NS.B.3 | Multiple choice

10. 6.7 L

Align decimal points; 6.96 - 0.26 = 6.7.

6.NS.B.3 | Constructed response

Day 011 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Solve Decimal Measurement Problems

Standards: 6.NS.B.3

Learning target: I can solve Decimal Measurement Problems, and explain and check my reasoning.

1. 147.63

Compute the whole-number product and restore decimal places: 77.7 × 1.9 = 147.63.

6.NS.B.3 | Constructed response

2. 49

Multiply both numbers by 10: 98 ÷ 2 = 49. Check: 0.2 × 49 = 9.8.

6.NS.B.3 | Constructed response

3. 218.96

Compute the whole-number product and restore decimal places: 32.2 × 6.8 = 218.96.

6.NS.B.3 | Constructed response

4. 2

Multiply both numbers by 10: 50 ÷ 25 = 2. Check: 2.5 × 2 = 5.

6.NS.B.3 | Constructed response

5. 16.827

Compute the whole-number product and restore decimal places: 2.37 × 7.1 = 16.827.

6.NS.B.3 | Constructed response

6. 8.4

Multiply both numbers by 10: 210 ÷ 25 = 8.4. Check: 2.5 × 8.4 = 21.

6.NS.B.3 | Constructed response

7. 1.72

Compute the whole-number product and restore decimal places: 4.3 × 0.4 = 1.72.

6.NS.B.3 | Constructed response

8. C - 49/10

Multiply both numbers by 100: 313.6 ÷ 64 = 4.9. Check: 0.64 × 4.9 = 3.136.

6.NS.B.3 | Multiple choice

9. C - 4681/10000

Compute the whole-number product and restore decimal places: 1.51 × 0.31 = 0.4681.

6.NS.B.3 | Multiple choice

10. 2.8 bags

Multiply both numbers by 100: 70 ÷ 25 = 2.8. Check: 0.25 × 2.8 = 0.7.

6.NS.B.3 | Constructed response

Day 012 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Interpret Whole-Number Remainders

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

1. 75

45 ÷ 6: write 7, subtract 42, remainder 3; bring down the next digit: 30 ÷ 6: write 5, subtract 30, remainder 0. Check: 6 × 75 = 450.

6.NS.B.2 | Constructed response

2. 56

44 ÷ 8: write 5, subtract 40, remainder 4; bring down the next digit: 48 ÷ 8: write 6, subtract 48, remainder 0. Check: 8 × 56 = 448.

6.NS.B.2 | Constructed response

3. 88

44 ÷ 5: write 8, subtract 40, remainder 4; bring down the next digit: 40 ÷ 5: write 8, subtract 40, remainder 0. Check: 5 × 88 = 440.

6.NS.B.2 | Constructed response

4. 289

66 ÷ 23: write 2, subtract 46, remainder 20; bring down the next digit: 204 ÷ 23: write 8, subtract 184, remainder 20; bring down the next digit: 207 ÷ 23: write 9, subtract 207, remainder 0. Check: 23 × 289 = 6647.

6.NS.B.2 | Constructed response

5. 275

63 ÷ 23: write 2, subtract 46, remainder 17; bring down the next digit: 172 ÷ 23: write 7, subtract 161, remainder 11; bring down the next digit: 115 ÷ 23: write 5, subtract 115, remainder 0. Check: 23 × 275 = 6325.

6.NS.B.2 | Constructed response

6. 179

23 ÷ 13: write 1, subtract 13, remainder 10; bring down the next digit: 102 ÷ 13: write 7, subtract 91, remainder 11; bring down the next digit: 117 ÷ 13: write 9, subtract 117, remainder 0. Check: 13 × 179 = 2327.

6.NS.B.2 | Constructed response

7. 384 R 13

115 ÷ 30: write 3, subtract 90, remainder 25; bring down the next digit: 253 ÷ 30: write 8, subtract 240, remainder 13; bring down the next digit: 133 ÷ 30: write 4, subtract 120, remainder 13. Check: 30 × 384 + 13 = 11533.

6.NS.B.2 | Constructed response

8. 376 R 19

135 ÷ 36: write 3, subtract 108, remainder 27; bring down the next digit: 275 ÷ 36: write 7, subtract 252, remainder 23; bring down the next digit: 235 ÷ 36: write 6, subtract 216, remainder 19. Check: 36 × 376 + 19 = 13555.

6.NS.B.2 | Model and constructed response

9. 125 R 12

42 ÷ 34: write 1, subtract 34, remainder 8; bring down the next digit: 86 ÷ 34: write 2, subtract 68, remainder 18; bring down the next digit: 182 ÷ 34: write 5, subtract 170, remainder 12. Check: 34 × 125 + 12 = 4262.

6.NS.B.2 | Model and constructed response

10. 264 full packets; 5 cards remain

6605 = 25 × 264 + 5.

6.NS.B.2 | Constructed response

Day 013 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Critique Decimal Calculations

Standards: 6.NS.B.3

Learning target: I can critique Decimal Calculations, and explain and check my reasoning.

1. 8.6

Align decimal points; 12.2 - 3.6 = 8.6.

6.NS.B.3 | Constructed response

2. 266.6

Compute the whole-number product and restore decimal places: 31 × 8.6 = 266.6.

6.NS.B.3 | Constructed response

3. 58

Multiply both numbers by 10: 522 ÷ 9 = 58. Check: 0.9 × 58 = 52.2.

6.NS.B.3 | Constructed response

4. 1.94

Align decimal points; 3.94 - 2 = 1.94.

6.NS.B.3 | Constructed response

5. 9.75

Compute the whole-number product and restore decimal places: 3.9 × 2.5 = 9.75.

6.NS.B.3 | Constructed response

6. 4.5

Multiply both numbers by 10: 54 ÷ 12 = 4.5. Check: 1.2 × 4.5 = 5.4.

6.NS.B.3 | Constructed response

7. 1.49

Align decimal points; 1.69 - 0.2 = 1.49.

6.NS.B.3 | Constructed response

8. C - 6783/10000

Compute the whole-number product and restore decimal places: 1.33 × 0.51 = 0.6783.

6.NS.B.3 | Multiple choice

9. D - 33/10

Multiply both numbers by 100: 39.6 ÷ 12 = 3.3. Check: 0.12 × 3.3 = 0.396.

6.NS.B.3 | Multiple choice

10. 3.63 L

Align decimal points; 4.46 - 0.83 = 3.63.

6.NS.B.3 | Constructed response

Day 014 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Model a Rational-Number Situation

Standards: 6.NS.B.2, 6.NS.B.3

Learning target: I can model a Rational-Number Situation, and explain and check my reasoning.

1. 43

12 ÷ 3: write 4, subtract 12, remainder 0; bring down the next digit: 9 ÷ 3: write 3, subtract 9, remainder 0. Check: 3 × 43 = 129.

6.NS.B.2 | Constructed response

2. 54.1

Align decimal points; 47.4 + 6.7 = 54.1.

6.NS.B.3 | Constructed response

3. 228.9

Compute the whole-number product and restore decimal places: 65.4 × 3.5 = 228.9.

6.NS.B.3 | Constructed response

4. 7.8

Multiply both numbers by 10: 93.6 ÷ 12 = 7.8. Check: 1.2 × 7.8 = 9.36.

6.NS.B.3 | Constructed response

5. 282

81 ÷ 29: write 2, subtract 58, remainder 23; bring down the next digit: 237 ÷ 29: write 8, subtract 232, remainder 5; bring down the next digit: 58 ÷ 29: write 2, subtract 58, remainder 0. Check: 29 × 282 = 8178.

6.NS.B.2 | Constructed response

6. 8.02

Align decimal points; 4.02 + 4 = 8.02.

6.NS.B.3 | Constructed response

7. 0.8355

Compute the whole-number product and restore decimal places: 5.57 × 0.15 = 0.8355.

6.NS.B.3 | Constructed response

8. B - 11/2

Multiply both numbers by 100: 82.5 ÷ 15 = 5.5. Check: 0.15 × 5.5 = 0.825.

6.NS.B.3 | Multiple choice

9. 367 R 2

84 ÷ 23: write 3, subtract 69, remainder 15; bring down the next digit: 154 ÷ 23: write 6, subtract 138, remainder 16; bring down the next digit: 163 ÷ 23: write 7, subtract 161, remainder 2. Check: 23 × 367 + 2 = 8443.

6.NS.B.2 | Model and constructed response

10. 1.58 km

Align decimal points; 1.31 + 0.27 = 1.58.

6.NS.B.3 | Constructed response

Day 015 | Unit 01 | Answer key

Positive Rational Numbers: Decimals and Division

Unit 1 Test

Standards: Grade 5 prerequisite, 6.NS.B.3, 6.NS.B.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 87

52 ÷ 6: write 8, subtract 48, remainder 4; bring down the next digit: 42 ÷ 6: write 7, subtract 42, remainder 0. Check: 6 × 87 = 522.

Grade 5 prerequisite | Constructed response

2. 71.9

Align decimal points; 66.2 + 5.7 = 71.9.

Grade 5 prerequisite | Constructed response

3. 60.8

Align decimal points; 68.2 - 7.4 = 60.8.

Grade 5 prerequisite | Constructed response

4. 1/2

1 × 1/2 = 1/2.

Grade 5 prerequisite | Constructed response

5. 20.2

Align decimal points; 18.7 + 1.5 = 20.2.

6.NS.B.3 | Constructed response

6. 28.2

Align decimal points; 31.6 - 3.4 = 28.2.

6.NS.B.3 | Constructed response

7. 39.9

Compute the whole-number product and restore decimal places: 26.6 × 1.5 = 39.9.

6.NS.B.3 | Constructed response

8. 281

70 ÷ 25: write 2, subtract 50, remainder 20; bring down the next digit: 202 ÷ 25: write 8, subtract 200, remainder 2; bring down the next digit: 25 ÷ 25: write 1, subtract 25, remainder 0. Check: 25 × 281 = 7025.

6.NS.B.2 | Constructed response

9. 168

60 ÷ 36: write 1, subtract 36, remainder 24; bring down the next digit: 244 ÷ 36: write 6, subtract 216, remainder 28; bring down the next digit: 288 ÷ 36: write 8, subtract 288, remainder 0. Check: 36 × 168 = 6048.

6.NS.B.2 | Constructed response

10. 109

38 ÷ 35: write 1, subtract 35, remainder 3; bring down the next digit: 31 ÷ 35: write 0, subtract 0, remainder 31; bring down the next digit: 315 ÷ 35: write 9, subtract 315, remainder 0. Check: 35 × 109 = 3815.

6.NS.B.2 | Constructed response

11. 5.6

Multiply both numbers by 10: 179.2 ÷ 32 = 5.6. Check: 3.2 × 5.6 = 17.92.

6.NS.B.3 | Constructed response

12. B - 17/10

Multiply both numbers by 10: 25.5 ÷ 15 = 1.7. Check: 1.5 × 1.7 = 2.55.

6.NS.B.3 | Multiple choice

13. C - 47/5

Align decimal points; 7.3 + 2.1 = 9.4.

6.NS.B.3 | Multiple choice

14. C - 313/50

Align decimal points; 5.58 + 0.68 = 6.26.

6.NS.B.3 | Multiple choice

15. A - 41/5

Multiply both numbers by 100: 196.8 ÷ 24 = 8.2. Check: 0.24 × 8.2 = 1.968.

6.NS.B.3 | Multiple choice

16. 264 R 1

71 ÷ 27: write 2, subtract 54, remainder 17; bring down the next digit: 172 ÷ 27: write 6, subtract 162, remainder 10; bring down the next digit: 109 ÷ 27: write 4, subtract 108, remainder 1. Check: 27 × 264 + 1 = 7129.

6.NS.B.2 | Model and constructed response

17. C - 3139/1000

Compute the whole-number product and restore decimal places: 4.3 × 0.73 = 3.139.

6.NS.B.3 | Multiple choice

18. A - 201/25

Align decimal points; 7.79 + 0.25 = 8.04.

6.NS.B.3 | Multiple choice

19. C - 671/100

Align decimal points; 6.12 + 0.59 = 6.71.

6.NS.B.3 | Multiple choice

20. B - 287/100

Align decimal points; 3.69 - 0.82 = 2.87.

6.NS.B.3 | Multiple choice

Day 016 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Multiply Fractions: Readiness

Standards: 6.NS.A.1

Learning target: I can use fraction multiplication to prepare for fraction division.

1. 1/2

1 × 1/2 = 1/2.

6.NS.A.1 | Constructed response

2. 1/16

1/4 × 1/4 = 1/16.

6.NS.A.1 | Constructed response

3. 1/4

1/2 × 1/2 = 1/4.

6.NS.A.1 | Constructed response

4. 1/2

2 × 1/4 = 1/2.

6.NS.A.1 | Constructed response

5. 8/3

8/5 × 5/3 = 8/3.

6.NS.A.1 | Constructed response

6. 1/10

1/2 × 1/5 = 1/10.

6.NS.A.1 | Constructed response

7. 1/8

1/2 × 1/4 = 1/8.

6.NS.A.1 | Constructed response

8. D - 1/5

1/2 × 2/5 = 1/5.

6.NS.A.1 | Multiple choice

9. D - 7/4

7/2 × 1/2 = 7/4.

6.NS.A.1 | Multiple choice

10. 9/20

3/4 × 3/5 = 9/20.

6.NS.A.1 | Constructed response

Day 017 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Multiply Mixed Numbers: Readiness

Standards: 6.NS.A.1

Learning target: I can use fraction multiplication to prepare for fraction division.

1. 25/8

5/4 × 5/2 = 25/8.

6.NS.A.1 | Constructed response

2. 10/3

5/2 × 4/3 = 10/3.

6.NS.A.1 | Constructed response

3. 21/4

7/2 × 3/2 = 21/4.

6.NS.A.1 | Constructed response

4. 5

15/4 × 4/3 = 5.

6.NS.A.1 | Constructed response

5. 14/3

7/2 × 4/3 = 14/3.

6.NS.A.1 | Constructed response

6. 9/4

3/2 × 3/2 = 9/4.

6.NS.A.1 | Constructed response

7. 10/3

5/2 × 4/3 = 10/3.

6.NS.A.1 | Constructed response

8. D - 5

15/4 × 4/3 = 5.

6.NS.A.1 | Multiple choice

9. B - 9/4

3/2 × 3/2 = 9/4.

6.NS.A.1 | Multiple choice

10. 13/3

13/4 × 4/3 = 13/3.

6.NS.A.1 | Constructed response

Day 018 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Model Fraction Division with Bars

Standards: 6.NS.A.1

Learning target: I can model fraction division as equal-sized groups.

1. 3/2

Each small part is 1/12; 9 shaded parts divided into groups of 6 gives 9/6 = 3/2 groups.

6.NS.A.1 | Constructed response

2. 1

Each small part is 1/4; 4 shaded parts divided into groups of 4 gives 4/4 = 1 groups.

6.NS.A.1 | Constructed response

3. 3/2

Each small part is 1/6; 3 shaded parts divided into groups of 2 gives 3/2 = 3/2 groups.

6.NS.A.1 | Constructed response

4. 1

Each small part is 1/6; 4 shaded parts divided into groups of 4 gives 4/4 = 1 groups.

6.NS.A.1 | Constructed response

5. 11/2

Each small part is 1/12; 11 shaded parts divided into groups of 2 gives 11/2 = 11/2 groups.

6.NS.A.1 | Constructed response

6. 9/5

Each small part is 1/12; 9 shaded parts divided into groups of 5 gives 9/5 = 9/5 groups.

6.NS.A.1 | Constructed response

7. 1

Each small part is 1/10; 4 shaded parts divided into groups of 4 gives 4/4 = 1 groups.

6.NS.A.1 | Constructed response

8. A - 4

Each small part is 1/6; 4 shaded parts divided into groups of 1 gives 4/1 = 4 groups.

6.NS.A.1 | Multiple choice

9. D - 8

Each small part is 1/8; 8 shaded parts divided into groups of 1 gives 8/1 = 8 groups.

6.NS.A.1 | Multiple choice

10. 6

Each small part is 1/8; 6 shaded parts divided into groups of 1 gives 6/1 = 6 groups.

6.NS.A.1 | Constructed response

Day 019 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Interpret a Partial Fraction Group

Standards: 6.NS.A.1

Learning target: I can model fraction division as equal-sized groups.

1. 2

Each small part is 1/12; 10 shaded parts divided into groups of 5 gives 10/5 = 2 groups.

6.NS.A.1 | Constructed response

2. 9/2

Each small part is 1/10; 9 shaded parts divided into groups of 2 gives 9/2 = 9/2 groups.

6.NS.A.1 | Constructed response

3. 2

Each small part is 1/4; 2 shaded parts divided into groups of 1 gives 2/1 = 2 groups.

6.NS.A.1 | Constructed response

4. 1

Each small part is 1/10; 7 shaded parts divided into groups of 7 gives 7/7 = 1 groups.

6.NS.A.1 | Constructed response

5. 2

Each small part is 1/12; 2 shaded parts divided into groups of 1 gives 2/1 = 2 groups.

6.NS.A.1 | Constructed response

6. 4

Each small part is 1/4; 4 shaded parts divided into groups of 1 gives 4/1 = 4 groups.

6.NS.A.1 | Constructed response

7. 8/3

Each small part is 1/10; 8 shaded parts divided into groups of 3 gives 8/3 = 8/3 groups.

6.NS.A.1 | Constructed response

8. D - 1

Each small part is 1/4; 4 shaded parts divided into groups of 4 gives 4/4 = 1 groups.

6.NS.A.1 | Multiple choice

9. B - 9/4

Each small part is 1/10; 9 shaded parts divided into groups of 4 gives 9/4 = 9/4 groups.

6.NS.A.1 | Multiple choice

10. 3/2

Each small part is 1/10; 6 shaded parts divided into groups of 4 gives 6/4 = 3/2 groups.

6.NS.A.1 | Constructed response

Day 020 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Connect Fraction Bars and Related Equations

Standards: 6.NS.A.1

Learning target: I can connect Fraction Bars and Related Equations, and explain and check my reasoning.

1. 11/8

Each small part is 1/12; 11 shaded parts divided into groups of 8 gives 11/8 = 11/8 groups.

6.NS.A.1 | Constructed response

2. n = 2; 1/2 ÷ 1/4 = 2

1/2 ÷ 1/4 = 1/2 × 4 = 2. Check: 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

3. 5/3

Each small part is 1/10; 10 shaded parts divided into groups of 6 gives 10/6 = 5/3 groups.

6.NS.A.1 | Constructed response

4. n = 8; 1 ÷ 1/8 = 8

1 ÷ 1/8 = 1 × 8 = 8. Check: 1/8 × 8 = 1.

6.NS.A.1 | Constructed response

5. 1

Each small part is 1/8; 3 shaded parts divided into groups of 3 gives 3/3 = 1 groups.

6.NS.A.1 | Constructed response

6. n = 2/3; 1/3 ÷ 1/2 = 2/3

1/3 ÷ 1/2 = 1/3 × 2 = 2/3. Check: 1/2 × 2/3 = 1/3.

6.NS.A.1 | Constructed response

7. 5/4

Each small part is 1/6; 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

6.NS.A.1 | Constructed response

8. n = 2; 4/5 ÷ 2/5 = 2

4/5 ÷ 2/5 = 4/5 × 5/2 = 2. Check: 2/5 × 2 = 4/5.

6.NS.A.1 | Model and constructed response

9. A - 7/4

Each small part is 1/8; 7 shaded parts divided into groups of 4 gives 7/4 = 7/4 groups.

6.NS.A.1 | Multiple choice

10. n = 6/5; 2 ÷ 5/3 = 6/5

2 ÷ 5/3 = 2 × 3/5 = 6/5. Check: 5/3 × 6/5 = 2.

6.NS.A.1 | Constructed response

Day 021 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Divide Fractions Using Reciprocals

Standards: 6.NS.A.1

Learning target: I can divide fractions and check using multiplication.

1. 3

3/4 ÷ 1/4 = 3/4 × 4 = 3.

6.NS.A.1 | Constructed response

2. 2

1/2 ÷ 1/4 = 1/2 × 4 = 2.

6.NS.A.1 | Constructed response

3. 2

1 ÷ 1/2 = 1 × 2 = 2.

6.NS.A.1 | Constructed response

4. 10

5/2 ÷ 1/4 = 5/2 × 4 = 10.

6.NS.A.1 | Constructed response

5. 16/3

4/3 ÷ 1/4 = 4/3 × 4 = 16/3.

6.NS.A.1 | Constructed response

6. 3/2

2 ÷ 4/3 = 2 × 3/4 = 3/2.

6.NS.A.1 | Constructed response

7. 1

1/2 ÷ 1/2 = 1/2 × 2 = 1.

6.NS.A.1 | Constructed response

8. C - 2

1 ÷ 1/2 = 1 × 2 = 2.

6.NS.A.1 | Multiple choice

9. A - 15/4

3/4 ÷ 1/5 = 3/4 × 5 = 15/4.

6.NS.A.1 | Multiple choice

10. 4/3

2/3 ÷ 1/2 = 2/3 × 2 = 4/3.

6.NS.A.1 | Constructed response

Day 022 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Check a Fraction Quotient

Standards: 6.NS.A.1

Learning target: I can check a Fraction Quotient, and explain and check my reasoning.

1. 2

1 ÷ 1/2 = 1 × 2 = 2.

6.NS.A.1 | Constructed response

2. n = 2; 1 ÷ 1/2 = 2

1 ÷ 1/2 = 1 × 2 = 2. Check: 1/2 × 2 = 1.

6.NS.A.1 | Constructed response

3. 2

1/2 ÷ 1/4 = 1/2 × 4 = 2.

6.NS.A.1 | Constructed response

4. n = 2; 1/2 ÷ 1/4 = 2

1/2 ÷ 1/4 = 1/2 × 4 = 2. Check: 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

5. 2/15

1/6 ÷ 5/4 = 1/6 × 4/5 = 2/15.

6.NS.A.1 | Constructed response

6. n = 9/4; 3/2 ÷ 2/3 = 9/4

3/2 ÷ 2/3 = 3/2 × 3/2 = 9/4. Check: 2/3 × 9/4 = 3/2.

6.NS.A.1 | Constructed response

7. 4/3

4/3 ÷ 1 = 4/3 × 1 = 4/3.

6.NS.A.1 | Constructed response

8. n = 8/5; 2 ÷ 5/4 = 8/5

2 ÷ 5/4 = 2 × 4/5 = 8/5. Check: 5/4 × 8/5 = 2.

6.NS.A.1 | Model and constructed response

9. B - 5/2

3/2 ÷ 3/5 = 3/2 × 5/3 = 5/2.

6.NS.A.1 | Multiple choice

10. n = 1/4; 1/3 ÷ 4/3 = 1/4

1/3 ÷ 4/3 = 1/3 × 3/4 = 1/4. Check: 4/3 × 1/4 = 1/3.

6.NS.A.1 | Constructed response

Day 023 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Divide Mixed Numbers

Standards: 6.NS.A.1

Learning target: I can divide mixed numbers after renaming them as fractions.

1. 3

3/2 ÷ 1/2 = 3/2 × 2 = 3.

6.NS.A.1 | Constructed response

2. 5/2

5/4 ÷ 1/2 = 5/4 × 2 = 5/2.

6.NS.A.1 | Constructed response

3. 10

5/2 ÷ 1/4 = 5/2 × 4 = 10.

6.NS.A.1 | Constructed response

4. 20/19

5/2 ÷ 19/8 = 5/2 × 8/19 = 20/19.

6.NS.A.1 | Constructed response

5. 11/18

11/6 ÷ 3 = 11/6 × 1/3 = 11/18.

6.NS.A.1 | Constructed response

6. 26/11

13/4 ÷ 11/8 = 13/4 × 8/11 = 26/11.

6.NS.A.1 | Constructed response

7. 9/20

3/2 ÷ 10/3 = 3/2 × 3/10 = 9/20.

6.NS.A.1 | Constructed response

8. B - 18/13

18/5 ÷ 13/5 = 18/5 × 5/13 = 18/13.

6.NS.A.1 | Multiple choice

9. A - 7/12

7/4 ÷ 3 = 7/4 × 1/3 = 7/12.

6.NS.A.1 | Multiple choice

10. 31/24

31/8 ÷ 3 = 31/8 × 1/3 = 31/24.

6.NS.A.1 | Constructed response

Day 024 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Interpret Mixed-Number Quotients

Standards: 6.NS.A.1

Learning target: I can divide mixed numbers after renaming them as fractions.

1. 9/2

9/4 ÷ 1/2 = 9/4 × 2 = 9/2.

6.NS.A.1 | Constructed response

2. 4

2 ÷ 1/2 = 2 × 2 = 4.

6.NS.A.1 | Constructed response

3. 5

5/2 ÷ 1/2 = 5/2 × 2 = 5.

6.NS.A.1 | Constructed response

4. 85/72

17/8 ÷ 9/5 = 17/8 × 5/9 = 85/72.

6.NS.A.1 | Constructed response

5. 3/2

3 ÷ 2 = 3 × 1/2 = 3/2.

6.NS.A.1 | Constructed response

6. 8/5

2 ÷ 5/4 = 2 × 4/5 = 8/5.

6.NS.A.1 | Constructed response

7. 10/27

5/3 ÷ 9/2 = 5/3 × 2/9 = 10/27.

6.NS.A.1 | Constructed response

8. D - 13/3

13/2 ÷ 3/2 = 13/2 × 2/3 = 13/3.

6.NS.A.1 | Multiple choice

9. A - 16/9

4 ÷ 9/4 = 4 × 4/9 = 16/9.

6.NS.A.1 | Multiple choice

10. 4/3

4 ÷ 3 = 4 × 1/3 = 4/3.

6.NS.A.1 | Constructed response

Day 025 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Explain Fraction Quotients in Context

Standards: 6.NS.A.1

Learning target: I can explain Fraction Quotients in Context, and explain and check my reasoning.

1. 1/2

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

2. 5/2

5/4 ÷ 1/2 = 5/4 × 2 = 5/2.

6.NS.A.1 | Constructed response

3. 2

1/2 ÷ 1/4 = 1/2 × 4 = 2.

6.NS.A.1 | Constructed response

4. 9/5

3 ÷ 5/3 = 3 × 3/5 = 9/5.

6.NS.A.1 | Constructed response

5. 3/10

1/5 ÷ 2/3 = 1/5 × 3/2 = 3/10.

6.NS.A.1 | Constructed response

6. 13/12

13/6 ÷ 2 = 13/6 × 1/2 = 13/12.

6.NS.A.1 | Constructed response

7. 4

1/2 ÷ 1/8 = 1/2 × 8 = 4.

6.NS.A.1 | Constructed response

8. A - 27/16

27/8 ÷ 2 = 27/8 × 1/2 = 27/16.

6.NS.A.1 | Multiple choice

9. D - 8

1 ÷ 1/8 = 1 × 8 = 8.

6.NS.A.1 | Multiple choice

10. 12/5

4 ÷ 5/3 = 4 × 3/5 = 12/5.

6.NS.A.1 | Constructed response

Day 026 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Write Fraction-Division Equations

Standards: 6.NS.A.1

Learning target: I can connect a fraction quotient to a missing factor.

1. n = 3/2; 3/4 ÷ 1/2 = 3/2

3/4 ÷ 1/2 = 3/4 × 2 = 3/2. Check: 1/2 × 3/2 = 3/4.

6.NS.A.1 | Constructed response

2. n = 1; 1/4 ÷ 1/4 = 1

1/4 ÷ 1/4 = 1/4 × 4 = 1. Check: 1/4 × 1 = 1/4.

6.NS.A.1 | Constructed response

3. n = 4; 1 ÷ 1/4 = 4

1 ÷ 1/4 = 1 × 4 = 4. Check: 1/4 × 4 = 1.

6.NS.A.1 | Constructed response

4. n = 5/6; 1/2 ÷ 3/5 = 5/6

1/2 ÷ 3/5 = 1/2 × 5/3 = 5/6. Check: 3/5 × 5/6 = 1/2.

6.NS.A.1 | Constructed response

5. n = 7/8; 7/8 ÷ 1 = 7/8

7/8 ÷ 1 = 7/8 × 1 = 7/8. Check: 1 × 7/8 = 7/8.

6.NS.A.1 | Constructed response

6. n = 1/2; 1 ÷ 2 = 1/2

1 ÷ 2 = 1 × 1/2 = 1/2. Check: 2 × 1/2 = 1.

6.NS.A.1 | Constructed response

7. n = 1; 1 ÷ 1 = 1

1 ÷ 1 = 1 × 1 = 1. Check: 1 × 1 = 1.

6.NS.A.1 | Constructed response

8. n = 6/5; 2/5 ÷ 1/3 = 6/5

2/5 ÷ 1/3 = 2/5 × 3 = 6/5. Check: 1/3 × 6/5 = 2/5.

6.NS.A.1 | Model and constructed response

9. n = 8/15; 2/3 ÷ 5/4 = 8/15

2/3 ÷ 5/4 = 2/3 × 4/5 = 8/15. Check: 5/4 × 8/15 = 2/3.

6.NS.A.1 | Model and constructed response

10. n = 3/2; 6/5 ÷ 4/5 = 3/2

6/5 ÷ 4/5 = 6/5 × 5/4 = 3/2. Check: 4/5 × 3/2 = 6/5.

6.NS.A.1 | Constructed response

Day 027 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Solve Fraction-Division Stories

Standards: 6.NS.A.1

Learning target: I can write and solve a fraction-division equation for a situation.

1. 1 piece-lengths

1/2 ÷ 1/2 = 1/2 × 2 = 1.

6.NS.A.1 | Constructed response

2. 4 piece-lengths

1 ÷ 1/4 = 1 × 4 = 4.

6.NS.A.1 | Constructed response

3. 1/2 piece-lengths

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

4. 32/15 piece-lengths

4/3 ÷ 5/8 = 4/3 × 8/5 = 32/15.

6.NS.A.1 | Constructed response

5. 35/8 piece-lengths

7/4 ÷ 2/5 = 7/4 × 5/2 = 35/8.

6.NS.A.1 | Constructed response

6. 3/4 piece-lengths

1/4 ÷ 1/3 = 1/4 × 3 = 3/4.

6.NS.A.1 | Constructed response

7. 13 bottles; 1/15 L left

8/3 ÷ 1/5 = 8/3 × 5 = 40/3. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response

8. 0 bottles; 1/2 L left

1/2 ÷ 4/3 = 1/2 × 3/4 = 3/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

9. 3 bottles; 1/10 L left

5/2 ÷ 4/5 = 5/2 × 5/4 = 25/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

10. 2 bottles; 1/5 L left

1 ÷ 2/5 = 1 × 5/2 = 5/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response

Day 028 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Find Full Containers and Leftovers

Standards: 6.NS.A.1

Learning target: I can write and solve a fraction-division equation for a situation.

1. 1/2 piece-lengths

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

2. 3 piece-lengths

3/4 ÷ 1/4 = 3/4 × 4 = 3.

6.NS.A.1 | Constructed response

3. 2 piece-lengths

1 ÷ 1/2 = 1 × 2 = 2.

6.NS.A.1 | Constructed response

4. 5/3 piece-lengths

2/3 ÷ 2/5 = 2/3 × 5/2 = 5/3.

6.NS.A.1 | Constructed response

5. 3/10 piece-lengths

1/2 ÷ 5/3 = 1/2 × 3/5 = 3/10.

6.NS.A.1 | Constructed response

6. 1 piece-lengths

3/4 ÷ 3/4 = 3/4 × 4/3 = 1.

6.NS.A.1 | Constructed response

7. 1 bottles; 1/2 L left

3/2 ÷ 1 = 3/2 × 1 = 3/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response

8. 2 bottles; 1/2 L left

7/2 ÷ 3/2 = 7/2 × 2/3 = 7/3. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

9. 12 bottles; 0 L left

4 ÷ 1/3 = 4 × 3 = 12. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

10. 2 bottles; 1/12 L left

3/4 ÷ 1/3 = 3/4 × 3 = 9/4. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Constructed response

Day 029 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Explain Fraction Models and Algorithms

Standards: 6.NS.A.1

Learning target: I can explain Fraction Models and Algorithms, and explain and check my reasoning.

1. 3/2

Each small part is 1/4; 3 shaded parts divided into groups of 2 gives 3/2 = 3/2 groups.

6.NS.A.1 | Constructed response

2. 3/2

3/4 ÷ 1/2 = 3/4 × 2 = 3/2.

6.NS.A.1 | Constructed response

3. 7/2

7/4 ÷ 1/2 = 7/4 × 2 = 7/2.

6.NS.A.1 | Constructed response

4. 3/5 piece-lengths

3/4 ÷ 5/4 = 3/4 × 4/5 = 3/5.

6.NS.A.1 | Constructed response

5. 1

Each small part is 1/8; 2 shaded parts divided into groups of 2 gives 2/2 = 1 groups.

6.NS.A.1 | Constructed response

6. 1/5

1/4 ÷ 5/4 = 1/4 × 4/5 = 1/5.

6.NS.A.1 | Constructed response

7. 33/64

11/8 ÷ 8/3 = 11/8 × 3/8 = 33/64.

6.NS.A.1 | Constructed response

8. 0 bottles; 3/8 L left

3/8 ÷ 1 = 3/8 × 1 = 3/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

9. D - 5/4

Each small part is 1/8; 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

6.NS.A.1 | Multiple choice

10. 16/5

2 ÷ 5/8 = 2 × 8/5 = 16/5.

6.NS.A.1 | Constructed response

Day 030 | Unit 02 | Answer key

Positive Rational Numbers: Fractions and Mixed Numbers

Unit 2 Test

Standards: 6.NS.B.3, 6.NS.B.2, 6.NS.A.1

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 29.5

Align decimal points; 20.8 + 8.7 = 29.5.

6.NS.B.3 | Constructed response

2. 33.8

Align decimal points; 35.3 - 1.5 = 33.8.

6.NS.B.3 | Constructed response

3. 783.72

Compute the whole-number product and restore decimal places: 93.3 × 8.4 = 783.72.

6.NS.B.3 | Constructed response

4. 50

35 ÷ 7: write 5, subtract 35, remainder 0; bring down the next digit: 0 ÷ 7: write 0, subtract 0, remainder 0. Check: 7 × 50 = 350.

6.NS.B.2 | Constructed response

5. 1/16

1/4 × 1/4 = 1/16.

6.NS.A.1 | Constructed response

6. 45/8

9/4 × 5/2 = 45/8.

6.NS.A.1 | Constructed response

7. 8

Each small part is 1/8; 8 shaded parts divided into groups of 1 gives 8/1 = 8 groups.

6.NS.A.1 | Constructed response

8. 3/2

Each small part is 1/8; 3 shaded parts divided into groups of 2 gives 3/2 = 3/2 groups.

6.NS.A.1 | Constructed response

9. 4/3

Each small part is 1/8; 4 shaded parts divided into groups of 3 gives 4/3 = 4/3 groups.

6.NS.A.1 | Constructed response

10. 32/5

8/5 ÷ 1/4 = 8/5 × 4 = 32/5.

6.NS.A.1 | Constructed response

11. 15/4

3/4 ÷ 1/5 = 3/4 × 5 = 15/4.

6.NS.A.1 | Constructed response

12. C - 38/15

19/6 ÷ 5/4 = 19/6 × 4/5 = 38/15.

6.NS.A.1 | Multiple choice

13. B - 52/33

13/3 ÷ 11/4 = 13/3 × 4/11 = 52/33.

6.NS.A.1 | Multiple choice

14. D - 3/4

1/2 ÷ 2/3 = 1/2 × 3/2 = 3/4.

6.NS.A.1 | Multiple choice

15. n = 35/9; 7/3 ÷ 3/5 = 35/9

7/3 ÷ 3/5 = 7/3 × 5/3 = 35/9. Check: 3/5 × 35/9 = 7/3.

6.NS.A.1 | Model and constructed response

16. 0 bottles; 1/3 L left

1/3 ÷ 2 = 1/3 × 1/2 = 1/6. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

17. 2 bottles; 0 L left

1 ÷ 1/2 = 1 × 2 = 2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

18. B - 1

2/3 ÷ 2/3 = 2/3 × 3/2 = 1.

6.NS.A.1 | Multiple choice

19. A - 1/4

1/2 × 1/2 = 1/4.

6.NS.A.1 | Multiple choice

20. B - 11/3

11/4 × 4/3 = 11/3.

6.NS.A.1 | Multiple choice

Day 031 | Unit 03 | Answer key

Integers and Rational Numbers

Represent Signed Quantities and Zero

Standards: 6.NS.C.5

Learning target: I can use positive numbers, negative numbers, and zero in context.

1. +14 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

2. -9 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

3. +10 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

4. -10 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. +35 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

6. -28 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

7. +15 m; 0 means at sea level; opposite -15 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

8. A, C - Its signed position is -2 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -2. Zero identifies the reference; the opposite is 2 m above it.

6.NS.C.5 | Multiple select

9. B, D - A reading of 0 m means at the reference point.; Its signed position is -5 m.

Below the chosen zero is negative, so the position is -5. Zero identifies the reference; the opposite is 5 m above it.

6.NS.C.5 | Multiple select

10. -15 m; 0 means at sea level; opposite +15 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

Day 032 | Unit 03 | Answer key

Integers and Rational Numbers

Recognize Opposites

Standards: 6.NS.C.6a

Learning target: I can identify opposites and the opposite of an opposite.

1. 3

-3 and 3 are each 3 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

2. 6

-6 and 6 are each 6 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

3. 5

-5 and 5 are each 5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

4. 4/5; opposite again -4/5

-4/5 and 4/5 are each 4/5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

5. 3/4; opposite again -3/4

-3/4 and 3/4 are each 3/4 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

6. 2; opposite again -2

-2 and 2 are each 2 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

7. 7; opposite again -7

-7 and 7 are each 7 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

8. A, D - The opposite of the opposite of -5/2 is -5/2.; The opposite of -5/2 is 5/2.

An opposite reverses direction without changing distance. Reversing twice returns to -5/2. Zero is its own opposite.

6.NS.C.6a | Multiple select

9. A, D - The opposite of the opposite of -7/2 is -7/2.; The opposite of -7/2 is 7/2.

An opposite reverses direction without changing distance. Reversing twice returns to -7/2. Zero is its own opposite.

6.NS.C.6a | Multiple select

10. 7/5; opposite again -7/5

-7/5 and 7/5 are each 7/5 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

Day 033 | Unit 03 | Answer key

Integers and Rational Numbers

Locate Integers on Number Lines

Standards: 6.NS.C.6c

Learning target: I can locate integers and negative fractions on a number line.

1. -8 < 0 < 7; unit steps

Negative -8 is left of or below 0; 7 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

2. -5 < 0 < 5; unit steps

Negative -5 is left of or below 0; 5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

3. -3 < 0 < 2; unit steps

Negative -3 is left of or below 0; 2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

4. -2 < 0 < 3/4; use steps of 1/4

Negative -2 is left of or below 0; 3/4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

5. -1/2 < 0 < 3/4; use steps of 1/4

Negative -1/2 is left of or below 0; 3/4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

6. -7/10 < 0 < 3/10; use steps of 1/10

Negative -7/10 is left of or below 0; 3/10 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

7. -5/4 < 0 < 1/2; use steps of 1/4

Negative -5/4 is left of or below 0; 1/2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

8. Leftmost -11/2; midpoint -11/2

-11/2 = -5 - 1/2 lies halfway between -6 and -5, left of both -5 and 0.

6.NS.C.6c | Graphing and two-part response

9. Leftmost -15/2; midpoint -15/2

-15/2 = -7 - 1/2 lies halfway between -8 and -7, left of both -7 and 0.

6.NS.C.6c | Graphing and two-part response

10. -3/4 < 0 < 1; use steps of 1/4

Negative -3/4 is left of or below 0; 1 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

Day 034 | Unit 03 | Answer key

Integers and Rational Numbers

Locate Negative Fractions

Standards: 6.NS.C.6c

Learning target: I can locate integers and negative fractions on a number line.

1. -3 < 0 < 2; unit steps

Negative -3 is left of or below 0; 2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

2. -8 < 0 < 3; unit steps

Negative -8 is left of or below 0; 3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

3. -2 < 0 < 7; unit steps

Negative -2 is left of or below 0; 7 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

4. -1 < 0 < 2/5; use steps of 1/5

Negative -1 is left of or below 0; 2/5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

5. -2 < 0 < 7/2; use steps of 1/2

Negative -2 is left of or below 0; 7/2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

6. -7/5 < 0 < 3/5; use steps of 1/5

Negative -7/5 is left of or below 0; 3/5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

7. -7/4 < 0 < 7/4; use steps of 1/4

Negative -7/4 is left of or below 0; 7/4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

8. Leftmost -13/2; midpoint -13/2

-13/2 = -6 - 1/2 lies halfway between -7 and -6, left of both -6 and 0.

6.NS.C.6c | Graphing and two-part response

9. Leftmost -7/2; midpoint -7/2

-7/2 = -3 - 1/2 lies halfway between -4 and -3, left of both -3 and 0.

6.NS.C.6c | Graphing and two-part response

10. -2/5 < 0 < 2/5; use steps of 1/5

Negative -2/5 is left of or below 0; 2/5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

Day 035 | Unit 03 | Answer key

Integers and Rational Numbers

Order Negative Fractions with Benchmarks

Standards: 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can order Negative Fractions with Benchmarks, and explain and check my reasoning.

1. -8 < 0 < 4; unit steps

Negative -8 is left of or below 0; 4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

2. =

-3 = -3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. -6 < 0 < 2; unit steps

Negative -6 is left of or below 0; 2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

4. <

-2 < -1/2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

5. -2/3 < 0 < 4/3; use steps of 1/3

Negative -2/3 is left of or below 0; 4/3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

6. >

-3/5 > -4/3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. -5/4 < 0 < 3/4; use steps of 1/4

Negative -5/4 is left of or below 0; 3/4 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

8. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

9. Leftmost -13/2; midpoint -13/2

-13/2 = -6 - 1/2 lies halfway between -7 and -6, left of both -6 and 0.

6.NS.C.6c | Graphing and two-part response

10. -7/5 < -2/5; -2/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

Day 036 | Unit 03 | Answer key

Integers and Rational Numbers

Compare and Order Rational Numbers

Standards: 6.NS.C.7a, 6.NS.C.7b

Learning target: I can order rational numbers using their positions.

1. <

-8 < -6; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

2. <

-8 < -2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. =

-7 = -7; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

4. >

-3/10 > -5/2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

5. <

-7/2 < -7/5; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

6. >

-1/5 > -2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. -3/10 = -3/10; equal temperatures

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

8. C, D - -4 < -2; -2 lies to the right of -4 on a number line.

Among negatives, the number closer to zero is greater. Thus -2 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

9. B, C - -5 < -3; -3 lies to the right of -5 on a number line.

Among negatives, the number closer to zero is greater. Thus -3 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

10. -7/4 > -5/2; -7/4°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

Day 037 | Unit 03 | Answer key

Integers and Rational Numbers

Interpret Order in Context

Standards: 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.5

Learning target: I can interpret Order in Context, and explain and check my reasoning.

1. <

-5 < -2; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

2. -12 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

3. >

-3 > -4; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

4. -32 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. <

-4 < -1; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

6. -8 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

7. -1/2 < -1/5; -1/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

8. A, B - Its signed position is -3 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -3. Zero identifies the reference; the opposite is 3 m above it.

6.NS.C.5 | Multiple select

9. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

10. -42 m; 0 means at sea level; opposite +42 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

Day 038 | Unit 03 | Answer key

Integers and Rational Numbers

Understand Absolute Value

Standards: 6.NS.C.7c, 6.NS.C.7d

Learning target: I can distinguish a number from its distance to zero.

1. 21

-21 is 21 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

2. 14

-14 is 14 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

3. 24

-24 is 24 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

4. 9/2; 9/2 > 3/2

-9/2 is 9/2 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

5. 5; 5 > 5/2

-5 is 5 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

6. 3; 3 > 1

-3 is 3 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

7. Diver: 6 m from sea level; bird has greater elevation

|-6| = 6 > |2| = 2, but -6 < 2.

6.NS.C.7c, 6.NS.C.7d | Constructed response

8. A, B - |-4| = 4; -4 is farther from zero than 2.

Absolute value measures distance: 4 versus 2. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

9. A, C - -6 is farther from zero than 4.; |-6| = 6

Absolute value measures distance: 6 versus 4. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

10. Diver: 8 m from sea level; bird has greater elevation

|-8| = 8 > |4| = 4, but -8 < 4.

6.NS.C.7c, 6.NS.C.7d | Constructed response

Day 039 | Unit 03 | Answer key

Integers and Rational Numbers

Compare Order and Distance from Zero

Standards: 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can compare Order and Distance from Zero, and explain and check my reasoning.

1. 40

-40 is 40 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

2. <

-5 < -4; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. 9

-9 is 9 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

4. >

-3/5 > -6/5; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

5. 9/2; 9/2 > 3/2

-9/2 is 9/2 units from 0; distance is nonnegative.

6.NS.C.7c, 6.NS.C.7d | Constructed response

6. >

-1 > -4/3; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. Diver: 15 m from sea level; bird has greater elevation

|-15| = 15 > |5/2| = 5/2, but -15 < 5/2.

6.NS.C.7c, 6.NS.C.7d | Constructed response

8. B, C - -6 < -4; -4 lies to the right of -6 on a number line.

Among negatives, the number closer to zero is greater. Thus -4 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

9. A, C - -6 is farther from zero than 4.; |-6| = 6

Absolute value measures distance: 6 versus 4. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

10. -2 < -2/5; -2/5°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

Day 040 | Unit 03 | Answer key

Integers and Rational Numbers

Interpret Points on Axes and in Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

1. Quadrant III

Move 7 left, then 5 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

2. Quadrant IV

Move 5 right, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

3. Quadrant II

Move 8 left, then 2 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

4. Quadrant IV

Move 5 right, then 6 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

5. Quadrant III

Move 8 left, then 4 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

6. Quadrant I

Move 3 right, then 4 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

7. Quadrant III; Q(0, -1)

Move 3/2 left, then 1 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

8. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

9. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

10. Quadrant IV; Q(0, -3/2)

Move 1 right, then 3/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

Day 041 | Unit 03 | Answer key

Integers and Rational Numbers

Plot Points in Four Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

1. Quadrant III

Move 8 left, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

2. Quadrant IV

Move 2 right, then 5 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

3. Quadrant II

Move 3 left, then 2 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

4. Quadrant IV

Move 6 right, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

5. Quadrant III

Move 6 left, then 7 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

6. Quadrant I

Move 2 right, then 3 up.

6.NS.C.6b, 6.NS.C.6c | Constructed response

7. Quadrant III; Q(0, -3/2)

Move 5/2 left, then 3/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

8. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

9. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

10. Quadrant IV; Q(0, -3/2)

Move 3 right, then 3/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

Day 042 | Unit 03 | Answer key

Integers and Rational Numbers

Reflect Points Across Axes

Standards: 6.NS.C.6b

Learning target: I can reflect points across an axis using signs.

1. (-3, -4)

Reverse the y sign: (-3, -4).

6.NS.C.6b | Constructed response

2. (4, -7)

Reverse the x sign: (4, -7).

6.NS.C.6b | Constructed response

3. (-8, -6)

Reverse the y sign: (-8, -6).

6.NS.C.6b | Constructed response

4. (7, -7)

Reverse the x sign: (7, -7).

6.NS.C.6b | Constructed response

5. (-4, -3)

Reverse the y sign: (-4, -3).

6.NS.C.6b | Constructed response

6. (7, -3)

Reverse the x sign: (7, -3).

6.NS.C.6b | Constructed response

7. (8, -4)

Reverse both signs: (8, -4).

6.NS.C.6b | Constructed response

8. A, C - Across the y-axis: (6, 4).; Across the x-axis: (-6, -4).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (6, -4).

6.NS.C.6b | Multiple select

9. B, C - Across the x-axis: (-2, -2).; Across the y-axis: (2, 2).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (2, -2).

6.NS.C.6b | Multiple select

10. (8, 4)

Reverse both signs: (8, 4).

6.NS.C.6b | Constructed response

Day 043 | Unit 03 | Answer key

Integers and Rational Numbers

Find Horizontal and Vertical Distance

Standards: 6.NS.C.8

Learning target: I can graph points and find horizontal or vertical distance.

1. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

2. 2 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 2.

6.NS.C.8 | Constructed response

3. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

4. 11 units

Only one coordinate changes. Opposite sides of 0: 5 + 6 = 11.

6.NS.C.8 | Constructed response

5. 9 units

Only one coordinate changes. Opposite sides of 0: 7 + 2 = 9.

6.NS.C.8 | Constructed response

6. 12 units

Only one coordinate changes. Opposite sides of 0: 7 + 5 = 12.

6.NS.C.8 | Constructed response

7. 5 units

Only one coordinate changes. Opposite sides of 0: 3 + 2 = 5.

6.NS.C.8 | Constructed response

8. AB = 9 units; AC = 10 units

AB is horizontal: 6 + 3 = 9. AC is vertical: 5 + 5 = 10. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

9. AB = 8 units; AC = 12 units

AB is horizontal: 5 + 3 = 8. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. 12 units

Only one coordinate changes. Opposite sides of 0: 5 + 7 = 12.

6.NS.C.8 | Constructed response

Day 044 | Unit 03 | Answer key

Integers and Rational Numbers

Solve Coordinate-Plane Problems

Standards: 6.NS.C.6b, 6.NS.C.6c, 6.NS.C.8

Learning target: I can solve Coordinate-Plane Problems, and explain and check my reasoning.

1. Quadrant III

Move 6 left, then 3 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

2. (5, -3)

Reverse the x sign: (5, -3).

6.NS.C.6b | Constructed response

3. 5 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 5.

6.NS.C.8 | Constructed response

4. Quadrant IV

Move 8 right, then 5 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

5. (-4, -2)

Reverse the y sign: (-4, -2).

6.NS.C.6b | Constructed response

6. 10 units

Only one coordinate changes. Opposite sides of 0: 8 + 2 = 10.

6.NS.C.8 | Constructed response

7. Quadrant III; Q(0, -2)

Move 5/2 left, then 2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

8. A, B - Across the y-axis: (5, 4).; Across the x-axis: (-5, -4).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (5, -4).

6.NS.C.6b | Multiple select

9. AB = 11 units; AC = 6 units

AB is horizontal: 6 + 5 = 11. AC is vertical: 3 + 3 = 6. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. Quadrant IV; Q(0, -7/2)

Move 3 right, then 7/2 down.

6.NS.C.6b, 6.NS.C.6c | Constructed response

Day 045 | Unit 03 | Answer key

Integers and Rational Numbers

Unit 3 Test

Standards: 6.NS.A.1, 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.6b, 6.NS.C.8

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 3/8

3/4 × 1/2 = 3/8.

6.NS.A.1 | Constructed response

2. 13/3

13/4 × 4/3 = 13/3.

6.NS.A.1 | Constructed response

3. 11/7

Each small part is 1/12; 11 shaded parts divided into groups of 7 gives 11/7 = 11/7 groups.

6.NS.A.1 | Constructed response

4. 2

Each small part is 1/6; 6 shaded parts divided into groups of 3 gives 6/3 = 2 groups.

6.NS.A.1 | Constructed response

5. +8 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

6. 8

-8 and 8 are each 8 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

7. -3 < 0 < 5; unit steps

Negative -3 is left of or below 0; 5 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

8. -8/3 < 0 < 4/3; use steps of 1/3

Negative -8/3 is left of or below 0; 4/3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

9. -1 < 0 < 3/2; use steps of 1/4

Negative -1 is left of or below 0; 3/2 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

10. >

-4/5 > -1; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

11. <

-4/5 < -7/10; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

12. C, D - |-8| = 8; -8 is farther from zero than 6.

Absolute value measures distance: 8 versus 6. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

13. B, C - |-5| = 5; -5 is farther from zero than 3.

Absolute value measures distance: 5 versus 3. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

14. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

15. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

16. A, D - Across the y-axis: (6, 3).; Across the x-axis: (-6, -3).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (6, -3).

6.NS.C.6b | Multiple select

17. AB = 7 units; AC = 12 units

AB is horizontal: 5 + 2 = 7. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

18. A, D - Across the x-axis: (-5, -5).; Across the y-axis: (5, 5).

The x-axis reflection changes y only; the y-axis reflection changes x only. Across both gives (5, -5).

6.NS.C.6b | Multiple select

19. A, D - Its signed position is -7 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -7. Zero identifies the reference; the opposite is 7 m above it.

6.NS.C.5 | Multiple select

20. A, D - The opposite of the opposite of -7/2 is -7/2.; The opposite of -7/2 is 7/2.

An opposite reverses direction without changing distance. Reversing twice returns to -7/2. Zero is its own opposite.

6.NS.C.6a | Multiple select

Day 046 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Write and Evaluate Powers

Standards: 6.EE.A.1

Learning target: I can write repeated multiplication using a whole-number exponent.

1. 3 × 3 × 3 = 27

The exponent counts 3 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response

2. 2 × 2 × 2 × 2 = 16

The exponent counts 4 factors of 2; multiply all factors.

6.EE.A.1 | Constructed response

3. 4 × 4 × 4 = 64

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

4. 6 × 6 × 6 × 6 = 1296

The exponent counts 4 factors of 6; multiply all factors.

6.EE.A.1 | Constructed response

5. 7 × 7 = 49

The exponent counts 2 factors of 7; multiply all factors.

6.EE.A.1 | Constructed response

6. 2 × 2 = 4

The exponent counts 2 factors of 2; multiply all factors.

6.EE.A.1 | Constructed response

7. 4 × 4 × 4 = 64; 4 × 3 = 12

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

8. B, D - 27; 3 × 3 × 3

The exponent 3 means three factors of 3; their product is 27. It does not mean 3 times 3.

6.EE.A.1 | Multiple select

9. B, D - 6 × 6 × 6; 216

The exponent 3 means three factors of 6; their product is 216. It does not mean 6 times 3.

6.EE.A.1 | Multiple select

10. 3 × 3 × 3 = 27; 3 × 3 = 9

The exponent counts 3 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response

Day 047 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Use the Order of Operations

Standards: 6.EE.A.1, 6.EE.A.2c

Learning target: I can evaluate expressions using the order of operations.

1. 15

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 15.

6.EE.A.1, 6.EE.A.2c | Constructed response

2. 33

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 33.

6.EE.A.1, 6.EE.A.2c | Constructed response

3. 40

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 40.

6.EE.A.1, 6.EE.A.2c | Constructed response

4. 468

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 468.

6.EE.A.1, 6.EE.A.2c | Constructed response

5. 117

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 117.

6.EE.A.1, 6.EE.A.2c | Constructed response

6. 96

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 96.

6.EE.A.1, 6.EE.A.2c | Constructed response

7. 21

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 21.

6.EE.A.1, 6.EE.A.2c | Constructed response

8. 20

Parentheses: 7 - 3 = 4. Exponent: $2^2 = 4$. Multiply: $4 \times 4 = 16$. Add: $4 + 16 = 20$. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

9. 46

Parentheses: 5 - 3 = 2. Exponent: $6^2 = 36$. Multiply: $5 \times 2 = 10$. Add: $36 + 10 = 46$. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

10. 66

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 66.

6.EE.A.1, 6.EE.A.2c | Constructed response

Day 048 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Find Greatest Common Factors

Standards: 6.NS.B.4a

Learning target: I can find the greatest common factor and interpret equal groups.

1. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

2. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

3. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

4. 15

Common factors: 1, 3, 5, 15; greatest is 15.

6.NS.B.4a | Constructed response

5. 10

Common factors: 1, 2, 5, 10; greatest is 10.

6.NS.B.4a | Constructed response

6. 14

Common factors: 1, 2, 7, 14; greatest is 14.

6.NS.B.4a | Constructed response

7. 12 kits; 2 pencils and 3 erasers each

GCF(24, 36) = 12; divide both counts by 12.

6.NS.B.4a | Constructed response

8. A, C - 9 is the largest number of identical no-leftover kits from 18 pencils and 27 erasers.; 9 divides both 18 and 27.

GCF(18, 27) = 9. Divide each count by 9: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

9. A, B - 15 is the largest number of identical no-leftover kits from 30 pencils and 45 erasers.; 15 divides both 30 and 45.

GCF(30, 45) = 15. Divide each count by 15: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

10. 6 kits; 3 pencils and 5 erasers each

GCF(18, 30) = 6; divide both counts by 6.

6.NS.B.4a | Constructed response

Day 049 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Find Least Common Multiples

Standards: 6.NS.B.4b

Learning target: I can find the least common multiple of two numbers no greater than 12.

1. 4

The first positive multiple shared by 2 and 4 is 4.

6.NS.B.4b | Constructed response

2. 12

The first positive multiple shared by 4 and 6 is 12.

6.NS.B.4b | Constructed response

3. 6

The first positive multiple shared by 2 and 6 is 6.

6.NS.B.4b | Constructed response

4. 6

The first positive multiple shared by 6 and 2 is 6.

6.NS.B.4b | Constructed response

5. 99

The first positive multiple shared by 9 and 11 is 99.

6.NS.B.4b | Constructed response

6. 63

The first positive multiple shared by 7 and 9 is 63.

6.NS.B.4b | Constructed response

7. 8 seconds

The first positive multiple shared by 8 and 2 is 8.

6.NS.B.4b | Constructed response

8. B, D - They next flash together after 24 seconds.; 48 seconds is also a time they flash together.

List multiples or use prime factors. The least shared positive multiple is 24; multiples of 24 are later shared times.

6.NS.B.4b | Multiple select

9. A, C - They next flash together after 90 seconds.; 180 seconds is also a time they flash together.

List multiples or use prime factors. The least shared positive multiple is 90; multiples of 90 are later shared times.

6.NS.B.4b | Multiple select

10. 60 seconds

The first positive multiple shared by 12 and 5 is 60.

6.NS.B.4b | Constructed response

Day 050 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Use Factors and Multiples to Explain Structure

Standards: 6.NS.B.4a, 6.NS.B.4b

Learning target: I can use Factors and Multiples to Explain Structure, and explain and check my reasoning.

1. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

2. 20

The first positive multiple shared by 5 and 4 is 20.

6.NS.B.4b | Constructed response

3. 3

Common factors: 1, 3; greatest is 3.

6.NS.B.4a | Constructed response

4. 12

The first positive multiple shared by 12 and 6 is 12.

6.NS.B.4b | Constructed response

5. 12

Common factors: 1, 2, 3, 4, 6, 12; greatest is 12.

6.NS.B.4a | Constructed response

6. 24

The first positive multiple shared by 3 and 8 is 24.

6.NS.B.4b | Constructed response

7. 10 kits; 7 pencils and 10 erasers each

GCF(70, 100) = 10; divide both counts by 10.

6.NS.B.4a | Constructed response

8. A, D - 66 seconds is also a time they flash together.; They next flash together after 33 seconds.

List multiples or use prime factors. The least shared positive multiple is 33; multiples of 33 are later shared times.

6.NS.B.4b | Multiple select

9. C, D - 18 divides both 36 and 54.; 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

GCF(36, 54) = 18. Divide each count by 18: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

10. 24 seconds

The first positive multiple shared by 8 and 3 is 24.

6.NS.B.4b | Constructed response

Day 051 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Factor Sums with the Distributive Property

Standards: 6.NS.B.4c

Learning target: I can factor a sum using the greatest common factor.

1. 3(2 + 3)

GCF = 3; 6 = 3 × 2 and 9 = 3 × 3.

6.NS.B.4c | Constructed response

2. 5(2 + 3)

GCF = 5; 10 = 5 × 2 and 15 = 5 × 3.

6.NS.B.4c | Constructed response

3. 4(3 + 4)

GCF = 4; 12 = 4 × 3 and 16 = 4 × 4.

6.NS.B.4c | Constructed response

4. 14(2 + 3)

GCF = 14; 28 = 14 × 2 and 42 = 14 × 3.

6.NS.B.4c | Constructed response

5. 24(2 + 3)

GCF = 24; 48 = 24 × 2 and 72 = 24 × 3.

6.NS.B.4c | Constructed response

6. 6(2 + 3)

GCF = 6; 12 = 6 × 2 and 18 = 6 × 3.

6.NS.B.4c | Constructed response

7. 6(3 + 5)

GCF = 6; 18 = 6 × 3 and 30 = 6 × 5.

6.NS.B.4c | Constructed response

8. B, D - 2(3 + 3); 6 + 6

Distribute 2 to both terms: 2 × 3 + 2 × 3 = 12. The other choices are 2 or 1 too large.

6.NS.B.4c | Multiple select

9. B, D - 5(5 + 3); 25 + 15

Distribute 5 to both terms: 5 × 5 + 5 × 3 = 40. The other choices are 5 or 1 too large.

6.NS.B.4c | Multiple select

10. 10(7 + 10)

GCF = 10; 70 = 10 × 7 and 100 = 10 × 10.

6.NS.B.4c | Constructed response

Day 052 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Identify Parts of Expressions

Standards: 6.EE.A.2b

Learning target: I can name terms, coefficients, factors, and grouped parts.

1. Terms 4x, 4y, 6; coefficients 4, 4; constant 6

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

2. Terms 3x, 7y, 5; coefficients 3, 7; constant 5

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

3. Outer quotient; dividend (6x + 7); divisor 3; inner sum 6x + 7

The grouped sum is one complete quantity being divided. Within it, the terms are 6x and 7.

6.EE.A.2b | Constructed response

4. Terms 2x, 5y, 2; coefficients 2, 5; constant 2

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

5. Outer quotient; dividend (4x + 4); divisor 4; inner sum 4x + 4

The grouped sum is one complete quantity being divided. Within it, the terms are 4x and 4.

6.EE.A.2b | Constructed response

6. Terms 6x, 5y, 5; coefficients 6, 5; constant 5

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

7. Outer quotient; dividend (2x + 3); divisor 5; inner sum 2x + 3

The grouped sum is one complete quantity being divided. Within it, the terms are 2x and 3.

6.EE.A.2b | Constructed response

8. B, D - 2 is a constant term.; 7 and (x + 4) are factors of the first term.

The outer sum has two terms. The first term is a product with two factors. Expanding gives 7x + 28 + 2, so the coefficient is 7.

6.EE.A.2b | Multiple select

9. B, C - 3 is a constant term.; 3 and (x + 4) are factors of the first term.

The outer sum has two terms. The first term is a product with two factors. Expanding gives 3x + 12 + 3, so the coefficient is 3.

6.EE.A.2b | Multiple select

10. Outer product; factors 7 and (x + 4); inner sum

The complete sum (x + 4) is one factor multiplied by 7.

6.EE.A.2b | Constructed response

Day 053 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Write Expressions from Words

Standards: 6.EE.A.2a, 6.EE.B.6

Learning target: I can translate words and situations into expressions.

1. n - 7

Start with n, then subtract the named amount.

6.EE.A.2a, 6.EE.B.6 | Constructed response

2. n - 5

Start with n, then subtract the named amount.

6.EE.A.2a, 6.EE.B.6 | Constructed response

3. n - 8

Start with n, then subtract the named amount.

6.EE.A.2a, 6.EE.B.6 | Constructed response

4. 6(x + 4)

Group the sum before multiplying.

6.EE.A.2a, 6.EE.B.6 | Constructed response

5. 7(x + 6)

Group the sum before multiplying.

6.EE.A.2a, 6.EE.B.6 | Constructed response

6. 3(x + 7)

Group the sum before multiplying.

6.EE.A.2a, 6.EE.B.6 | Constructed response

7. n = ticket count; 4n + 7; $23

Ticket cost is 4 × n; add the one-time fee 7.

6.EE.A.2a, 6.EE.B.6 | Constructed response

8. 6x + 4y - 3

Marker cost is 6x; folder cost is 4y. Add these subtotals, then subtract the one fixed coupon of 3. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

9. 4x + 2y - 3

Marker cost is 4x; folder cost is 2y. Add these subtotals, then subtract the one fixed coupon of 3. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

10. n = ticket count; 7n + 3; $24

Ticket cost is 7 × n; add the one-time fee 3.

6.EE.A.2a, 6.EE.B.6 | Constructed response

Day 054 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Evaluate Expressions and Formulas

Standards: 6.EE.A.2c

Learning target: I can substitute values and evaluate expressions or formulas.

1. 21

Substitute first: 8 × 2 + 5 = 21.

6.EE.A.2c | Constructed response

2. 22

Substitute first: 8 × 2 + 6 = 22.

6.EE.A.2c | Constructed response

3. 18

Substitute first: 4 × 3 + 6 = 18.

6.EE.A.2c | Constructed response

4. 56

Substitute first: 6^2 + 5 × 4 = 56.

6.EE.A.2c | Constructed response

5. 18

Substitute first: 3^2 + 3 × 3 = 18.

6.EE.A.2c | Constructed response

6. 22

Substitute first: 4^2 + 2 × 3 = 22.

6.EE.A.2c | Constructed response

7. A = 4(x + 6); 48 square units

Substitute x = 6: 4(6 + 6) = 48.

6.EE.A.2c | Constructed response

8. 59/2

Substitute: 3 × (3/2) + 5². The product is 9/2 and the square is 25. Add to get 59/2.

6.EE.A.2c | Numeric entry and reasoning

9. 32

Substitute: 7 × (1) + 5². The product is 7 and the square is 25. Add to get 32.

6.EE.A.2c | Numeric entry and reasoning

10. A = 6(x + 6); 48 square units

Substitute x = 2: 6(2 + 6) = 48.

6.EE.A.2c | Constructed response

Day 055 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Connect Distribution and Expression Structure

Standards: 6.EE.A.3, 6.EE.A.2c

Learning target: I can connect Distribution and Expression Structure, and explain and check my reasoning.

1. 2x + 12

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

2. 20

Substitute first: 8 × 2 + 4 = 20.

6.EE.A.2c | Constructed response

3. 5x + 10

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

4. 56

Substitute first: 4^2 + 8 × 5 = 56.

6.EE.A.2c | Constructed response

5. 42x + 21

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

6. 20

Substitute first: 4^2 + 2 × 2 = 20.

6.EE.A.2c | Constructed response

7. 49x + 42; both give 140 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

8. 19

Substitute: 3 × (1) + 4². The product is 3 and the square is 16. Add to get 19.

6.EE.A.2c | Numeric entry and reasoning

9. A, C - 6 + 3x; 3x + 6

Distribute 3 to both terms to get 3x + 6. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

10. A = 6(x + 3); 48 square units

Substitute x = 5: 6(5 + 3) = 48.

6.EE.A.2c | Constructed response

Day 056 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Generate Equivalent Expressions

Standards: 6.EE.A.3

Learning target: I can use distribution to generate an equivalent expression.

1. 2x + 12

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

2. 3x + 18

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

3. 3x + 21

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

4. 8x + 8

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

5. 12x + 6

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

6. 49x + 42

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

7. 12x + 10; both give 34 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

8. A, D - 12 + 3x; 3x + 12

Distribute 3 to both terms to get 3x + 12. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

9. A, D - 12 + 4x; 4x + 12

Distribute 4 to both terms to get 4x + 12. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

10. 35x + 10; both give 80 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

Day 057 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Combine Like Terms

Standards: 6.EE.A.3

Learning target: I can combine terms with the same variable part.

1. 9x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

2. 7x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

3. 7x

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

4. 8x + 7

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

5. 9x + 8

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

6. 7x + 7

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

7. 5x + 10

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

8. A, D - (7 + 5)x + 4; 12x + 4

Combine coefficients of like x terms: 7 + 5 = 12. The constant 4 stays separate.

6.EE.A.3 | Multiple select

9. B, C - 8x + 4; (4 + 4)x + 4

Combine coefficients of like x terms: 4 + 4 = 8. The constant 4 stays separate.

6.EE.A.3 | Multiple select

10. 14x + 14

Add coefficients of x terms; add only constants to constants. Distribute first if needed.

6.EE.A.3 | Constructed response

Day 058 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Identify Equivalent Expressions

Standards: 6.EE.A.4

Learning target: I can justify whether expressions are equivalent.

1. No; at x = 0, values are 14 and 2

7(x + 2) = 7x + 14. The constants 14 and 2 differ.

6.EE.A.4 | Constructed response

2. Yes; distribution gives 2x + 10

2(x + 5) = 2x + 10. This matches.

6.EE.A.4 | Constructed response

3. No; at x = 0, values are 56 and 7

8(x + 7) = 8x + 56. The constants 56 and 7 differ.

6.EE.A.4 | Constructed response

4. Yes; distribution gives 8x + 16

8(x + 2) = 8x + 16. This matches.

6.EE.A.4 | Constructed response

5. No; at x = 0, values are 6 and 2

3(x + 2) = 3x + 6. The constants 6 and 2 differ.

6.EE.A.4 | Constructed response

6. Yes; distribution gives 2x + 14

2(x + 7) = 2x + 14. This matches.

6.EE.A.4 | Constructed response

7. No; at x = 0, values are 36 and 6

6(x + 6) = 6x + 36. The constants 36 and 6 differ.

6.EE.A.4 | Constructed response

8. B, D - 7x + 28; 28 + 7x

Distribute 7 to both terms to get 7x + 28. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

9. C, D - 4x + 24; 24 + 4x

Distribute 4 to both terms to get 4x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

10. Yes; distribution gives 3x + 18

3(x + 6) = 3x + 18. This matches.

6.EE.A.4 | Constructed response

Day 059 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Apply Expression Structure

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4c, 6.EE.A.4

Learning target: I can apply Expression Structure, and explain and check my reasoning.

1. 27

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 27.

6.EE.A.1, 6.EE.A.2c | Constructed response

2. 11

Substitute first: 4 × 2 + 3 = 11.

6.EE.A.2c | Constructed response

3. 4(2 + 3)

GCF = 4; 8 = 4 × 2 and 12 = 4 × 3.

6.NS.B.4c | Constructed response

4. Yes; distribution gives 7x + 21

7(x + 3) = 7x + 21. This matches.

6.EE.A.4 | Constructed response

5. 126

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 126.

6.EE.A.1, 6.EE.A.2c | Constructed response

6. 81

Substitute first: 5^2 + 8 × 7 = 81.

6.EE.A.2c | Constructed response

7. 16(2 + 3)

GCF = 16; 32 = 16 × 2 and 48 = 16 × 3.

6.NS.B.4c | Constructed response

8. A, C - 6 + 3x; 3x + 6

Distribute 3 to both terms to get 3x + 6. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

9. 26

Parentheses: 8 - 3 = 5. Exponent: $4^2 = 16$. Multiply: 2 × 5 = 10. Add: 16 + 10 = 26. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

10. A = 7(x + 4); 56 square units

Substitute x = 4: 7(4 + 4) = 56.

6.EE.A.2c | Constructed response

Day 060 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Unit 4 Test

Standards: 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.NS.B.4c, 6.EE.A.2b, 6.EE.A.2a, 6.EE.B.6, 6.EE.A.3, 6.EE.A.4

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. +28 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

2. 8

-8 and 8 are each 8 units from 0 on opposite sides.

6.NS.C.6a | Constructed response

3. -8 < 0 < 7; unit steps

Negative -8 is left of or below 0; 7 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

4. -5 < 0 < 3; unit steps

Negative -5 is left of or below 0; 3 is right of or above 0. Use equal intervals.

6.NS.C.6c | Constructed response

5. 4 × 4 = 16

The exponent counts 2 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

6. 21

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 21.

6.EE.A.1, 6.EE.A.2c | Constructed response

7. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

8. 72

The first positive multiple shared by 8 and 9 is 72.

6.NS.B.4b | Constructed response

9. 12

Common factors: 1, 2, 3, 4, 6, 12; greatest is 12.

6.NS.B.4a | Constructed response

10. 14(2 + 3)

GCF = 14; 28 = 14 × 2 and 42 = 14 × 3.

6.NS.B.4c | Constructed response

11. Outer quotient; dividend (5x + 7); divisor 2; inner sum 5x + 7

The grouped sum is one complete quantity being divided. Within it, the terms are 5x and 7.

6.EE.A.2b | Constructed response

12. 4x + 3y - 2

Marker cost is 4x; folder cost is 3y. Add these subtotals, then subtract the one fixed coupon of 2. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

13. 13

Substitute: 4 × (1) + 3^2. The product is 4 and the square is 9. Add to get 13.

6.EE.A.2c | Numeric entry and reasoning

14. A, D - 42 + 7x; 7x + 42

Distribute 7 to both terms to get 7x + 42. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

15. C, D - 30 + 6x; 6x + 30

Distribute 6 to both terms to get 6x + 30. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

16. A, B - 7x + 4; (2 + 5)x + 4

Combine coefficients of like x terms: 2 + 5 = 7. The constant 4 stays separate.

6.EE.A.3 | Multiple select

17. C, D - 4x + 24; 24 + 4x

Distribute 4 to both terms to get 4x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

18. 23/2

Substitute: 5 × (3/2) + 2². The product is 15/2 and the square is 4. Add to get 23/2.

6.EE.A.2c | Numeric entry and reasoning

19. B, D - 6 × 6 × 6; 216

The exponent 3 means three factors of 6; their product is 216. It does not mean 6 times 3.

6.EE.A.1 | Multiple select

20. 18

Parentheses: 6 - 3 = 3. Exponent: 3² = 9. Multiply: 3 × 3 = 9. Add: 9 + 9 = 18. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

Day 061 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Test Solutions by Substitution

Standards: 6.EE.B.5

Learning target: I can test a proposed solution by substitution.

1. No; left side = 29

4 × 6 + 5 = 29; compare with 25.

6.EE.B.5 | Constructed response

2. Yes; left side = 17

3 × 5 + 2 = 17; compare with 17.

6.EE.B.5 | Constructed response

3. No; left side = 18

3 × 4 + 6 = 18; compare with 15.

6.EE.B.5 | Constructed response

4. Yes; left side = 37

8 × 4 + 5 = 37; compare with 37.

6.EE.B.5 | Constructed response

5. No; left side = 42

8 × 5 + 2 = 42; compare with 34.

6.EE.B.5 | Constructed response

6. Yes; left side = 21

5 × 3 + 6 = 21; compare with 21.

6.EE.B.5 | Constructed response

7. No; left side = 25

5 × 4 + 5 = 25; compare with 20.

6.EE.B.5 | Constructed response

8. A, B - 3; 4

Subtract 3 conceptually: x must exceed 2. Equality at 2 fails because the inequality is strict.

6.EE.B.5 | Multiple select

9. B, D - 6; 5

Subtract 6 conceptually: x must exceed 4. Equality at 4 fails because the inequality is strict.

6.EE.B.5 | Multiple select

10. Yes; left side = 21

7 × 2 + 7 = 21; compare with 21.

6.EE.B.5 | Constructed response

Day 062 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Solve Addition Equations

Standards: 6.EE.B.7

Learning target: I can solve x + p = q and check.

1. x = 2

Subtract 2 on both sides. Then x = 2; substitution gives 4 on each side.

6.EE.B.7 | Constructed response

2. x = 4

Subtract 5 on both sides. Then x = 4; substitution gives 9 on each side.

6.EE.B.7 | Constructed response

3. x = 3

Subtract 5 on both sides. Then x = 3; substitution gives 8 on each side.

6.EE.B.7 | Constructed response

4. x = 3/2

Subtract 8 on both sides. Then x = 3/2; substitution gives 19/2 on each side.

6.EE.B.7 | Constructed response

5. x = 1

Subtract 3 on both sides. Then x = 1; substitution gives 4 on each side.

6.EE.B.7 | Constructed response

6. x = 5/2

Subtract 8 on both sides. Then x = 5/2; substitution gives 21/2 on each side.

6.EE.B.7 | Constructed response

7. x = 3/2

Subtract 2 on both sides. Then x = 3/2; substitution gives 7/2 on each side.

6.EE.B.7 | Constructed response

8. B, C - 11/2; 22/4

Use the inverse operation on both sides. The solution is 11/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. A, C - 26/4; 13/2

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 5/2

Subtract 4 on both sides. Then x = 5/2; substitution gives 13/2 on each side.

6.EE.B.7 | Constructed response

Day 063 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Solve Subtraction Equations

Standards: 6.EE.B.7

Learning target: I can solve x - p = q and check.

1. x = 5

Add 4 on both sides. Then x = 5; substitution gives 1 on each side.

6.EE.B.7 | Constructed response

2. x = 6

Add 4 on both sides. Then x = 6; substitution gives 2 on each side.

6.EE.B.7 | Constructed response

3. x = 6

Add 5 on both sides. Then x = 6; substitution gives 1 on each side.

6.EE.B.7 | Constructed response

4. x = 7/2

Add 3 on both sides. Then x = 7/2; substitution gives 1/2 on each side.

6.EE.B.7 | Constructed response

5. x = 11/2

Add 3 on both sides. Then x = 11/2; substitution gives 5/2 on each side.

6.EE.B.7 | Constructed response

6. x = 9

Add 6 on both sides. Then x = 9; substitution gives 3 on each side.

6.EE.B.7 | Constructed response

7. x = 2

Add 3/2 on both sides. Then x = 2; substitution gives 1/2 on each side.

6.EE.B.7 | Constructed response

8. C, D - 5/2; 10/4

Use the inverse operation on both sides. The solution is 5/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. B, D - 26/4; 13/2

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 3

Add 1 on both sides. Then x = 3; substitution gives 2 on each side.

6.EE.B.7 | Constructed response

Day 064 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Solve Multiplication Equations

Standards: 6.EE.B.7

Learning target: I can solve px = q using division.

1. x = 4

Divide by 5 on both sides. Then x = 4; substitution gives 20 on each side.

6.EE.B.7 | Constructed response

2. x = 3

Divide by 2 on both sides. Then x = 3; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

3. x = 2

Divide by 2 on both sides. Then x = 2; substitution gives 4 on each side.

6.EE.B.7 | Constructed response

4. x = 7/2

Divide by 4 on both sides. Then x = 7/2; substitution gives 14 on each side.

6.EE.B.7 | Constructed response

5. x = 5/2

Divide by 4 on both sides. Then x = 5/2; substitution gives 10 on each side.

6.EE.B.7 | Constructed response

6. x = 1

Divide by 6 on both sides. Then x = 1; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

7. x = 3

Divide by 1 on both sides. Then x = 3; substitution gives 3 on each side.

6.EE.B.7 | Constructed response

8. A, D - 14/4; 7/2

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. B, C - 5/2; 10/4

Use the inverse operation on both sides. The solution is 5/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 7/2

Divide by 2 on both sides. Then x = 7/2; substitution gives 7 on each side.

6.EE.B.7 | Constructed response

Day 065 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Explain Balance and Inverse Operations

Standards: 6.EE.B.7

Learning target: I can explain Balance and Inverse Operations, and explain and check my reasoning.

1. x = 2

Subtract 5 on both sides. Then x = 2; substitution gives 7 on each side.

6.EE.B.7 | Constructed response

2. x = 5

Add 3 on both sides. Then x = 5; substitution gives 2 on each side.

6.EE.B.7 | Constructed response

3. x = 3

Divide by 6 on both sides. Then x = 3; substitution gives 18 on each side.

6.EE.B.7 | Constructed response

4. x = 3/2

Subtract 8 on both sides. Then x = 3/2; substitution gives 19/2 on each side.

6.EE.B.7 | Constructed response

5. x = 15/2

Add 6 on both sides. Then x = 15/2; substitution gives 3/2 on each side.

6.EE.B.7 | Constructed response

6. x = 1

Divide by 8 on both sides. Then x = 1; substitution gives 8 on each side.

6.EE.B.7 | Constructed response

7. x = 3/2

Subtract 4 on both sides. Then x = 3/2; substitution gives 11/2 on each side.

6.EE.B.7 | Constructed response

8. B, D - 7/2; 14/4

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. A, D - 9/2; 18/4

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 7/2

Subtract 1 on both sides. Then x = 7/2; substitution gives 9/2 on each side.

6.EE.B.7 | Constructed response

Day 066 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Solve Division Equations

Standards: 6.EE.B.7

Learning target: I can solve x/p = q using multiplication.

1. x = 3

Multiply by 5 on both sides. Then x = 3; substitution gives 3/5 on each side.

6.EE.B.7 | Constructed response

2. x = 7

Multiply by 4 on both sides. Then x = 7; substitution gives 7/4 on each side.

6.EE.B.7 | Constructed response

3. x = 7

Multiply by 8 on both sides. Then x = 7; substitution gives 7/8 on each side.

6.EE.B.7 | Constructed response

4. x = 5/2

Multiply by 7 on both sides. Then x = 5/2; substitution gives 5/14 on each side.

6.EE.B.7 | Constructed response

5. x = 5/2

Multiply by 6 on both sides. Then x = 5/2; substitution gives 5/12 on each side.

6.EE.B.7 | Constructed response

6. x = 3/2

Multiply by 5 on both sides. Then x = 3/2; substitution gives 3/10 on each side.

6.EE.B.7 | Constructed response

7. x = 7/2

Multiply by 1 on both sides. Then x = 7/2; substitution gives 7/2 on each side.

6.EE.B.7 | Constructed response

8. A, C - 10/4; 5/2

Use the inverse operation on both sides. The solution is 5/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

9. B, C - 13/2; 26/4

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

10. x = 3/2

Multiply by 5/2 on both sides. Then x = 3/2; substitution gives 3/5 on each side.

6.EE.B.7 | Constructed response

Day 067 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Write One-Step Equations from Context

Standards: 6.EE.B.6, 6.EE.B.7

Learning target: I can model a situation with a one-step equation and solve it.

1. x + 7 = 35; x = 28 cards

Subtract 7 from both sides; 28 + 7 = 35.

6.EE.B.6, 6.EE.B.7 | Constructed response

2. x + 2 = 6; x = 4 cards

Subtract 2 from both sides; 4 + 2 = 6.

6.EE.B.6, 6.EE.B.7 | Constructed response

3. x + 7 = 49; x = 42 cards

Subtract 7 from both sides; 42 + 7 = 49.

6.EE.B.6, 6.EE.B.7 | Constructed response

4. 7x = 14; x = 2 m

Divide the total by 7: 14 ÷ 7 = 2.

6.EE.B.6, 6.EE.B.7 | Constructed response

5. 2x = 16; x = 8 m

Divide the total by 2: 16 ÷ 2 = 8.

6.EE.B.6, 6.EE.B.7 | Constructed response

6. 5x = 35; x = 7 m

Divide the total by 5: 35 ÷ 5 = 7.

6.EE.B.6, 6.EE.B.7 | Constructed response

7. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

8. 2x = 3; x = 3/2 meters

Total = number of ropes × one length. Divide both sides by 2; 3 ÷ 2 = 3/2. Check 2 × 3/2 = 3.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

9. 3x = 21/2; x = 7/2 meters

Total = number of ropes × one length. Divide both sides by 3; 21/2 ÷ 3 = 7/2. Check 3 × 7/2 = 21/2.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

10. 6x = 18; x = 3 m

Divide the total by 6: 18 ÷ 6 = 3.

6.EE.B.6, 6.EE.B.7 | Constructed response

Day 068 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Write and Graph Inequalities

Standards: 6.EE.B.8

Learning target: I can write and graph a constraint using an inequality.

1. x ≤ 49; closed circle at 49, shade left

“at most” means ≤; include the endpoint.

6.EE.B.8 | Constructed response

2. x ≥ 42; closed circle at 42, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

3. x ≥ 35; closed circle at 35, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

4. x ≤ 16; closed circle at 16, shade left

“at most” means ≤; include the endpoint.

6.EE.B.8 | Constructed response

5. x > 8; open circle at 8, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

6. x > 25; open circle at 25, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

7. x ≥ 24; closed circle at 24, shade right

“at least” means ≥; include the endpoint. Example solution: 25.

6.EE.B.8 | Constructed response

8. B, D - 10 is an allowed membership count.; n ≥ 10

At least includes the boundary, so n ≥ 10. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

9. B, C - n ≥ 8; 8 is an allowed membership count.

At least includes the boundary, so n ≥ 8. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

10. p ≥ 25; closed circle at 25, shade right; participant counts are whole numbers 25, 26, …

“at least” means ≥. The real-number ray has infinitely many solutions, but participant counts must be whole numbers.

6.EE.B.8 | Constructed response

Day 069 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Test Inequality Solutions

Standards: 6.EE.B.5, 6.EE.B.8

Learning target: I can test values and explain the many solutions of an inequality.

1. 6

Values below 7 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

2. 11, 12

Values below 12 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

3. 14

Values below 15 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

4. 10, 11

Values below 11 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

5. 5

Values below 6 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

6. 8, 9

Values below 9 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

7. 8; no, infinitely many real values satisfy it

Values below 9 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

8. C, D - n ≥ 6; 6 is an allowed membership count.

At least includes the boundary, so n ≥ 6. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

9. B, D - 10 is an allowed membership count.; n ≥ 10

At least includes the boundary, so n ≥ 10. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

10. 10, 11; no, infinitely many real values satisfy it

Values below 11 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

Day 070 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Connect Constraints and Solution Sets

Standards: 6.EE.B.8, 6.EE.B.5

Learning target: I can connect Constraints and Solution Sets, and explain and check my reasoning.

1. x ≥ 21; closed circle at 21, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

2. 9, 10

Values below 10 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

3. x > 4; open circle at 4, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

4. 11, 12

Values below 12 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

5. x > 28; open circle at 28, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

6. 7, 8

Values below 8 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

7. x ≤ 30; closed circle at 30, shade left

“at most” means ≤; include the endpoint. Example solution: 29.

6.EE.B.8 | Constructed response

8. A, B - 28 is an allowed membership count.; n ≥ 28

At least includes the boundary, so n ≥ 28. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

9. A, D - n ≥ 16; 16 is an allowed membership count.

At least includes the boundary, so n ≥ 16. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

10. 4, 5; no, infinitely many real values satisfy it

Values below 5 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

Day 071 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Interpret Endpoints and Infinitely Many Solutions

Standards: 6.EE.B.8, 6.EE.B.5

Learning target: I can interpret Endpoints and Infinitely Many Solutions, and explain and check my reasoning.

1. x > 24; open circle at 24, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

2. 7, 8

Values below 8 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

3. x ≥ 8; closed circle at 8, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

4. 4, 5

Values below 5 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

5. x > 14; open circle at 14, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

6. 11, 12

Values below 12 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

7. x > 24; open circle at 24, shade right

“more than” means >; exclude the endpoint. Example solution: 25.

6.EE.B.8 | Constructed response

8. A, D - n ≥ 18; 18 is an allowed membership count.

At least includes the boundary, so n ≥ 18. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

9. C, D - n ≥ 30; 30 is an allowed membership count.

At least includes the boundary, so n ≥ 30. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

10. 10, 11; no, infinitely many real values satisfy it

Values below 11 work; the endpoint works too.

6.EE.B.5, 6.EE.B.8 | Constructed response

Day 072 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Identify Independent and Dependent Variables

Standards: 6.EE.C.9

Learning target: I can identify quantities that change together and write a rule.

1. Independent n = notebooks; dependent C = cost; C = 5n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

2. Independent n = notebooks; dependent C = cost; C = 4n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

3. Independent n = notebooks; dependent C = cost; C = 3n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

4. Independent n = notebooks; dependent C = cost; C = 2n + 3

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

5. Independent n = notebooks; dependent C = cost; C = 2n + 4

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

6. Independent n = notebooks; dependent C = cost; C = 8n + 2

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

7. Independent n = notebooks; dependent C = cost; C = 4n + 3; C(0) = $3, the fee

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

8. (0,0), (1,6), (4,24); independent h; L = 6h

Hours are the input. Multiply each hour value by 6 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

9. (0,0), (1,3), (4,12); independent h; L = 3h

Hours are the input. Multiply each hour value by 3 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

10. Independent n = notebooks; dependent C = cost; C = 5n + 3; C(0) = $3, the fee

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

Day 073 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Connect Rules, Tables, and Graphs

Standards: 6.EE.C.9

Learning target: I can connect an equation, a table, and a graph.

1. (0, 0); (1, 2); (2, 4); (3, 6)

Substitute each input into y = 2x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

2. (0, 0); (1, 2); (2, 4); (6, 12)

Substitute each input into y = 2x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

3. (0, 0); (1, 6); (2, 12); (6, 36)

Substitute each input into y = 6x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

4. (0, 3); (1, 9); (2, 15); (4, 27)

Substitute each input into y = 6x + 3; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

5. (0, 5); (1, 7); (2, 9); (5, 15)

Substitute each input into y = 2x + 5; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

6. (0, 7); (1, 15); (2, 23); (4, 39)

Substitute each input into y = 8x + 7; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

7. (0, 4); (1, 9); (2, 14); y increases by 5

Substitute each input into y = 5x + 4; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

8. (0,0), (1,2), (2,4); independent h; L = 2h

Hours are the input. Multiply each hour value by 2 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

9. (0,0), (1,4), (4,16); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

10. (0, 2); (1, 10); (2, 18); (6, 50); y increases by 8

Substitute each input into y = 8x + 2; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

Day 074 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Interpret Relationships in Context

Standards: 6.EE.C.9

Learning target: I can interpret Relationships in Context, and explain and check my reasoning.

1. Independent n = notebooks; dependent C = cost; C = 6n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

2. (0, 0); (1, 2); (2, 4); (3, 6)

Substitute each input into y = 2x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

3. Independent n = notebooks; dependent C = cost; C = 8n

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

4. (0, 7); (1, 15); (2, 23); (5, 47)

Substitute each input into y = 8x + 7; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

5. Independent n = notebooks; dependent C = cost; C = 4n + 7

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

6. (0, 3); (1, 9); (2, 15); (5, 33)

Substitute each input into y = 6x + 3; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

7. Independent n = notebooks; dependent C = cost; C = 8n + 3; C(0) = $3, the fee

The chosen notebook count is the input; multiply it by price and add any fixed fee to get the cost.

6.EE.C.9 | Constructed response

8. (0,0), (1,4), (3,12); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

9. (0,0), (1,2), (5,10); independent h; L = 2h

Hours are the input. Multiply each hour value by 2 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

10. (0, 6); (1, 13); (2, 20); (3, 27); y increases by 7

Substitute each input into y = 7x + 6; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

Day 075 | Unit 05 | Answer key

Represent and Solve Equations and Inequalities

Unit 5 Test

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.EE.B.5, 6.EE.B.7, 6.EE.B.6, 6.EE.B.8, 6.EE.C.9

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 3 × 3 = 9

The exponent counts 2 factors of 3; multiply all factors.

6.EE.A.1 | Constructed response

2. 18

Use parentheses and exponents first, then × and ÷ left to right, then + and -. Result: 18.

6.EE.A.1, 6.EE.A.2c | Constructed response

3. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

4. 6

The first positive multiple shared by 2 and 6 is 6.

6.NS.B.4b | Constructed response

5. No; left side = 28

3 × 7 + 7 = 28; compare with 25.

6.EE.B.5 | Constructed response

6. x = 2

Subtract 3 on both sides. Then x = 2; substitution gives 5 on each side.

6.EE.B.7 | Constructed response

7. x = 19/2

Add 7 on both sides. Then x = 19/2; substitution gives 5/2 on each side.

6.EE.B.7 | Constructed response

8. x = 3/2

Divide by 7 on both sides. Then x = 3/2; substitution gives 21/2 on each side.

6.EE.B.7 | Constructed response

9. x = 7/2

Subtract 4 on both sides. Then x = 7/2; substitution gives 15/2 on each side.

6.EE.B.7 | Constructed response

10. x = 7/2

Multiply by 8 on both sides. Then x = 7/2; substitution gives 7/16 on each side.

6.EE.B.7 | Constructed response

11. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

12. C, D - 12 is an allowed membership count.; n ≥ 12

At least includes the boundary, so n ≥ 12. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

13. C, D - n ≥ 15; 15 is an allowed membership count.

At least includes the boundary, so n ≥ 15. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

14. A, C - n ≥ 42; 42 is an allowed membership count.

At least includes the boundary, so n ≥ 42. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

15. C, D - 25 is an allowed membership count.; n ≥ 25

At least includes the boundary, so n ≥ 25. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.5, 6.EE.B.8 | Multiple select

16. (0,0), (1,4), (4,16); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

17. (0,0), (1,3), (6,18); independent h; L = 3h

Hours are the input. Multiply each hour value by 3 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

18. (0,0), (1,6), (4,24); independent h; L = 6h

Hours are the input. Multiply each hour value by 6 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

19. C, D - 6; 7

Subtract 7 conceptually: x must exceed 5. Equality at 5 fails because the inequality is strict.

6.EE.B.5 | Multiple select

20. B, D - 18/4; 9/2

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

Day 076 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Describe Ratios with Ordered Labels

Standards: 6.RP.A.1

Learning target: I can describe Ratios with Ordered Labels, and explain and check my reasoning.

1. 3:2

Keep red first; divide 3:2 by 1.

6.RP.A.1 | Constructed response

2. 2/5

Total = 25; green fraction = 10/25 = 2/5.

6.RP.A.1 | Constructed response

3. 5:4

Keep red first; divide 5:4 by 1.

6.RP.A.1 | Constructed response

4. 5/9

Total = 45; green fraction = 25/45 = 5/9.

6.RP.A.1 | Constructed response

5. 1:1

Keep red first; divide 25:25 by 25.

6.RP.A.1 | Constructed response

6. 2/3

Total = 60; green fraction = 40/60 = 2/3.

6.RP.A.1 | Constructed response

7. 1:3; 1/3 times as many

Keep red first; divide 4:12 by 4.

6.RP.A.1 | Constructed response

8. A, D - The red-to-blue ratio is 2:3.; The fraction of all tiles that are red is 2/5.

Divide both color counts by 5. For the fraction of the whole, use total 25 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

9. B, C - The red-to-blue ratio is 2:6.; The fraction of all tiles that are red is 1/4.

Divide both color counts by 5. For the fraction of the whole, use total 40 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

10. 1/3; 1/2 or 1:2; the whole includes both colors

Total = 54; green fraction = 18/54 = 1/3.

6.RP.A.1 | Constructed response

Day 077 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Generate Equivalent Ratios

Standards: 6.RP.A.3a

Learning target: I can complete an equivalent-ratio table.

1. 14; 14

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

2. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

3. 6; 6

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

4. 8; 20

Multiply both parts by 2 and by 5.

6.RP.A.3a | Constructed response

5. 4; 4

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

6. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

7. 6; 12

Multiply both parts by 2 and by 4.

6.RP.A.3a | Constructed response

8. B, D - (15, 9); (10, 6)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

9. B, C - (6, 6); (4, 4)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

10. 10; 35

Multiply both parts by 2 and by 7.

6.RP.A.3a | Constructed response

Day 078 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Compare Ratios Fairly

Standards: 6.RP.A.3a

Learning target: I can compare ratios using a common quantity.

1. Mix B

For 35 cups water, A uses 25 cups juice; B uses 26. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

2. Mix B

For 12 cups water, A uses 30 cups juice; B uses 33. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

3. Mix B

For 24 cups water, A uses 42 cups juice; B uses 45. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

4. Mix B

For 36 cups water, A uses 36 cups juice; B uses 39. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

5. Mix B

For 30 cups water, A uses 35 cups juice; B uses 38. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

6. Mix B

For 6 cups water, A uses 18 cups juice; B uses 20. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

7. Mix B

For 30 cups water, A uses 42 cups juice; B uses 44. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

8. A, C - For 25 cups water, A uses 25 cups concentrate.; Mix B is stronger.

Scale A by 5 to the same water amount. B has 1 more cup of concentrate with equal water, so B is stronger.

6.RP.A.3a | Multiple select

9. A, D - For 30 cups water, A uses 25 cups concentrate.; Mix B is stronger.

Scale A by 5 to the same water amount. B has 1 more cup of concentrate with equal water, so B is stronger.

6.RP.A.3a | Multiple select

10. Mix B

For 25 cups water, A uses 35 cups juice; B uses 39. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

Day 079 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Represent Ratios with Tape Diagrams

Standards: 6.RP.A.3a

Learning target: I can use equal-size tape sections to solve a ratio problem.

1. 4 oaks and 12 pines

8 equal sections total; each section is 16 ÷ 8 = 2.

6.RP.A.3a | Constructed response

2. 36 oaks and 18 pines

9 equal sections total; each section is 54 ÷ 9 = 6.

6.RP.A.3a | Constructed response

3. 10 oaks and 25 pines

7 equal sections total; each section is 35 ÷ 7 = 5.

6.RP.A.3a | Constructed response

4. 16 oaks and 12 pines

14 equal sections total; each section is 28 ÷ 14 = 2.

6.RP.A.3a | Constructed response

5. 30 oaks and 18 pines

8 equal sections total; each section is 48 ÷ 8 = 6.

6.RP.A.3a | Constructed response

6. 10 oaks and 14 pines

12 equal sections total; each section is 24 ÷ 12 = 2.

6.RP.A.3a | Constructed response

7. 24 oaks and 28 pines; difference 4

13 equal sections total; each section is 52 ÷ 13 = 4.

6.RP.A.3a | Constructed response

8. 8 paper; 8 fabric; the claim is 1 paper flag too many.

There are 8 equal sections. Each is 16 ÷ 8 = 2. Multiply the section counts by 2.

6.RP.A.3a | Model and error analysis

9. 21 paper; 15 fabric; the claim is 1 paper flag too many.

There are 12 equal sections. Each is 36 ÷ 12 = 3. Multiply the section counts by 3.

6.RP.A.3a | Model and error analysis

10. 20 oaks and 12 pines; difference 8

8 equal sections total; each section is 32 ÷ 8 = 4.

6.RP.A.3a | Constructed response

Day 080 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Explain Equivalent Ratios in Two Models

Standards: 6.RP.A.3a

Learning target: I can explain Equivalent Ratios in Two Models, and explain and check my reasoning.

1. 6; 9

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

2. 28 oaks and 24 pines

13 equal sections total; each section is 52 ÷ 13 = 4.

6.RP.A.3a | Constructed response

3. 12; 12

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

4. 16 oaks and 6 pines

11 equal sections total; each section is 22 ÷ 11 = 2.

6.RP.A.3a | Constructed response

5. 4; 12

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

6. 16 oaks and 28 pines

11 equal sections total; each section is 44 ÷ 11 = 4.

6.RP.A.3a | Constructed response

7. 4; 6

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

8. 15 paper; 6 fabric; the claim is 1 paper flag too many.

There are 7 equal sections. Each is 21 ÷ 7 = 3. Multiply the section counts by 3.

6.RP.A.3a | Model and error analysis

9. B, D - (12, 8); (18, 12)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

10. 12 oaks and 24 pines; difference 12

9 equal sections total; each section is 36 ÷ 9 = 4.

6.RP.A.3a | Constructed response

Day 081 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Use Double Number Lines

Standards: 6.RP.A.3a

Learning target: I can represent a ratio on a double number line.

1. 35 square meters; aligned pairs (0,0), (3,7), (15,35)

15 ÷ 3 = 5 batches; 5 × 7 = 35.

6.RP.A.3a | Constructed response

2. 6 square meters; aligned pairs (0,0), (8,2), (24,6)

24 ÷ 8 = 3 batches; 3 × 2 = 6.

6.RP.A.3a | Constructed response

3. 42 square meters; aligned pairs (0,0), (2,7), (12,42)

12 ÷ 2 = 6 batches; 6 × 7 = 42.

6.RP.A.3a | Constructed response

4. 15 square meters; aligned pairs (0,0), (4,3), (20,15)

20 ÷ 4 = 5 batches; 5 × 3 = 15.

6.RP.A.3a | Constructed response

5. 25 square meters; aligned pairs (0,0), (6,5), (30,25)

30 ÷ 6 = 5 batches; 5 × 5 = 25.

6.RP.A.3a | Constructed response

6. 6 square meters; aligned pairs (0,0), (4,3), (8,6)

8 ÷ 4 = 2 batches; 2 × 3 = 6.

6.RP.A.3a | Constructed response

7. 8 square meters; aligned pairs (0,0), (6,2), (24,8)

24 ÷ 6 = 4 batches; 4 × 2 = 8.

6.RP.A.3a | Constructed response

8. Aligned pairs (0,0), (6,4), (12,8), (42,28); 28 m²

Each aligned jump adds 6 liters and 4 m². Both quantities scale by 7; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

9. Aligned pairs (0,0), (7,6), (14,12), (42,36); 36 m²

Each aligned jump adds 7 liters and 6 m². Both quantities scale by 6; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

10. 28 square meters; aligned pairs (0,0), (6,7), (24,28)

24 ÷ 6 = 4 batches; 4 × 7 = 28.

6.RP.A.3a | Constructed response

Day 082 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Represent and Graph Ratios

Standards: 6.RP.A.3a

Learning target: I can plot equivalent-ratio pairs and explain a point.

1. (0,0), (6,5), (12,10), (36,30); 36 tickets earns $30

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

2. (0,0), (2,7), (4,14), (10,35); 10 tickets earns $35

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

3. (0,0), (7,6), (14,12), (14,12); 14 tickets earns $12

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

4. (0,0), (2,6), (4,12), (8,24); 8 tickets earns $24

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

5. (0,0), (2,3), (4,6), (6,9); 6 tickets earns $9

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

6. (0,0), (5,6), (10,12), (10,12); 10 tickets earns $12

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

7. (0,0), (2,2), (4,4), (6,6); 6 tickets earns $6

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

8. (0,0), (7,2), (14,4), (35,10); 35 tickets earn $10.

For each batch count, multiply both 7 and 2 by that count. Keep tickets horizontal and dollars vertical.

6.RP.A.3a | Table, graph, and interpretation

9. (0,0), (4,6), (8,12), (16,24); 16 tickets earn $24.

For each batch count, multiply both 4 and 6 by that count. Keep tickets horizontal and dollars vertical.

6.RP.A.3a | Table, graph, and interpretation

10. (0,0), (5,4), (10,8), (30,24); 30 tickets earns $24

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

Day 083 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Understand Rates and Unit Rates

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can find a unit rate and explain its units.

1. $3 per notebook

Unit rate = 18 ÷ 6 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

2. $3 per notebook

Unit rate = 12 ÷ 4 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

3. $6 per notebook

Unit rate = 42 ÷ 7 = 6 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

4. $3 per notebook

Unit rate = 21 ÷ 7 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

5. $6 per notebook

Unit rate = 12 ÷ 2 = 6 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

6. $7 per notebook

Unit rate = 42 ÷ 6 = 7 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

7. $2 per notebook; $20

Unit rate = 14 ÷ 7 = 2 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

8. A, B - Each notebook costs $2.; 4 notebooks cost $8.

Cost per notebook = 12 ÷ 6 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

9. A, C - 5 notebooks cost $10.; Each notebook costs $2.

Cost per notebook = 10 ÷ 5 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

10. $2.5 per notebook; $22.5

Unit rate = 15 ÷ 6 = 2.5 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

Day 084 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Compare Unit Rates and Prices

Standards: 6.RP.A.3b

Learning target: I can compare unit prices to find the better buy.

1. A $4/marker; B $4.5/marker; A is cheaper

Divide each price by its marker count; compare 4 with 4.5.

6.RP.A.3b | Constructed response

2. A $8/marker; B $8.5/marker; A is cheaper

Divide each price by its marker count; compare 8 with 8.5.

6.RP.A.3b | Constructed response

3. A $4/marker; B $4.5/marker; A is cheaper

Divide each price by its marker count; compare 4 with 4.5.

6.RP.A.3b | Constructed response

4. A $1.5/marker; B $2/marker; A is cheaper

Divide each price by its marker count; compare 1.5 with 2.

6.RP.A.3b | Constructed response

5. A $1/marker; B $1.5/marker; A is cheaper

Divide each price by its marker count; compare 1 with 1.5.

6.RP.A.3b | Constructed response

6. A $4/marker; B $4.5/marker; A is cheaper

Divide each price by its marker count; compare 4 with 4.5.

6.RP.A.3b | Constructed response

7. A $1/marker; B $1.5/marker; A is cheaper; A $36, B $54

Divide each price by its marker count; compare 1 with 1.5.

6.RP.A.3b | Constructed response

8. A: $3 each; B: $4 each; A is the better buy.

Divide each cost by its own notebook count: 9/3 = 3; 16/4 = 4. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

9. A: $2 each; B: $3 each; A is the better buy.

Divide each cost by its own notebook count: 4/2 = 2; 9/3 = 3. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

10. A $3/marker; B $3.5/marker; A is cheaper; A $165, B $192.5

Divide each price by its marker count; compare 3 with 3.5.

6.RP.A.3b | Constructed response

Day 085 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Compare Rate Models and Units

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can compare Rate Models and Units, and explain and check my reasoning.

1. $4 per notebook

Unit rate = 16 ÷ 4 = 4 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

2. A $6/marker; B $6.5/marker; A is cheaper

Divide each price by its marker count; compare 6 with 6.5.

6.RP.A.3b | Constructed response

3. 40 km/h

Speed = 80 ÷ 2 = 40; distance = speed × time.

6.RP.A.3b | Constructed response

4. $7 per notebook

Unit rate = 28 ÷ 4 = 7 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

5. A $4/marker; B $4.5/marker; A is cheaper

Divide each price by its marker count; compare 4 with 4.5.

6.RP.A.3b | Constructed response

6. 3 hours

Time = distance ÷ speed = 30 ÷ 10 = 3.

6.RP.A.3b | Constructed response

7. $3 per notebook; $15

Unit rate = 6 ÷ 2 = 3 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

8. A: $2 each; B: $3 each; A is the better buy.

Divide each cost by its own notebook count: 12/6 = 2; 21/7 = 3. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

9. A, B - The speed is 35 km per hour.; In 4 hours it travels 140 km.

Divide distance by time: 70 ÷ 2 = 35 km/h. Multiply this unit rate by 4 for the later distance.

6.RP.A.3b | Multiple select

10. $3.5 per notebook; $24.5

Unit rate = 14 ÷ 4 = 3.5 dollars per notebook.

6.RP.A.2, 6.RP.A.3b | Constructed response

Day 086 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Solve Constant-Speed Problems

Standards: 6.RP.A.3b

Learning target: I can solve constant-speed problems using a rate per hour.

1. 10 km/h

Speed = 60 ÷ 6 = 10; distance = speed × time.

6.RP.A.3b | Constructed response

2. 7 hours

Time = distance ÷ speed = 140 ÷ 20 = 7.

6.RP.A.3b | Constructed response

3. 30 km/h

Speed = 210 ÷ 7 = 30; distance = speed × time.

6.RP.A.3b | Constructed response

4. 5 hours

Time = distance ÷ speed = 50 ÷ 10 = 5.

6.RP.A.3b | Constructed response

5. 40 km/h

Speed = 200 ÷ 5 = 40; distance = speed × time.

6.RP.A.3b | Constructed response

6. 5 hours

Time = distance ÷ speed = 200 ÷ 40 = 5.

6.RP.A.3b | Constructed response

7. 30 km/h; 150 km

Speed = 45 ÷ 3/2 = 30; distance = speed × time.

6.RP.A.3b | Constructed response

8. B, C - In 5 hours it travels 125 km.; The speed is 25 km per hour.

Divide distance by time: 50 ÷ 2 = 25 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

9. B, D - In 5 hours it travels 50 km.; The speed is 10 km per hour.

Divide distance by time: 40 ÷ 4 = 10 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

10. 1 hours

Time = distance ÷ speed = 25 ÷ 25 = 1.

6.RP.A.3b | Constructed response

Day 087 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Convert Customary Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

1. 18 feet

6 × 3 = 18. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

2. 72 inches

6 × 12 = 72. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

3. 15 feet

5 × 3 = 15. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

4. 7 feet

84 ÷ 12 = 7; 1 feet = 12 inches.

6.RP.A.3d | Constructed response

5. 36 inches

3 × 12 = 36. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

6. 3 feet

36 ÷ 12 = 3; 1 feet = 12 inches.

6.RP.A.3d | Constructed response

7. 18 inches

3/2 × 12 = 18. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

8. A, C - 42 inches; 7/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 7/2 by 12, giving 42 inches. The physical length does not change.

6.RP.A.3d | Multiple select

9. A, C - 78 inches; 13/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 13/2 by 12, giving 78 inches. The physical length does not change.

6.RP.A.3d | Multiple select

10. 2 yards

6 ÷ 3 = 2; 1 yards = 3 feet.

6.RP.A.3d | Constructed response

Day 088 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Convert Metric Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

1. 7000 milliliters

7 × 1000 = 7000. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

2. 3000 milliliters

3 × 1000 = 3000. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

3. 500 centimeters

5 × 100 = 500. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

4. 8 liters

8000 ÷ 1000 = 8; 1 liters = 1000 milliliters.

6.RP.A.3d | Constructed response

5. 700 centimeters

7 × 100 = 700. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

6. 6 meters

600 ÷ 100 = 6; 1 meters = 100 centimeters.

6.RP.A.3d | Constructed response

7. 300 centimeters

3 × 100 = 300. Smaller units require more units for the same amount.

6.RP.A.3d | Constructed response

8. B, D - 650 centimeters; 13/2 × 100 centimeters

There are 100 smaller units in each larger one. Multiply 13/2 by 100, giving 650 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

9. C, D - 7/2 × 100 centimeters; 350 centimeters

There are 100 smaller units in each larger one. Multiply 7/2 by 100, giving 350 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

10. 2 liters

2000 ÷ 1000 = 2; 1 liters = 1000 milliliters.

6.RP.A.3d | Constructed response

Day 089 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Apply Ratio and Rate Reasoning

Standards: 6.RP.A.3a, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can apply Ratio and Rate Reasoning, and explain and check my reasoning.

1. 10; 15

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

2. 7 hours

Time = distance ÷ speed = 175 ÷ 25 = 7.

6.RP.A.3b | Constructed response

3. A $6/marker; B $6.5/marker; A is cheaper

Divide each price by its marker count; compare 6 with 6.5.

6.RP.A.3b | Constructed response

4. 2 hours

120 ÷ 60 = 2; 1 hours = 60 minutes.

6.RP.A.3d | Constructed response

5. 10; 15

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

6. 4 hours

Time = distance ÷ speed = 160 ÷ 40 = 4.

6.RP.A.3b | Constructed response

7. A $3.5/marker; B $4/marker; A is cheaper; A $49, B $56

Divide each price by its marker count; compare 3.5 with 4.

6.RP.A.3b | Constructed response

8. A, D - 54 inches; 9/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 9/2 by 12, giving 54 inches. The physical length does not change.

6.RP.A.3d | Multiple select

9. B, D - (14, 8); (21, 12)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

10. 3 hours

Time = distance ÷ speed = 60 ÷ 20 = 3.

6.RP.A.3b | Constructed response

Day 090 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Unit 6 Test

Standards: 6.EE.B.5, 6.EE.B.7, 6.RP.A.1, 6.RP.A.3a, 6.RP.A.2, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. No; left side = 13

2 × 3 + 7 = 13; compare with 11.

6.EE.B.5 | Constructed response

2. x = 2

Subtract 4 on both sides. Then x = 2; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

3. x = 21/2

Add 8 on both sides. Then x = 21/2; substitution gives 5/2 on each side.

6.EE.B.7 | Constructed response

4. x = 6

Divide by 2 on both sides. Then x = 6; substitution gives 12 on each side.

6.EE.B.7 | Constructed response

5. 7:2

Keep red first; divide 7:2 by 1.

6.RP.A.1 | Constructed response

6. 6; 18

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

7. Mix B

For 16 cups water, A uses 8 cups juice; B uses 12. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

8. 30 oaks and 15 pines

9 equal sections total; each section is 45 ÷ 9 = 5.

6.RP.A.3a | Constructed response

9. 6; 6

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

10. 30 square meters; aligned pairs (0,0), (3,6), (15,30)

15 ÷ 3 = 5 batches; 5 × 6 = 30.

6.RP.A.3a | Constructed response

11. (0,0), (6,3), (12,6), (30,15); 30 tickets earns $15

Scale both quantities by each batch count; plot horizontal tickets and vertical dollars.

6.RP.A.3a | Constructed response

12. A, D - 4 notebooks cost $24.; Each notebook costs $6.

Cost per notebook = 18 ÷ 3 = 6. Multiply $6 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

13. A: $4 each; B: $5 each; A is the better buy.

Divide each cost by its own notebook count: 20/5 = 4; 30/6 = 5. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

14. B, C - 2 notebooks cost $4.; Each notebook costs $2.

Cost per notebook = 6 ÷ 3 = 2. Multiply $2 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

15. B, D - The speed is 30 km per hour.; In 3 hours it travels 90 km.

Divide distance by time: 120 ÷ 4 = 30 km/h. Multiply this unit rate by 3 for the later distance.

6.RP.A.3b | Multiple select

16. C, D - 30 inches; 5/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 5/2 by 12, giving 30 inches. The physical length does not change.

6.RP.A.3d | Multiple select

17. B, C - 5/2 × 100 centimeters; 250 centimeters

There are 100 smaller units in each larger one. Multiply 5/2 by 100, giving 250 centimeters. The physical length does not change.

6.RP.A.3d | Multiple select

18. A, B - In 5 hours it travels 125 km.; The speed is 25 km per hour.

Divide distance by time: 75 ÷ 3 = 25 km/h. Multiply this unit rate by 5 for the later distance.

6.RP.A.3b | Multiple select

19. B, D - The fraction of all tiles that are red is 2/7.; The red-to-blue ratio is 2:5.

Divide both color counts by 2. For the fraction of the whole, use total 14 as denominator; red-to-blue compares just the two colors.

6.RP.A.1 | Multiple select

20. A, C - (15, 6); (10, 4)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

Day 091 | Unit 07 | Answer key

Understand and Use Percent

Understand Percent as a Rate per 100

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

1. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

2. 1/5; 0.2

20% = 20/100 = 1/5; divide by 100 for 0.2.

6.RP.A.3c | Constructed response

3. 1/2; 0.5

50% = 50/100 = 1/2; divide by 100 for 0.5.

6.RP.A.3c | Constructed response

4. 2/5; 0.4

40% = 40/100 = 2/5; divide by 100 for 0.4.

6.RP.A.3c | Constructed response

5. 3/10; 0.3

30% = 30/100 = 3/10; divide by 100 for 0.3.

6.RP.A.3c | Constructed response

6. 3/20; 0.15

15% = 15/100 = 3/20; divide by 100 for 0.15.

6.RP.A.3c | Constructed response

7. 3/4; 0.75

75% = 75/100 = 3/4; divide by 100 for 0.75.

6.RP.A.3c | Constructed response

8. A, B - 75/100; 0.75

Percent means per 100. Divide 75 by 100 to get 0.75; simplify 75/100 to 3/4.

6.RP.A.3c | Multiple select

9. A, D - 325/100; 3.25

Percent means per 100. Divide 325 by 100 to get 3.25; simplify 325/100 to 13/4.

6.RP.A.3c | Multiple select

10. 3/5; 0.6

60% = 60/100 = 3/5; divide by 100 for 0.6.

6.RP.A.3c | Constructed response

Day 092 | Unit 07 | Answer key

Understand and Use Percent

Relate Fractions, Decimals, and Percents

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

1. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

2. 1/5; 0.2

20% = 20/100 = 1/5; divide by 100 for 0.2.

6.RP.A.3c | Constructed response

3. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

4. 1/20; 0.05

5% = 5/100 = 1/20; divide by 100 for 0.05.

6.RP.A.3c | Constructed response

5. 3/5; 0.6

60% = 60/100 = 3/5; divide by 100 for 0.6.

6.RP.A.3c | Constructed response

6. 3/4; 0.75

75% = 75/100 = 3/4; divide by 100 for 0.75.

6.RP.A.3c | Constructed response

7. 3/10; 0.3

30% = 30/100 = 3/10; divide by 100 for 0.3.

6.RP.A.3c | Constructed response

8. A, B - 0.4; 40/100

Percent means per 100. Divide 40 by 100 to get 0.4; simplify 40/100 to 2/5.

6.RP.A.3c | Multiple select

9. C, D - 3.5; 350/100

Percent means per 100. Divide 350 by 100 to get 3.5; simplify 350/100 to 7/2.

6.RP.A.3c | Multiple select

10. 3/20; 0.15

15% = 15/100 = 3/20; divide by 100 for 0.15.

6.RP.A.3c | Constructed response

Day 093 | Unit 07 | Answer key

Understand and Use Percent

Represent Percents Above 100

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

1. 3/2; 1.5

150% = 150/100 = 3/2. Dividing by 100 gives 1.5, which exceeds 1 because 150 exceeds 100.

6.RP.A.3c | Constructed response

2. 21/10; 2.1

210% = 210/100 = 21/10. Dividing by 100 gives 2.1, which exceeds 1 because 210 exceeds 100.

6.RP.A.3c | Constructed response

3. 31/20; 1.55

155% = 155/100 = 31/20. Dividing by 100 gives 1.55, which exceeds 1 because 155 exceeds 100.

6.RP.A.3c | Constructed response

4. 11/10; 1.1

110% = 110/100 = 11/10. Dividing by 100 gives 1.1, which exceeds 1 because 110 exceeds 100.

6.RP.A.3c | Constructed response

5. 29/10; 2.9

290% = 290/100 = 29/10. Dividing by 100 gives 2.9, which exceeds 1 because 290 exceeds 100.

6.RP.A.3c | Constructed response

6. 47/20; 2.35

235% = 235/100 = 47/20. Dividing by 100 gives 2.35, which exceeds 1 because 235 exceeds 100.

6.RP.A.3c | Constructed response

7. 8/5; 1.6

160% = 160/100 = 8/5. Dividing by 100 gives 1.6, which exceeds 1 because 160 exceeds 100.

6.RP.A.3c | Constructed response

8. B, C - 275/100; 2.75

Percent means per 100. Divide 275 by 100 to get 2.75; simplify 275/100 to 11/4.

6.RP.A.3c | Multiple select

9. A, D - 325/100; 3.25

Percent means per 100. Divide 325 by 100 to get 3.25; simplify 325/100 to 13/4.

6.RP.A.3c | Multiple select

10. 23/20; 1.15

115% = 115/100 = 23/20. Dividing by 100 gives 1.15, which exceeds 1 because 115 exceeds 100.

6.RP.A.3c | Constructed response

Day 094 | Unit 07 | Answer key

Understand and Use Percent

Use Benchmark Percents

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

1. 20

10/100 × 200 = 20.

6.RP.A.3c | Constructed response

2. 90

50/100 × 180 = 90.

6.RP.A.3c | Constructed response

3. 40

20/100 × 200 = 40.

6.RP.A.3c | Constructed response

4. 125

125/100 × 100 = 125.

6.RP.A.3c | Constructed response

5. 84/5

12/100 × 140 = 84/5.

6.RP.A.3c | Constructed response

6. 11

5/100 × 220 = 11.

6.RP.A.3c | Constructed response

7. 40

40/100 × 100 = 40.

6.RP.A.3c | Constructed response

8. 125 cm

125/100 = 5/4 > 1. New width = 5/4 × 100 = 125 cm.

6.RP.A.3c | Numeric entry and explanation

9. 12 participants; n = (15/100) × 80; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 80 to obtain 12. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

10. 125

125/100 × 100 = 125.

6.RP.A.3c | Constructed response

Day 095 | Unit 07 | Answer key

Understand and Use Percent

Connect Percent Models and Benchmarks

Standards: 6.RP.A.3c

Learning target: I can connect Percent Models and Benchmarks, and explain and check my reasoning.

1. 1/2; 0.5

50% = 50/100 = 1/2; divide by 100 for 0.5.

6.RP.A.3c | Constructed response

2. 24

10/100 × 240 = 24.

6.RP.A.3c | Constructed response

3. 1/5; 0.2

20% = 20/100 = 1/5; divide by 100 for 0.2.

6.RP.A.3c | Constructed response

4. 175

125/100 × 140 = 175.

6.RP.A.3c | Constructed response

5. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

6. 40

20/100 × 200 = 40.

6.RP.A.3c | Constructed response

7. 5/4; 1.25

125% = 125/100 = 5/4; divide by 100 for 1.25.

6.RP.A.3c | Constructed response

8. 6 participants; n = (15/100) × 40; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 40 to obtain 6. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. B, D - 375/100; 3.75

Percent means per 100. Divide 375 by 100 to get 3.75; simplify 375/100 to 15/4.

6.RP.A.3c | Multiple select

10. 18

15/100 × 120 = 18.

6.RP.A.3c | Constructed response

Day 096 | Unit 07 | Answer key

Understand and Use Percent

Find the Percent of a Number

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

1. 110

50/100 × 220 = 110.

6.RP.A.3c | Constructed response

2. 40

20/100 × 200 = 40.

6.RP.A.3c | Constructed response

3. 30

25/100 × 120 = 30.

6.RP.A.3c | Constructed response

4. 108

60/100 × 180 = 108.

6.RP.A.3c | Constructed response

5. 30

15/100 × 200 = 30.

6.RP.A.3c | Constructed response

6. 150

75/100 × 200 = 150.

6.RP.A.3c | Constructed response

7. 50

125/100 × 40 = 50.

6.RP.A.3c | Constructed response

8. 20 participants; n = (20/100) × 100; less than the whole because the percent is below 100

Rename 20% as 1/5. Multiply by 100 to obtain 20. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. 32 participants; n = (40/100) × 80; less than the whole because the percent is below 100

Rename 40% as 2/5. Multiply by 80 to obtain 32. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

10. 36

15/100 × 240 = 36.

6.RP.A.3c | Constructed response

Day 097 | Unit 07 | Answer key

Understand and Use Percent

Use a Percent Model in Context

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

1. 110

50/100 × 220 = 110.

6.RP.A.3c | Constructed response

2. 16

10/100 × 160 = 16.

6.RP.A.3c | Constructed response

3. 12

20/100 × 60 = 12.

6.RP.A.3c | Constructed response

4. 30

15/100 × 200 = 30.

6.RP.A.3c | Constructed response

5. 11

5/100 × 220 = 11.

6.RP.A.3c | Constructed response

6. 30

30/100 × 100 = 30.

6.RP.A.3c | Constructed response

7. 120

75/100 × 160 = 120.

6.RP.A.3c | Constructed response

8. 30 participants; n = (25/100) × 120; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 120 to obtain 30. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. 10 participants; n = (25/100) × 40; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 40 to obtain 10. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

10. 75

75/100 × 100 = 75.

6.RP.A.3c | Constructed response

Day 098 | Unit 07 | Answer key

Understand and Use Percent

Find the Whole from Part and Percent

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

1. 140

Whole = 70 ÷ 1/2 = 140. Check 50% × 140 = 70.

6.RP.A.3c | Constructed response

2. 220

Whole = 22 ÷ 1/10 = 220. Check 10% × 220 = 22.

6.RP.A.3c | Constructed response

3. 160

Whole = 40 ÷ 1/4 = 160. Check 25% × 160 = 40.

6.RP.A.3c | Constructed response

4. 60

Whole = 15 ÷ 1/4 = 60. Check 25% × 60 = 15.

6.RP.A.3c | Constructed response

5. 180

Whole = 27 ÷ 3/20 = 180. Check 15% × 180 = 27.

6.RP.A.3c | Constructed response

6. 100

Whole = 75 ÷ 3/4 = 100. Check 75% × 100 = 75.

6.RP.A.3c | Constructed response

7. 220

Whole = 33 ÷ 3/20 = 220. Check 15% × 220 = 33.

6.RP.A.3c | Constructed response

8. A, B - (30/100)w = 12; w = 12 ÷ (30/100)

The part is percent rate times whole. Divide 12 by 3/10; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. A, B - w = 60 ÷ (60/100); (60/100)w = 60

The part is percent rate times whole. Divide 60 by 3/5; w = 100. Check by multiplication.

6.RP.A.3c | Multiple select

10. 140

Whole = 56 ÷ 2/5 = 140. Check 40% × 140 = 56.

6.RP.A.3c | Constructed response

Day 099 | Unit 07 | Answer key

Understand and Use Percent

Check the Reference Whole

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

1. 180

Whole = 36 ÷ 1/5 = 180. Check 20% × 180 = 36.

6.RP.A.3c | Constructed response

2. 60

Whole = 30 ÷ 1/2 = 60. Check 50% × 60 = 30.

6.RP.A.3c | Constructed response

3. 100

Whole = 50 ÷ 1/2 = 100. Check 50% × 100 = 50.

6.RP.A.3c | Constructed response

4. 240

Whole = 144/5 ÷ 3/25 = 240. Check 12% × 240 = 144/5.

6.RP.A.3c | Constructed response

5. 120

Whole = 30 ÷ 1/4 = 120. Check 25% × 120 = 30.

6.RP.A.3c | Constructed response

6. 160

Whole = 200 ÷ 5/4 = 160. Check 125% × 160 = 200.

6.RP.A.3c | Constructed response

7. 160

Whole = 48 ÷ 3/10 = 160. Check 30% × 160 = 48.

6.RP.A.3c | Constructed response

8. C, D - w = 24 ÷ (60/100); (60/100)w = 24

The part is percent rate times whole. Divide 24 by 3/5; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. A, B - (30/100)w = 12; w = 12 ÷ (30/100)

The part is percent rate times whole. Divide 12 by 3/10; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

10. 200

Whole = 80 ÷ 2/5 = 200. Check 40% × 200 = 80.

6.RP.A.3c | Constructed response

Day 100 | Unit 07 | Answer key

Understand and Use Percent

Explain the Reference Whole in Percent Models

Standards: 6.RP.A.3c

Learning target: I can explain the Reference Whole in Percent Models, and explain and check my reasoning.

1. 180

Whole = 45 ÷ 1/4 = 180. Check 25% × 180 = 45.

6.RP.A.3c | Constructed response

2. 55

25/100 × 220 = 55.

6.RP.A.3c | Constructed response

3. 140

Whole = 35 ÷ 1/4 = 140. Check 25% × 140 = 35.

6.RP.A.3c | Constructed response

4. 18

15/100 × 120 = 18.

6.RP.A.3c | Constructed response

5. 40

Whole = 50 ÷ 5/4 = 40. Check 125% × 40 = 50.

6.RP.A.3c | Constructed response

6. 88

40/100 × 220 = 88.

6.RP.A.3c | Constructed response

7. 220

Whole = 11 ÷ 1/20 = 220. Check 5% × 220 = 11.

6.RP.A.3c | Constructed response

8. 25 participants; n = (25/100) × 100; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 100 to obtain 25. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. C, D - (20/100)w = 24; w = 24 ÷ (20/100)

The part is percent rate times whole. Divide 24 by 1/5; w = 120. Check by multiplication.

6.RP.A.3c | Multiple select

10. 108/5

12/100 × 180 = 108/5.

6.RP.A.3c | Constructed response

Day 101 | Unit 07 | Answer key

Understand and Use Percent

Find the Unknown Percent

Standards: 6.RP.A.3c

Learning target: I can find the percent that one quantity is of another.

1. 20%

40 ÷ 200 × 100 = 20%; the reference whole is 200.

6.RP.A.3c | Constructed response

2. 10%

12 ÷ 120 × 100 = 10%; the reference whole is 120.

6.RP.A.3c | Constructed response

3. 50%

80 ÷ 160 × 100 = 50%; the reference whole is 160.

6.RP.A.3c | Constructed response

4. 125%

125 ÷ 100 × 100 = 125%; the reference whole is 100.

6.RP.A.3c | Constructed response

5. 75%

75 ÷ 100 × 100 = 75%; the reference whole is 100.

6.RP.A.3c | Constructed response

6. 40%

40 ÷ 100 × 100 = 40%; the reference whole is 100.

6.RP.A.3c | Constructed response

7. 75%

30 ÷ 40 × 100 = 75%; the reference whole is 40.

6.RP.A.3c | Constructed response

8. 15%; denominator 100

Percent = (15 ÷ 100) × 100 = 15%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

9. 60%; denominator 60

Percent = (36 ÷ 60) × 100 = 60%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 40%

24 ÷ 60 × 100 = 40%; the reference whole is 60.

6.RP.A.3c | Constructed response

Day 102 | Unit 07 | Answer key

Understand and Use Percent

Solve Percent Problems with Any Unknown

Standards: 6.RP.A.3c

Learning target: I can solve Percent Problems with Any Unknown, and explain and check my reasoning.

1. 50

25/100 × 200 = 50.

6.RP.A.3c | Constructed response

2. 180

Whole = 18 ÷ 1/10 = 180. Check 10% × 180 = 18.

6.RP.A.3c | Constructed response

3. 25%

10 ÷ 40 × 100 = 25%; the reference whole is 40.

6.RP.A.3c | Constructed response

4. 60

30/100 × 200 = 60.

6.RP.A.3c | Constructed response

5. 200

Whole = 150 ÷ 3/4 = 200. Check 75% × 200 = 150.

6.RP.A.3c | Constructed response

6. 40%

40 ÷ 100 × 100 = 40%; the reference whole is 100.

6.RP.A.3c | Constructed response

7. 12

20/100 × 60 = 12.

6.RP.A.3c | Constructed response

8. B, D - w = 32 ÷ (40/100); (40/100)w = 32

The part is percent rate times whole. Divide 32 by 2/5; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

9. 15%; denominator 60

Percent = (9 ÷ 60) × 100 = 15%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 225

125/100 × 180 = 225.

6.RP.A.3c | Constructed response

Day 103 | Unit 07 | Answer key

Understand and Use Percent

Explain and Critique Percent Reasoning

Standards: 6.RP.A.3c

Learning target: I can explain and Critique Percent Reasoning, and explain and check my reasoning.

1. 50%

40 ÷ 80 × 100 = 50%; the reference whole is 80.

6.RP.A.3c | Constructed response

2. 120

Whole = 24 ÷ 1/5 = 120. Check 20% × 120 = 24.

6.RP.A.3c | Constructed response

3. 20%

20 ÷ 100 × 100 = 20%; the reference whole is 100.

6.RP.A.3c | Constructed response

4. 40

Whole = 24/5 ÷ 3/25 = 40. Check 12% × 40 = 24/5.

6.RP.A.3c | Constructed response

5. 60%

24 ÷ 40 × 100 = 60%; the reference whole is 40.

6.RP.A.3c | Constructed response

6. 180

Whole = 108 ÷ 3/5 = 180. Check 60% × 180 = 108.

6.RP.A.3c | Constructed response

7. 60%

84 ÷ 140 × 100 = 60%; the reference whole is 140.

6.RP.A.3c | Constructed response

8. A, C - w = 16 ÷ (40/100); (40/100)w = 16

The part is percent rate times whole. Divide 16 by 2/5; w = 40. Check by multiplication.

6.RP.A.3c | Multiple select

9. 40%; denominator 100

Percent = (40 ÷ 100) × 100 = 40%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 120

Whole = 150 ÷ 5/4 = 120. Check 125% × 120 = 150.

6.RP.A.3c | Constructed response

Day 104 | Unit 07 | Answer key

Understand and Use Percent

Model a Percent Situation

Standards: 6.RP.A.3c

Learning target: I can model a Percent Situation, and explain and check my reasoning.

1. 40

50/100 × 80 = 40.

6.RP.A.3c | Constructed response

2. 80

Whole = 16 ÷ 1/5 = 80. Check 20% × 80 = 16.

6.RP.A.3c | Constructed response

3. 50%

70 ÷ 140 × 100 = 50%; the reference whole is 140.

6.RP.A.3c | Constructed response

4. 12

30/100 × 40 = 12.

6.RP.A.3c | Constructed response

5. 180

Whole = 225 ÷ 5/4 = 180. Check 125% × 180 = 225.

6.RP.A.3c | Constructed response

6. 30%

60 ÷ 200 × 100 = 30%; the reference whole is 200.

6.RP.A.3c | Constructed response

7. 60

30/100 × 200 = 60.

6.RP.A.3c | Constructed response

8. A, B - (15/100)w = 9; w = 9 ÷ (15/100)

The part is percent rate times whole. Divide 9 by 3/20; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

9. 40%; denominator 40

Percent = (16 ÷ 40) × 100 = 40%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

10. 9

15/100 × 60 = 9.

6.RP.A.3c | Constructed response

Day 105 | Unit 07 | Answer key

Understand and Use Percent

Unit 7 Test

Standards: 6.RP.A.1, 6.RP.A.3a, 6.RP.A.3c

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 1:1

Keep red first; divide 6:6 by 6.

6.RP.A.1 | Constructed response

2. 12; 12

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

3. Mix B

For 12 cups water, A uses 10 cups juice; B uses 14. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

4. 8 oaks and 24 pines

8 equal sections total; each section is 32 ÷ 8 = 4.

6.RP.A.3a | Constructed response

5. 1/2; 0.5

50% = 50/100 = 1/2; divide by 100 for 0.5.

6.RP.A.3c | Constructed response

6. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

7. 3; 3

300% = 300/100 = 3. Dividing by 100 gives 3, which exceeds 1 because 300 exceeds 100.

6.RP.A.3c | Constructed response

8. 40

40/100 × 100 = 40.

6.RP.A.3c | Constructed response

9. 1/20; 0.05

5% = 5/100 = 1/20; divide by 100 for 0.05.

6.RP.A.3c | Constructed response

10. 24

15/100 × 160 = 24.

6.RP.A.3c | Constructed response

11. 24

40/100 × 60 = 24.

6.RP.A.3c | Constructed response

12. C, D - (125/100)w = 100; w = 100 ÷ (125/100)

The part is percent rate times whole. Divide 100 by 5/4; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

13. B, D - (20/100)w = 20; w = 20 ÷ (20/100)

The part is percent rate times whole. Divide 20 by 1/5; w = 100. Check by multiplication.

6.RP.A.3c | Multiple select

14. C, D - w = 15 ÷ (25/100); (25/100)w = 15

The part is percent rate times whole. Divide 15 by 1/4; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

15. 125%; denominator 60

Percent = (75 ÷ 60) × 100 = 125%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

16. A, C - (15/100)w = 12; w = 12 ÷ (15/100)

The part is percent rate times whole. Divide 12 by 3/20; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

17. A, D - (40/100)w = 24; w = 24 ÷ (40/100)

The part is percent rate times whole. Divide 24 by 2/5; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

18. A, B - (60/100)w = 48; w = 48 ÷ (60/100)

The part is percent rate times whole. Divide 48 by 3/5; w = 80. Check by multiplication.

6.RP.A.3c | Multiple select

19. B, D - 250/100; 2.5

Percent means per 100. Divide 250 by 100 to get 2.5; simplify 250/100 to 5/2.

6.RP.A.3c | Multiple select

20. C, D - 3.5; 350/100

Percent means per 100. Divide 350 by 100 to get 3.5; simplify 350/100 to 7/2.

6.RP.A.3c | Multiple select

Day 106 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Triangle Area

Standards: 6.G.A.1

Learning target: I can find the area of any triangle using a perpendicular height.

1. 12 cm²

A = bh/2 = 8 × 3 ÷ 2 = 12. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

2. 40 cm²

A = bh/2 = 16 × 5 ÷ 2 = 40. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

3. 12 cm²

A = bh/2 = 4 × 6 ÷ 2 = 12. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

4. 35 cm²

A = bh/2 = 14 × 5 ÷ 2 = 35. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

5. 8 cm²

A = bh/2 = 8 × 2 ÷ 2 = 8. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

6. 30 cm²

A = bh/2 = 12 × 5 ÷ 2 = 30. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

7. 6 cm²

A = bh/2 = 8 × 3/2 ÷ 2 = 6. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

8. B, C - 4 × 6; (8 × 6) ÷ 2

Triangle area is half the matching parallelogram: (8 × 6)/2 = 24 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

9. B, C - 5 × 5; (10 × 5) ÷ 2

Triangle area is half the matching parallelogram: (10 × 5)/2 = 25 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

10. 35/2 cm²

A = bh/2 = 10 × 7/2 ÷ 2 = 35/2. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

Day 107 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Parallelogram and Trapezoid Area

Standards: 6.G.A.1

Learning target: I can find parallelogram and trapezoid areas by rearranging shapes.

1. 32 m²

Cut and shift a corner; A = bh = 8 × 4 = 32.

6.G.A.1 | Constructed response

2. 24 m²

Cut and shift a corner; A = bh = 6 × 4 = 24.

6.G.A.1 | Constructed response

3. 56 m²

Cut and shift a corner; A = bh = 8 × 7 = 56.

6.G.A.1 | Constructed response

4. 45 m²

A = (4 + 14) × 5 ÷ 2 = 45.

6.G.A.1 | Constructed response

5. 84 m²

A = (8 + 16) × 7 ÷ 2 = 84.

6.G.A.1 | Constructed response

6. 40 m²

A = (8 + 12) × 4 ÷ 2 = 40.

6.G.A.1 | Constructed response

7. 50 m²

A = (6 + 14) × 5 ÷ 2 = 50.

6.G.A.1 | Constructed response

8. 72 m²

Use average base × height: (14 + 22)/2 × 4 = 72. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis

9. 12 m²

Use average base × height: (4 + 8)/2 × 2 = 12. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis

10. 36 m²

A = (12 + 24) × 2 ÷ 2 = 36.

6.G.A.1 | Constructed response

Day 108 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Decompose Composite Polygons

Standards: 6.G.A.1

Learning target: I can decompose a polygon into familiar shapes to find area.

1. 95 m²

Outer area 143 minus cutout 48 = 95.

6.G.A.1 | Constructed response

2. 84 m²

Outer area 99 minus cutout 15 = 84.

6.G.A.1 | Constructed response

3. 114 m²

Outer area 154 minus cutout 40 = 114.

6.G.A.1 | Constructed response

4. 60 m²

Outer area 72 minus cutout 12 = 60.

6.G.A.1 | Constructed response

5. 72 m²

Outer area 81 minus cutout 9 = 72.

6.G.A.1 | Constructed response

6. 96 m²

Outer area 112 minus cutout 16 = 96.

6.G.A.1 | Constructed response

7. 23 m²

Whole rectangle 28; remove triangle 5; difference 23.

6.G.A.1 | Constructed response

8. Removed 12 m²; remaining 98 m²

Outer area = 110. Triangle = 4 × 6/2 = 12. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

9. Removed 15/2 m²; remaining 113/2 m²

Outer area = 64. Triangle = 5 × 3/2 = 15/2. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

10. 99/2 m²

Whole rectangle 54; remove triangle 9/2; difference 99/2.

6.G.A.1 | Constructed response

Day 109 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Use Coordinates to Find Polygon Lengths and Area

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can draw polygons from coordinates and use side lengths to find area.

1. Sides 8 and 7; area 56 square units

Horizontal length 8 - (0) = 8; vertical length 7 - (0) = 7; area = 8 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

2. Sides 8 and 5; area 40 square units

Horizontal length 8 - (0) = 8; vertical length 5 - (0) = 5; area = 8 × 5.

6.G.A.3, 6.G.A.1 | Constructed response

3. Sides 7 and 12; area 84 square units

Horizontal length 7 - (0) = 7; vertical length 12 - (0) = 12; area = 7 × 12.

6.G.A.3, 6.G.A.1 | Constructed response

4. Sides 9 and 11; area 99 square units

Horizontal length 6 - (-3) = 9; vertical length 6 - (-5) = 11; area = 9 × 11.

6.G.A.3, 6.G.A.1 | Constructed response

5. Sides 6 and 4; area 24 square units

Horizontal length 2 - (-4) = 6; vertical length 2 - (-2) = 4; area = 6 × 4.

6.G.A.3, 6.G.A.1 | Constructed response

6. Sides 5 and 6; area 30 square units

Horizontal length 2 - (-3) = 5; vertical length 2 - (-4) = 6; area = 5 × 6.

6.G.A.3, 6.G.A.1 | Constructed response

7. Sides 14 and 9; area 126 square units; perimeter 46 units

Horizontal length 6 - (-8) = 14; vertical length 6 - (-3) = 9; area = 14 × 9.

6.G.A.3, 6.G.A.1 | Constructed response

8. 10 by 8 units; area 80 square units

Horizontal length = 6 + 4; vertical length = 4 + 4. Multiply 10 × 8 = 80.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

9. 10 by 8 units; area 80 square units

Horizontal length = 5 + 5; vertical length = 4 + 4. Multiply 10 × 8 = 80.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. Sides 11 and 7; area 77 square units; perimeter 36 units

Horizontal length 4 - (-7) = 11; vertical length 4 - (-3) = 7; area = 11 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

Day 110 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Justify Polygon Area with Coordinates

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can justify Polygon Area with Coordinates, and explain and check my reasoning.

1. Sides 11 and 5; area 55 square units

Horizontal length 11 - (0) = 11; vertical length 5 - (0) = 5; area = 11 × 5.

6.G.A.3, 6.G.A.1 | Constructed response

2. 90 m²

Outer area 130 minus cutout 40 = 90.

6.G.A.1 | Constructed response

3. Sides 11 and 10; area 110 square units

Horizontal length 11 - (0) = 11; vertical length 10 - (0) = 10; area = 11 × 10.

6.G.A.3, 6.G.A.1 | Constructed response

4. 39 m²

Outer area 60 minus cutout 21 = 39.

6.G.A.1 | Constructed response

5. Sides 11 and 7; area 77 square units

Horizontal length 3 - (-8) = 11; vertical length 3 - (-4) = 7; area = 11 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

6. 24 m²

Outer area 30 minus cutout 6 = 24.

6.G.A.1 | Constructed response

7. Sides 8 and 10; area 80 square units; perimeter 36 units

Horizontal length 6 - (-2) = 8; vertical length 6 - (-4) = 10; area = 8 × 10.

6.G.A.3, 6.G.A.1 | Constructed response

8. Removed 9 m²; remaining 72 m²

Outer area = 81. Triangle = 3 × 6/2 = 9. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

9. 6 by 4 units; area 24 square units

Horizontal length = 2 + 4; vertical length = 2 + 2. Multiply 6 × 4 = 24.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. 96 m²

Whole rectangle 110; remove triangle 14; difference 96.

6.G.A.1 | Constructed response

Day 111 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Build and Interpret Nets

Standards: 6.G.A.4

Learning target: I can draw a net and account for every face of a solid.

1. 2 faces each of 4×4, 4×4, 4×4; 96 cm²

Add 2(16) + 2(16) + 2(16) = 96.

6.G.A.4 | Constructed response

2. 2 faces each of 8×3, 8×2, 3×2; 92 cm²

Add 2(24) + 2(16) + 2(6) = 92.

6.G.A.4 | Constructed response

3. 2 faces each of 7×5, 7×2, 5×2; 118 cm²

Add 2(35) + 2(14) + 2(10) = 118.

6.G.A.4 | Constructed response

4. 2 faces each of 3×7, 3×5, 7×5; 142 cm²

Add 2(21) + 2(15) + 2(35) = 142.

6.G.A.4 | Constructed response

5. 2 faces each of 8×7, 8×4, 7×4; 232 cm²

Add 2(56) + 2(32) + 2(28) = 232.

6.G.A.4 | Constructed response

6. 2 faces each of 2×4, 2×3, 4×3; 52 cm²

Add 2(8) + 2(6) + 2(12) = 52.

6.G.A.4 | Constructed response

7. 2 triangles (area 6 each), rectangles 3×4, 4×4, 5×4; SA 60 cm²

Triangle pair: 12 cm²; rectangles: (3+4+5) × 4 = 48; total 60.

6.G.A.4 | Constructed response

8. A, C - Two faces are 6 by 4.; Two faces are 6 by 4 and two are 4 by 4.

A closed rectangular prism has 6 rectangular faces in 3 matching pairs. Count both copies of each face.

6.G.A.4 | Multiple select

9. C, D - Two faces are 6 by 5.; Two faces are 6 by 3 and two are 5 by 3.

A closed rectangular prism has 6 rectangular faces in 3 matching pairs. Count both copies of each face.

6.G.A.4 | Multiple select

10. 2 triangles (area 6 each), rectangles 3×6, 4×6, 5×6; SA 84 cm²

Triangle pair: 12 cm²; rectangles: (3+4+5) × 6 = 72; total 84.

6.G.A.4 | Constructed response

Day 112 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Prism Surface Area

Standards: 6.G.A.4

Learning target: I can use a rectangular-prism net to find surface area.

1. 122 cm²

Closed surface area = 2(28+12+21) = 122.

6.G.A.4 | Constructed response

2. 180 cm²

Closed surface area = 2(24+48+18) = 180.

6.G.A.4 | Constructed response

3. 104 cm²

Closed surface area = 2(30+12+10) = 104.

6.G.A.4 | Constructed response

4. 52 cm²

Closed surface area = 2(12+6+8) = 52.

6.G.A.4 | Constructed response

5. 126 cm²

Closed surface area = 2(30+15+18) = 126.

6.G.A.4 | Constructed response

6. 184 cm²

Closed surface area = 2(32+40+20) = 184.

6.G.A.4 | Constructed response

7. 84 cm²

Closed surface area = 2(36+12+12) = 120. Remove the top area 36: 84.

6.G.A.4 | Constructed response

8. 76 cm²; open box 68 cm²

Closed area = 2(8+20+10) = 76. Subtract exactly one top face, 8, for the open box.

6.G.A.4 | Two-part numeric entry

9. 76 cm²; open box 66 cm²

Closed area = 2(10+8+20) = 76. Subtract exactly one top face, 10, for the open box.

6.G.A.4 | Two-part numeric entry

10. 61 cm²

Closed surface area = 2(21+14+6) = 82. Remove the top area 21: 61.

6.G.A.4 | Constructed response

Day 113 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Pyramid Surface Area

Standards: 6.G.A.4

Learning target: I can find pyramid surface area from its triangular faces and base.

1. 336 cm²

Base = 144; each triangle = 12 × 8 ÷ 2 = 48; add four triangles to get 336.

6.G.A.4 | Constructed response

2. 340 cm²

Base = 100; each triangle = 10 × 12 ÷ 2 = 60; add four triangles to get 340.

6.G.A.4 | Constructed response

3. 456 cm²

Base = 144; each triangle = 12 × 13 ÷ 2 = 78; add four triangles to get 456.

6.G.A.4 | Constructed response

4. 64 cm²

Base = 16; each triangle = 4 × 6 ÷ 2 = 12; add four triangles to get 64.

6.G.A.4 | Constructed response

5. 192 cm²

Base = 64; each triangle = 8 × 8 ÷ 2 = 32; add four triangles to get 192.

6.G.A.4 | Constructed response

6. 176 cm²

Base = 64; each triangle = 8 × 7 ÷ 2 = 28; add four triangles to get 176.

6.G.A.4 | Constructed response

7. 120 cm²

Base = 36; each triangle = 6 × 7 ÷ 2 = 21; add four triangles to get 120.

6.G.A.4 | Constructed response

8. Face 48 cm²; total 336 cm²

One face = (12 × 8)/2 = 48. Four faces total 192. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

9. Face 60 cm²; total 384 cm²

One face = (12 × 10)/2 = 60. Four faces total 240. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

10. 408 cm²

Base = 144; each triangle = 12 × 11 ÷ 2 = 66; add four triangles to get 408.

6.G.A.4 | Constructed response

Day 114 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Find Volume with Fractional Edge Lengths

Standards: 6.G.A.2

Learning target: I can calculate prism volume with fractional edge lengths.

1. 20 cm³

B = 2 × 2 = 4 cm²; V = B × 5 = 20 cm³.

6.G.A.2 | Constructed response

2. 20 cm³

B = 4 × 1 = 4 cm²; V = B × 5 = 20 cm³.

6.G.A.2 | Constructed response

3. 18 cm³

B = 2 × 3 = 6 cm²; V = B × 3 = 18 cm³.

6.G.A.2 | Constructed response

4. 25/2 cm³

B = 5/2 × 5/2 = 25/4 cm²; V = B × 2 = 25/2 cm³.

6.G.A.2 | Constructed response

5. 45/4 cm³

B = 3/2 × 5/2 = 15/4 cm²; V = B × 3 = 45/4 cm³.

6.G.A.2 | Constructed response

6. 15 cm³

B = 2 × 3/2 = 3 cm²; V = B × 5 = 15 cm³.

6.G.A.2 | Constructed response

7. 15/2 cm³

B = 3 × 5/2 = 15/2 cm²; V = B × 1 = 15/2 cm³.

6.G.A.2 | Constructed response

8. 105/8 m³; 105 cubes

Multiply edges: 7/2 × 3/2 × 5/2 = 105/8. The edge counts are 7, 3, 5, giving 105 cubes. Each cube is 1/8 m³; 105/8 = 105/8.

6.G.A.2 | Numeric entry and model reasoning

9. 45/4 m³; 90 cubes

Multiply edges: 15/2 × 3/2 × 1 = 45/4. The edge counts are 15, 3, 2, giving 90 cubes. Each cube is 1/8 m³; 90/8 = 45/4.

6.G.A.2 | Numeric entry and model reasoning

10. 15/2 cm³

B = 1 × 5/2 = 5/2 cm²; V = B × 3 = 15/2 cm³.

6.G.A.2 | Constructed response

Day 115 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Connect Surface Area and Volume Models

Standards: 6.G.A.4, 6.G.A.2

Learning target: I can connect Surface Area and Volume Models, and explain and check my reasoning.

1. 180 cm²

Closed surface area = 2(48+24+18) = 180.

6.G.A.4 | Constructed response

2. 28 cm³

B = 4 × 7/2 = 14 cm²; V = B × 2 = 28 cm³.

6.G.A.2 | Constructed response

3. 162 cm²

Closed surface area = 2(42+21+18) = 162.

6.G.A.4 | Constructed response

4. 30 cm³

B = 3 × 2 = 6 cm²; V = B × 5 = 30 cm³.

6.G.A.2 | Constructed response

5. 128 cm²

Closed surface area = 2(16+24+24) = 128.

6.G.A.4 | Constructed response

6. 21 cm³

B = 7/2 × 3/2 = 21/4 cm²; V = B × 4 = 21 cm³.

6.G.A.2 | Constructed response

7. 95 cm²

Closed surface area = 2(15+25+15) = 110. Remove the top area 15: 95.

6.G.A.4 | Constructed response

8. 14 m³; 112 cubes

Multiply edges: 7/2 × 2 × 2 = 14. The edge counts are 7, 4, 4, giving 112 cubes. Each cube is 1/8 m³; 112/8 = 14.

6.G.A.2 | Numeric entry and model reasoning

9. 48 cm²; open box 44 cm²

Closed area = 2(4+10+10) = 48. Subtract exactly one top face, 4, for the open box.

6.G.A.4 | Two-part numeric entry

10. 9/2 cm³

B = 2 × 3/2 = 3 cm²; V = B × 3/2 = 9/2 cm³.

6.G.A.2 | Constructed response

Day 116 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Connect Fractional Unit Cubes and Volume

Standards: 6.G.A.2

Learning target: I can connect fractional unit cubes with a prism volume formula.

1. 90 cubes; each 1/8 cubic unit; prism 45/4 cubic units

Counts along edges: 6, 5, 3. Product 90; 90 × 1/8 = 45/4, also lwh.

6.G.A.2 | Constructed response

2. 45 cubes; each 1/8 cubic unit; prism 45/8 cubic units

Counts along edges: 3, 3, 5. Product 45; 45 × 1/8 = 45/8, also lwh.

6.G.A.2 | Constructed response

3. 140 cubes; each 1/8 cubic unit; prism 35/2 cubic units

Counts along edges: 5, 7, 4. Product 140; 140 × 1/8 = 35/2, also lwh.

6.G.A.2 | Constructed response

4. 125 cubes; each 1/8 cubic unit; prism 125/8 cubic units

Counts along edges: 5, 5, 5. Product 125; 125 × 1/8 = 125/8, also lwh.

6.G.A.2 | Constructed response

5. 168 cubes; each 1/8 cubic unit; prism 21 cubic units

Counts along edges: 7, 4, 6. Product 168; 168 × 1/8 = 21, also lwh.

6.G.A.2 | Constructed response

6. 96 cubes; each 1/8 cubic unit; prism 12 cubic units

Counts along edges: 8, 3, 4. Product 96; 96 × 1/8 = 12, also lwh.

6.G.A.2 | Constructed response

7. 100 cubes; each 1/27 cubic unit; prism 100/27 cubic units

Counts along edges: 5, 4, 5. Product 100; 100 × 1/27 = 100/27, also lwh.

6.G.A.2 | Constructed response

8. 39/2 m³; 156 cubes

Multiply edges: 13/2 × 3/2 × 2 = 39/2. The edge counts are 13, 3, 4, giving 156 cubes. Each cube is 1/8 m³; 156/8 = 39/2.

6.G.A.2 | Numeric entry and model reasoning

9. 45/2 m³; 180 cubes

Multiply edges: 15/2 × 2 × 3/2 = 45/2. The edge counts are 15, 4, 3, giving 180 cubes. Each cube is 1/8 m³; 180/8 = 45/2.

6.G.A.2 | Numeric entry and model reasoning

10. 90 cubes; each 1/27 cubic unit; prism 10/3 cubic units

Counts along edges: 5, 3, 6. Product 90; 90 × 1/27 = 10/3, also lwh.

6.G.A.2 | Constructed response

Day 117 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Use Base Area to Find Prism Volume

Standards: 6.G.A.2

Learning target: I can calculate prism volume with fractional edge lengths.

1. 147/2 cm³

B = 7 × 7/2 = 49/2 cm²; V = B × 3 = 147/2 cm³.

6.G.A.2 | Constructed response

2. 56 cm³

B = 7 × 2 = 14 cm²; V = B × 4 = 56 cm³.

6.G.A.2 | Constructed response

3. 9 cm³

B = 2 × 3/2 = 3 cm²; V = B × 3 = 9 cm³.

6.G.A.2 | Constructed response

4. 18 cm³

B = 2 × 3/2 = 3 cm²; V = B × 6 = 18 cm³.

6.G.A.2 | Constructed response

5. 75/2 cm³

B = 5/2 × 5/2 = 25/4 cm²; V = B × 6 = 75/2 cm³.

6.G.A.2 | Constructed response

6. 18 cm³

B = 3 × 1 = 3 cm²; V = B × 6 = 18 cm³.

6.G.A.2 | Constructed response

7. 8 cm³

B = 4 × 1 = 4 cm²; V = B × 2 = 8 cm³.

6.G.A.2 | Constructed response

8. 15/2 m³; 60 cubes

Multiply edges: 5/2 × 3/2 × 2 = 15/2. The edge counts are 5, 3, 4, giving 60 cubes. Each cube is 1/8 m³; 60/8 = 15/2.

6.G.A.2 | Numeric entry and model reasoning

9. 21/4 m³; 42 cubes

Multiply edges: 7/2 × 3/2 × 1 = 21/4. The edge counts are 7, 3, 2, giving 42 cubes. Each cube is 1/8 m³; 42/8 = 21/4.

6.G.A.2 | Numeric entry and model reasoning

10. 10 cm³

B = 5/2 × 2 = 5 cm²; V = B × 2 = 10 cm³.

6.G.A.2 | Constructed response

Day 118 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Distinguish Area, Surface Area, and Volume

Standards: 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can distinguish Area, Surface Area, and Volume, and explain and check my reasoning.

1. 28 cm²

A = bh/2 = 8 × 7 ÷ 2 = 28. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

2. 144 cm²

Closed surface area = 2(28+28+16) = 144.

6.G.A.4 | Constructed response

3. 72 cm³

B = 6 × 3 = 18 cm²; V = B × 4 = 72 cm³.

6.G.A.2 | Constructed response

4. 21 cm²

A = bh/2 = 6 × 7 ÷ 2 = 21. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

5. 72 cm²

Closed surface area = 2(12+18+6) = 72.

6.G.A.4 | Constructed response

6. 6 cm³

B = 2 × 1 = 2 cm²; V = B × 3 = 6 cm³.

6.G.A.2 | Constructed response

7. 35/2 cm²

A = bh/2 = 10 × 7/2 ÷ 2 = 35/2. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

8. 94 cm²; open box 79 cm²

Closed area = 2(15+20+12) = 94. Subtract exactly one top face, 15, for the open box.

6.G.A.4 | Two-part numeric entry

9. 15/2 m³; 60 cubes

Multiply edges: 5/2 × 2 × 3/2 = 15/2. The edge counts are 5, 4, 3, giving 60 cubes. Each cube is 1/8 m³; 60/8 = 15/2.

6.G.A.2 | Numeric entry and model reasoning

10. 6 cm²

A = bh/2 = 4 × 3 ÷ 2 = 6. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

Day 119 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Model a Geometry Design

Standards: 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can model a Geometry Design, and explain and check my reasoning.

1. Sides 11 and 8; area 88 square units

Horizontal length 11 - (0) = 11; vertical length 8 - (0) = 8; area = 11 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

2. 48 m²

Outer area 90 minus cutout 42 = 48.

6.G.A.1 | Constructed response

3. 158 cm²

Closed surface area = 2(40+24+15) = 158.

6.G.A.4 | Constructed response

4. 4 cm³

B = 2 × 1 = 2 cm²; V = B × 2 = 4 cm³.

6.G.A.2 | Constructed response

5. Sides 7 and 7; area 49 square units

Horizontal length 2 - (-5) = 7; vertical length 2 - (-5) = 7; area = 7 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

6. 72 m²

Outer area 81 minus cutout 9 = 72.

6.G.A.1 | Constructed response

7. 26 cm²

Closed surface area = 2(6+4+6) = 32. Remove the top area 6: 26.

6.G.A.4 | Constructed response

8. 63/8 m³; 63 cubes

Multiply edges: 7/2 × 3/2 × 3/2 = 63/8. The edge counts are 7, 3, 3, giving 63 cubes. Each cube is 1/8 m³; 63/8 = 63/8.

6.G.A.2 | Numeric entry and model reasoning

9. 11 by 8 units; area 88 square units

Horizontal length = 7 + 4; vertical length = 4 + 4. Multiply 11 × 8 = 88.

6.G.A.3, 6.G.A.1 | Coordinate plotting and two-part response

10. 89 m²

Whole rectangle 99; remove triangle 10; difference 89.

6.G.A.1 | Constructed response

Day 120 | Unit 08 | Answer key

Solve Area, Surface Area, and Volume Problems

Unit 8 Test

Standards: 6.RP.A.3c, 6.G.A.1, 6.G.A.3, 6.G.A.4, 6.G.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 1/4; 0.25

25% = 25/100 = 1/4; divide by 100 for 0.25.

6.RP.A.3c | Constructed response

2. 1/10; 0.1

10% = 10/100 = 1/10; divide by 100 for 0.1.

6.RP.A.3c | Constructed response

3. 29/10; 2.9

290% = 290/100 = 29/10. Dividing by 100 gives 2.9, which exceeds 1 because 290 exceeds 100.

6.RP.A.3c | Constructed response

4. 50

50/100 × 100 = 50.

6.RP.A.3c | Constructed response

5. 36 cm²

A = bh/2 = 12 × 6 ÷ 2 = 36. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

6. 10 m²

Cut and shift a corner; A = bh = 5 × 2 = 10.

6.G.A.1 | Constructed response

7. 18 m²

Outer area 28 minus cutout 10 = 18.

6.G.A.1 | Constructed response

8. Sides 7 and 7; area 49 square units

Horizontal length 2 - (-5) = 7; vertical length 2 - (-5) = 7; area = 7 × 7.

6.G.A.3, 6.G.A.1 | Constructed response

9. Sides 11 and 8; area 88 square units

Horizontal length 5 - (-6) = 11; vertical length 5 - (-3) = 8; area = 11 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

10. 2 faces each of 4×7, 4×5, 7×5; 166 cm²

Add 2(28) + 2(20) + 2(35) = 166.

6.G.A.4 | Constructed response

11. 170 cm²

Closed surface area = 2(30+25+30) = 170.

6.G.A.4 | Constructed response

12. Face 60 cm²; total 384 cm²

One face = (12 × 10)/2 = 60. Four faces total 240. Add square base area 144.

6.G.A.4 | Net reasoning and numeric entry

13. 45/4 m³; 90 cubes

Multiply edges: 9/2 × 1 × 5/2 = 45/4. The edge counts are 9, 2, 5, giving 90 cubes. Each cube is 1/8 m³; 90/8 = 45/4.

6.G.A.2 | Numeric entry and model reasoning

14. 148 cm²; open box 124 cm²

Closed area = 2(24+30+20) = 148. Subtract exactly one top face, 24, for the open box.

6.G.A.4 | Two-part numeric entry

15. 165/8 m³; 165 cubes

Multiply edges: 11/2 × 3/2 × 5/2 = 165/8. The edge counts are 11, 3, 5, giving 165 cubes. Each cube is 1/8 m³; 165/8 = 165/8.

6.G.A.2 | Numeric entry and model reasoning

16. 21/4 m³; 42 cubes

Multiply edges: 7/2 × 1 × 3/2 = 21/4. The edge counts are 7, 2, 3, giving 42 cubes. Each cube is 1/8 m³; 42/8 = 21/4.

6.G.A.2 | Numeric entry and model reasoning

17. 126 cm²; open box 96 cm²

Closed area = 2(30+15+18) = 126. Subtract exactly one top face, 30, for the open box.

6.G.A.4 | Two-part numeric entry

18. Removed 15 m²; remaining 95 m²

Outer area = 110. Triangle = 5 × 6/2 = 15. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

19. A, C - 4 × 3; (8 × 3) ÷ 2

Triangle area is half the matching parallelogram: (8 × 3)/2 = 12 cm². Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

20. 30 m²

Use average base × height: (6 + 14)/2 × 3 = 30. A full rectangle at the longer base includes extra corner regions.

6.G.A.1 | Numeric entry and error analysis

Day 121 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Recognize Statistical Questions

Standards: 6.SP.A.1

Learning target: I can recognize a statistical question and the expected variability.

1. Yes; pages read vary across students.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

2. No; one fixed conversion, 100 cm.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

3. Yes; plant heights vary.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

4. Group: class; attribute: sleep duration; units: minutes.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

5. Example: What are the lengths in centimeters of the pencils in our classroom?

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. It seeks one fixed value, rather than anticipating variation in a data collection.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

7. Example: How many minutes do students in our class take to reach school? Group: class; travel time in minutes varies with routes and travel methods.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

8. B, D - What is the reading time in minutes of each of the students?; How does reading time vary among the students?

Different students can have different reading time. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

9. A, C - How does mass vary among the backpacks?; What is the mass in kilograms of each of the backpacks?

Different backpacks can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

10. Counterexample: How many sides does a triangle have? It has a fixed answer. A statistical question anticipates variability in a collection of data.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

Day 122 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Summarize Data with Mean and Median

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c

Learning target: I can summarize Data with Mean and Median, and explain and check my reasoning.

1. n = 6; minutes; mean 11/2 min

Sum = 33; mean = 33 ÷ 6 = 11/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

2. Median 11/2 min

6 ordered values: average positions 3 and 4: (5 + 6) ÷ 2 = 11/2.

6.SP.B.5c | Constructed response

3. n = 6; minutes; mean 9/2 min

Sum = 27; mean = 27 ÷ 6 = 9/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

4. Median 14 min

6 ordered values: average positions 3 and 4: (12 + 16) ÷ 2 = 14.

6.SP.B.5c | Constructed response

5. n = 6; minutes; mean 19/2 min

Sum = 57; mean = 57 ÷ 6 = 19/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

6. Median 15/2 min

6 ordered values: average positions 3 and 4: (6 + 9) ÷ 2 = 15/2.

6.SP.B.5c | Constructed response

7. n = 6; minutes; mean 31/6 min; new mean 37/7 min

Sum = 31; mean = 31 ÷ 6 = 31/6. New sum = 37; divide by 7.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

8. A, D - The median is the average of the 3rd and 4th ordered values.; The median is 21/2 minutes.

Six ordered observations have middle positions 3 and 4, so median = (8 + 13)/2 = 21/2.

6.SP.B.5c | Multiple select

9. 12 min; new mean 99/7 min

Original sum 72 divided by 6 gives 12. Add 27, then divide the new sum by 7 because there are now 7 observations.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Two-part numeric entry

10. Median 17/2 min; mean 61/6 min; the high value pulls the mean upward

6 ordered values: average positions 3 and 4: (8 + 9) ÷ 2 = 17/2.

6.SP.B.5c | Constructed response

Day 123 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Display and Interpret Dot Plots

Standards: 6.SP.B.4

Learning target: I can construct and interpret a dot plot on a number line.

1. 8: 1 dot; 9: 2 dots; 10: 2 dots; 11: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

2. 2: 1 dot; 3: 2 dots; 4: 2 dots; 5: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

3. 4: 1 dot; 5: 2 dots; 6: 2 dots; 7: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

4. 4: 1 dot; 6: 2 dots; 8: 2 dots; 10: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

5. 3: 1 dot; 8: 2 dots; 13: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

6. 5: 1 dot; 11: 2 dots; 17: 2 dots; 23: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

7. 5: 1 dot; 6: 2 dots; 7: 1 dot; 8: 1 dot; 17: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

8. 12 students; stacks at 0, 1, 2, 3 have 3, 5, 3, 1 dots.

Place one dot per student, not one per distinct value. Total dots = 3 + 5 + 3 + 1 = 12. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

9. 18 students; stacks at 0, 1, 2, 3 have 4, 4, 5, 5 dots.

Place one dot per student, not one per distinct value. Total dots = 4 + 4 + 5 + 5 = 18. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

10. 6: 1 dot; 7: 2 dots; 8: 1 dot; 9: 1 dot; 18: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

Day 124 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Display Data in Box Plots

Standards: 6.SP.B.4, 6.SP.B.5c

Learning target: I can create a box plot from a five-number summary.

1. (3, 4, 9/2, 5, 6)

Box runs from 4 to 5, median line at 9/2, whiskers to 3 and 6.

6.SP.B.4, 6.SP.B.5c | Constructed response

2. (2, 3, 7/2, 4, 5)

Box runs from 3 to 4, median line at 7/2, whiskers to 2 and 5.

6.SP.B.4, 6.SP.B.5c | Constructed response

3. (5, 6, 13/2, 7, 8)

Box runs from 6 to 7, median line at 13/2, whiskers to 5 and 8.

6.SP.B.4, 6.SP.B.5c | Constructed response

4. (2, 6, 8, 10, 14)

Box runs from 6 to 10, median line at 8, whiskers to 2 and 14.

6.SP.B.4, 6.SP.B.5c | Constructed response

5. (6, 10, 12, 14, 18)

Box runs from 10 to 14, median line at 12, whiskers to 6 and 18.

6.SP.B.4, 6.SP.B.5c | Constructed response

6. (2, 5, 13/2, 8, 11)

Box runs from 5 to 8, median line at 13/2, whiskers to 2 and 11.

6.SP.B.4, 6.SP.B.5c | Constructed response

7. (2, 3, 7/2, 5, 14); IQR 2 min

Box runs from 3 to 5, median line at 7/2, whiskers to 2 and 14.

6.SP.B.4, 6.SP.B.5c | Constructed response

8. (7, 11, 13, 15, 19); IQR 4 min

Minimum and maximum anchor whiskers. Box ends are 11, 15; median line is 13. IQR = 15 - 11.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

9. (2, 7, 19/2, 12, 17); IQR 5 min

Minimum and maximum anchor whiskers. Box ends are 7, 12; median line is 19/2. IQR = 12 - 7.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

10. (7, 8, 17/2, 10, 19); IQR 2 min

Box runs from 8 to 10, median line at 17/2, whiskers to 7 and 19.

6.SP.B.4, 6.SP.B.5c | Constructed response

Day 125 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Connect Data Values and Data Displays

Standards: 6.SP.B.4, 6.SP.B.5c

Learning target: I can connect Data Values and Data Displays, and explain and check my reasoning.

1. 3: 1 dot; 4: 2 dots; 5: 2 dots; 6: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

2. (8, 9, 19/2, 10, 11)

Box runs from 9 to 10, median line at 19/2, whiskers to 8 and 11.

6.SP.B.4, 6.SP.B.5c | Constructed response

3. 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

4. (3, 7, 9, 11, 15)

Box runs from 7 to 11, median line at 9, whiskers to 3 and 15.

6.SP.B.4, 6.SP.B.5c | Constructed response

5. 8: 1 dot; 10: 2 dots; 12: 2 dots; 14: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

6. (8, 10, 11, 12, 14)

Box runs from 10 to 12, median line at 11, whiskers to 8 and 14.

6.SP.B.4, 6.SP.B.5c | Constructed response

7. 5: 1 dot; 6: 2 dots; 7: 1 dot; 8: 1 dot; 17: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

8. (3, 5, 6, 7, 9); IQR 2 min

Minimum and maximum anchor whiskers. Box ends are 5, 7; median line is 6. IQR = 7 - 5.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

9. 13 students; stacks at 0, 1, 2, 3 have 4, 5, 2, 2 dots.

Place one dot per student, not one per distinct value. Total dots = 4 + 5 + 2 + 2 = 13. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

10. (7, 8, 17/2, 10, 19); IQR 2 min

Box runs from 8 to 10, median line at 17/2, whiskers to 7 and 19.

6.SP.B.4, 6.SP.B.5c | Constructed response

Day 126 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Use Frequency Tables and Histograms

Standards: 6.SP.B.4

Learning target: I can construct and interpret a histogram with equal-width bins.

1. Frequencies 0, 2, 1, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

2. Frequencies 1, 1, 1, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

3. Frequencies 1, 1, 2, 2

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

4. Frequencies 4, 4, 0, 2

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

5. Frequencies 1, 4, 1, 4

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

6. Frequencies 4, 2, 3, 1

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

7. Frequencies 2, 3, 2, 3; greatest: 5–9, 15–19

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

8. 15 observations; highest intervals 10–14, 15–19

Use four equal-width touching bars with heights 2, 3, 5, 5. Frequencies sum to 15. Include every interval with frequency 5.

6.SP.B.4 | Data display and explanation

9. 9 observations; highest intervals 0–4

Use four equal-width touching bars with heights 4, 2, 1, 2. Frequencies sum to 9. Include every interval with frequency 4.

6.SP.B.4 | Data display and explanation

10. Frequencies 5, 1, 2, 2; greatest: 0–4

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

Day 127 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Find Quartiles and Interquartile Range

Standards: 6.SP.A.3, 6.SP.B.5c

Learning target: I can calculate quartiles and interquartile range.

1. Q1 9, Q3 10; IQR 1 min; range 3 min

Lower half [8, 9, 9] has median 9; upper half [10, 10, 11] has median 10. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

2. Q1 4, Q3 5; IQR 1 min; range 3 min

Lower half [3, 4, 4] has median 4; upper half [5, 5, 6] has median 5. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

3. Q1 7, Q3 8; IQR 1 min; range 3 min

Lower half [6, 7, 7] has median 7; upper half [8, 8, 9] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

4. Q1 11, Q3 17; IQR 6 min; range 18 min

Lower half [5, 11, 11] has median 11; upper half [17, 17, 23] has median 17. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

5. Q1 5, Q3 8; IQR 3 min; range 9 min

Lower half [2, 5, 5] has median 5; upper half [8, 8, 11] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

6. Q1 7, Q3 12; IQR 5 min; range 15 min

Lower half [2, 7, 7] has median 7; upper half [12, 12, 17] has median 12. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

7. Q1 6, Q3 8; IQR 2 min; range 12 min; IQR is the span of the middle half

Lower half [5, 6, 6] has median 6; upper half [7, 8, 17] has median 8. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

8. Q1 = 11; Q3 = 15; IQR = 4 min

Lower-half median is 11; upper-half median is 15. IQR = 15 - 11 = 4, the span of the middle half of the ordered data.

6.SP.A.3, 6.SP.B.5c | Multi-field numeric entry and explanation

9. Q1 = 11; Q3 = 16; IQR = 5 min

Lower-half median is 11; upper-half median is 16. IQR = 16 - 11 = 5, the span of the middle half of the ordered data.

6.SP.A.3, 6.SP.B.5c | Multi-field numeric entry and explanation

10. Q1 4, Q3 6; IQR 2 min; range 12 min; IQR is the span of the middle half

Lower half [3, 4, 4] has median 4; upper half [5, 6, 15] has median 6. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

Day 128 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Find Mean Absolute Deviation

Standards: 6.SP.B.5c

Learning target: I can calculate and interpret mean absolute deviation.

1. Mean 13/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

2. Mean 17/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

3. Mean 7/2 min; MAD 5/6 min

Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2; sum 5; divide by 6 = 5/6.

6.SP.B.5c | Constructed response

4. Mean 15/2 min; MAD 5/2 min

Distances: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2; sum 15; divide by 6 = 5/2.

6.SP.B.5c | Constructed response

5. Mean 11 min; MAD 10/3 min

Distances: 6, 2, 2, 2, 2, 6; sum 20; divide by 6 = 10/3.

6.SP.B.5c | Constructed response

6. Mean 17 min; MAD 5 min

Distances: 9, 3, 3, 3, 3, 9; sum 30; divide by 6 = 5.

6.SP.B.5c | Constructed response

7. Mean 31/6 min; MAD 53/18 min; a typical distance from the mean is 53/18 min

Distances: 19/6, 13/6, 13/6, 7/6, 1/6, 53/6; sum 53/3; divide by 6 = 53/18.

6.SP.B.5c | Constructed response

8. Mean 11 min; MAD 5 min

Mean = 66/6 = 11. Absolute deviations: 9, 3, 3, 3, 3, 9. Their sum divided by 6 is 5. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

9. Mean 7 min; MAD 5/3 min

Mean = 42/6 = 7. Absolute deviations: 3, 1, 1, 1, 1, 3. Their sum divided by 6 is 5/3. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

10. Mean 49/6 min; MAD 53/18 min; a typical distance from the mean is 53/18 min

Distances: 19/6, 13/6, 13/6, 7/6, 1/6, 53/6; sum 53/3; divide by 6 = 53/18.

6.SP.B.5c | Constructed response

Day 129 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Choose Appropriate Statistical Measures

Standards: 6.SP.B.5d

Learning target: I can choose a measure of center and explain why.

1. Mean 11/2 min and MAD 5/6 min; roughly symmetric data

Mean = 11/2, median = 11/2, IQR = 6 - 5 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

2. Mean 15/2 min and MAD 5/6 min; roughly symmetric data

Mean = 15/2, median = 15/2, IQR = 8 - 7 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

3. Mean 9/2 min and MAD 5/6 min; roughly symmetric data

Mean = 9/2, median = 9/2, IQR = 5 - 4 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

4. Mean 11 min and MAD 5 min; roughly symmetric data

Mean = 11, median = 11, IQR = 14 - 8 = 6, MAD = 5. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

5. Mean 12 min and MAD 10/3 min; roughly symmetric data

Mean = 12, median = 12, IQR = 14 - 10 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

6. Mean 11 min and MAD 10/3 min; roughly symmetric data

Mean = 11, median = 11, IQR = 13 - 9 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

7. Median 9/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 37/6, median = 9/2, IQR = 6 - 4 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

8. 6 observations; minutes; median 10, IQR 4; mean 10, MAD 10/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 10 and half medians give IQR 4. Mean is 60/6; averaging distances gives MAD 10/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

9. 6 observations; minutes; median 13/2, IQR 3; mean 47/6, MAD 23/6. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 13/2 and half medians give IQR 3. Mean is 47/6; averaging distances gives MAD 23/6. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

10. Median 17/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 61/6, median = 17/2, IQR = 10 - 8 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

Day 130 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Critique Statistical Summaries

Standards: 6.SP.B.5d, 6.SP.A.2, 6.SP.A.3

Learning target: I can critique Statistical Summaries, and explain and check my reasoning.

1. Mean 17/2 min and MAD 5/6 min; roughly symmetric data

Mean = 17/2, median = 17/2, IQR = 9 - 8 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

2. Center near 19/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 11 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

3. Mean 15/2 min and MAD 5/6 min; roughly symmetric data

Mean = 15/2, median = 15/2, IQR = 8 - 7 = 1, MAD = 5/6. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

4. Center near 8 min; range 12 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 14 - 2.

6.SP.A.2, 6.SP.A.3 | Constructed response

5. Mean 13 min and MAD 10/3 min; roughly symmetric data

Mean = 13, median = 13, IQR = 15 - 11 = 4, MAD = 10/3. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

6. Center near 21/2 min; range 15 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 18 - 3.

6.SP.A.2, 6.SP.A.3 | Constructed response

7. Median 15/2 min and IQR 2 min; high value makes these resistant measures useful

Mean = 55/6, median = 15/2, IQR = 9 - 7 = 2, MAD = 53/18. Accept another justified choice tied to shape and purpose.

6.SP.B.5d | Constructed response

8. 6 observations; minutes; median 6, IQR 2; mean 6, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 6 and half medians give IQR 2. Mean is 36/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

9. 6 observations; minutes; median 10, IQR 4; mean 34/3, MAD 14/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 10 and half medians give IQR 4. Mean is 68/6; averaging distances gives MAD 14/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

10. Center near 19/2 min; range 12 min; right-skewed with high value 20 min

The values cluster low with a long high tail. Range = 20 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

Day 131 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Describe Center, Spread, and Shape

Standards: 6.SP.A.2, 6.SP.A.3

Learning target: I can describe a distribution by center, spread, and shape.

1. Center near 17/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 10 - 7.

6.SP.A.2, 6.SP.A.3 | Constructed response

2. Center near 13/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 8 - 5.

6.SP.A.2, 6.SP.A.3 | Constructed response

3. Center near 11/2 min; range 3 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 7 - 4.

6.SP.A.2, 6.SP.A.3 | Constructed response

4. Center near 11 min; range 6 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 14 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

5. Center near 21/2 min; range 15 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 18 - 3.

6.SP.A.2, 6.SP.A.3 | Constructed response

6. Center near 25/2 min; range 9 min; roughly symmetric; no isolated extreme

The values balance around the middle. Range = 17 - 8.

6.SP.A.2, 6.SP.A.3 | Constructed response

7. Center near 15/2 min; range 12 min; right-skewed with high value 18 min

The values cluster low with a long high tail. Range = 18 - 6.

6.SP.A.2, 6.SP.A.3 | Constructed response

8. 6 observations; minutes; median 9, IQR 2; mean 9, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 9 and half medians give IQR 2. Mean is 54/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

9. 6 observations; minutes; median 25/2, IQR 5; mean 83/6, MAD 11/2. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 25/2 and half medians give IQR 5. Mean is 83/6; averaging distances gives MAD 11/2. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

10. Center near 13/2 min; range 12 min; right-skewed with high value 17 min

The values cluster low with a long high tail. Range = 17 - 5.

6.SP.A.2, 6.SP.A.3 | Constructed response

Day 132 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Summarize Data in Context

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can write a statistical summary tied to its context.

1. 6 observations of reading time in minutes. Mean 7/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

2. 6 observations of reading time in minutes. Mean 11/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

3. 6 observations of reading time in minutes. Mean 13/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

4. 6 observations of reading time in minutes. Mean 12 min, MAD 10/3 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

5. 6 observations of reading time in minutes. Mean 11 min, MAD 5 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

6. 6 observations of reading time in minutes. Mean 12 min, MAD 5 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

7. 6 observations of reading time in minutes. Median 7/2 min, IQR 2 min; right-skewed with a high value 14 min, so median/IQR resist that value.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

8. 6 observations; minutes; median 15/2, IQR 3; mean 15/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 15/2 and half medians give IQR 3. Mean is 45/6; averaging distances gives MAD 5/2. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

9. 6 observations; minutes; median 11, IQR 4; mean 37/3, MAD 14/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 11 and half medians give IQR 4. Mean is 74/6; averaging distances gives MAD 14/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

10. 6 observations of reading time in minutes. Median 19/2 min, IQR 2 min; right-skewed with a high value 20 min, so median/IQR resist that value.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

Day 133 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Compare Data Distributions

Standards: 6.SP.A.2, 6.SP.B.5d

Learning target: I can compare centers and variability in two data sets.

1. A median 19/2, B median 23/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

2. A median 11/2, B median 15/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

3. A median 17/2, B median 21/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

4. A median 14, B median 16 min; both IQR 4 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

5. A median 21/2, B median 25/2 min; both IQR 3 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

6. A median 9, B median 11 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

7. A median 11/2, B median 15/2 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

8. First median 25/2; second 33/2 min; both IQR 5 min.

Adding 4 shifts every position and both quartiles by 4. The median increases by 4; Q3 - Q1 remains 5.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

9. First median 13/2; second 17/2 min; both IQR 3 min.

Adding 2 shifts every position and both quartiles by 2. The median increases by 2; Q3 - Q1 remains 3.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

10. A median 15/2, B median 19/2 min; both IQR 2 min. B is typically 2 min higher; equal middle-half variability.

Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

6.SP.A.2, 6.SP.B.5d | Constructed response

Day 134 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Plan and Explain a Statistical Investigation

Standards: 6.SP.A.1, 6.SP.B.4, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can plan and Explain a Statistical Investigation, and explain and check my reasoning.

1. Yes; pages read vary across students.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

2. 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

3. n = 6; minutes; mean 15/2 min

Sum = 45; mean = 45 ÷ 6 = 15/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

4. 6 observations of reading time in minutes. Mean 19/2 min, MAD 25/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

5. Example: What are the lengths in centimeters of the pencils in our classroom?

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. 3: 1 dot; 8: 2 dots; 13: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

7. n = 6; minutes; mean 43/6 min; new mean 51/7 min

Sum = 43; mean = 43 ÷ 6 = 43/6. New sum = 51; divide by 7.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

8. 6 observations; minutes; median 13/2, IQR 3; mean 13/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 13/2 and half medians give IQR 3. Mean is 39/6; averaging distances gives MAD 5/2. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

9. A, C - How does mass vary among the backpacks?; What is the mass in kilograms of each of the backpacks?

Different backpacks can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

10. 2: 1 dot; 3: 2 dots; 4: 1 dot; 5: 1 dot; 14: 1 dot; n = 6; low cluster and high isolated value

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

Day 135 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Unit 9 Test

Standards: 6.G.A.1, 6.G.A.3, 6.SP.A.1, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.4, 6.SP.A.3, 6.SP.B.5d, 6.SP.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 16 cm²

A = bh/2 = 8 × 4 ÷ 2 = 16. Two such triangles make a parallelogram.

6.G.A.1 | Constructed response

2. 6 m²

Cut and shift a corner; A = bh = 2 × 3 = 6.

6.G.A.1 | Constructed response

3. 32 m²

Outer area 36 minus cutout 4 = 32.

6.G.A.1 | Constructed response

4. Sides 9 and 9; area 81 square units

Horizontal length 9 - (0) = 9; vertical length 9 - (0) = 9; area = 9 × 9.

6.G.A.3, 6.G.A.1 | Constructed response

5. Yes; plant heights vary.

Look for a collection of data and anticipated variation; identify the population and variable.

6.SP.A.1 | Constructed response

6. n = 6; minutes; mean 9/2 min

Sum = 27; mean = 27 ÷ 6 = 9/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

7. 8: 1 dot; 9: 2 dots; 10: 2 dots; 11: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

8. (7, 9, 10, 11, 13)

Box runs from 9 to 11, median line at 10, whiskers to 7 and 13.

6.SP.B.4, 6.SP.B.5c | Constructed response

9. 6: 1 dot; 10: 2 dots; 14: 2 dots; 18: 1 dot; n = 6

Each observation contributes one dot; stack repeated values at their number-line position.

6.SP.B.4 | Constructed response

10. Frequencies 0, 3, 4, 3

Count every value once. Bars touch and have equal width; horizontal axis is minutes, vertical axis is frequency.

6.SP.B.4 | Constructed response

11. Q1 11, Q3 17; IQR 6 min; range 18 min

Lower half [5, 11, 11] has median 11; upper half [17, 17, 23] has median 17. Subtract Q1 from Q3.

6.SP.A.3, 6.SP.B.5c | Constructed response

12. Mean 13/2 min; MAD 5/2 min

Mean = 39/6 = 13/2. Absolute deviations: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2. Their sum divided by 6 is 5/2. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

13. 6 observations; minutes; median 8, IQR 2; mean 8, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 8 and half medians give IQR 2. Mean is 48/6; averaging distances gives MAD 5/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

14. 6 observations; minutes; median 16, IQR 6; mean 16, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 16 and half medians give IQR 6. Mean is 96/6; averaging distances gives MAD 5. Tie the chosen summary to the displayed shape and context.

6.SP.B.5d | Statistical reasoning and explanation

15. 6 observations; minutes; median 14, IQR 6; mean 14, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 14 and half medians give IQR 6. Mean is 84/6; averaging distances gives MAD 5. Tie the chosen summary to the displayed shape and context.

6.SP.A.2, 6.SP.A.3 | Statistical reasoning and explanation

16. 6 observations; minutes; median 27/2, IQR 5; mean 27/2, MAD 25/6. Roughly balanced; mean/MAD or median/IQR with justification.

Order the observations. Middle positions give median 27/2 and half medians give IQR 5. Mean is 81/6; averaging distances gives MAD 25/6. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

17. First median 9; second 13 min; both IQR 2 min.

Adding 4 shifts every position and both quartiles by 4. The median increases by 4; Q3 - Q1 remains 2.

6.SP.A.2, 6.SP.B.5d | Two-part comparison and reasoning

18. 13 students; stacks at 0, 1, 2, 3 have 1, 3, 4, 5 dots.

Place one dot per student, not one per distinct value. Total dots = 1 + 3 + 4 + 5 = 13. Label the number line books read.

6.SP.B.4 | Data display and numeric entry

19. B, D - What is the reading time in minutes of each of the students?; How does reading time vary among the students?

Different students can have different reading time. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

20. A, D - The median is the average of the 3rd and 4th ordered values.; The median is 9 minutes.

Six ordered observations have middle positions 3 and 4, so median = (8 + 10)/2 = 9.

6.SP.B.5c | Multiple select

Day 136 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Connect Rational Numbers, Expressions, and Ratios

Standards: 6.NS.A.1, 6.NS.C.7a, 6.NS.C.7b, 6.EE.A.2c, 6.RP.A.3a

Learning target: I can connect Rational Numbers, Expressions, and Ratios, and explain and check my reasoning.

1. 1/2

1/4 ÷ 1/2 = 1/4 × 2 = 1/2.

6.NS.A.1 | Constructed response

2. >

-2 > -7; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

3. 19

Substitute first: 4 × 4 + 3 = 19.

6.EE.A.2c | Constructed response

4. 14; 28

Multiply both parts by 2 and by 4.

6.RP.A.3a | Constructed response

5. 1/2

1/2 ÷ 1 = 1/2 × 1 = 1/2.

6.NS.A.1 | Constructed response

6. <

-3/2 < -7/5; greater numbers lie farther right.

6.NS.C.7a, 6.NS.C.7b | Constructed response

7. A = 2(x + 8); 26 square units

Substitute x = 5: 2(5 + 8) = 26.

6.EE.A.2c | Constructed response

8. A, C - (9, 12); (6, 8)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

9. A - 10/3

2/3 ÷ 1/5 = 2/3 × 5 = 10/3.

6.NS.A.1 | Multiple choice

10. -7/3 > -5/2; -7/3°C is warmer

The greater value is farther right on a number line, even when both are negative.

6.NS.C.7a, 6.NS.C.7b | Constructed response

Day 137 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Connect Equations, Rates, and Percent

Standards: 6.EE.B.6, 6.EE.B.7, 6.RP.A.3b, 6.RP.A.3c

Learning target: I can connect Equations, Rates, and Percent, and explain and check my reasoning.

1. x + 5 = 35; x = 30 cards

Subtract 5 from both sides; 30 + 5 = 35.

6.EE.B.6, 6.EE.B.7 | Constructed response

2. 5 hours

Time = distance ÷ speed = 200 ÷ 40 = 5.

6.RP.A.3b | Constructed response

3. 80

Whole = 40 ÷ 1/2 = 80. Check 50% × 80 = 40.

6.RP.A.3c | Constructed response

4. 60%

144 ÷ 240 × 100 = 60%; the reference whole is 240.

6.RP.A.3c | Constructed response

5. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

6. 3 hours

Time = distance ÷ speed = 120 ÷ 40 = 3.

6.RP.A.3b | Constructed response

7. 180

Whole = 225 ÷ 5/4 = 180. Check 125% × 180 = 225.

6.RP.A.3c | Constructed response

8. 75%; denominator 100

Percent = (75 ÷ 100) × 100 = 75%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

9. 2x = 3; x = 3/2 meters

Total = number of ropes × one length. Divide both sides by 2; 3 ÷ 2 = 3/2. Check 2 × 3/2 = 3.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

10. 5/2 hours

Time = distance ÷ speed = 175/2 ÷ 35 = 5/2.

6.RP.A.3b | Constructed response

Day 138 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Connect Coordinates and Geometry Models

Standards: 6.NS.C.8, 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can connect Coordinates and Geometry Models, and explain and check my reasoning.

1. 4 units

Only one coordinate changes. Same side of 0: subtract magnitudes to get 4.

6.NS.C.8 | Constructed response

2. Sides 12 and 8; area 96 square units

Horizontal length 12 - (0) = 12; vertical length 8 - (0) = 8; area = 12 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

3. 170 cm²

Closed surface area = 2(25+30+30) = 170.

6.G.A.4 | Constructed response

4. 27 cm³

B = 3 × 3 = 9 cm²; V = B × 3 = 27 cm³.

6.G.A.2 | Constructed response

5. 9 units

Only one coordinate changes. Opposite sides of 0: 4 + 5 = 9.

6.NS.C.8 | Constructed response

6. Sides 10 and 8; area 80 square units

Horizontal length 4 - (-6) = 10; vertical length 4 - (-4) = 8; area = 10 × 8.

6.G.A.3, 6.G.A.1 | Constructed response

7. 68 cm²

Closed surface area = 2(8+20+10) = 76. Remove the top area 8: 68.

6.G.A.4 | Constructed response

8. 25/4 m³; 50 cubes

Multiply edges: 5/2 × 1 × 5/2 = 25/4. The edge counts are 5, 2, 5, giving 50 cubes. Each cube is 1/8 m³; 50/8 = 25/4.

6.G.A.2 | Numeric entry and model reasoning

9. AB = 12 units; AC = 12 units

AB is horizontal: 7 + 5 = 12. AC is vertical: 6 + 6 = 12. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

10. Sides 6 and 9; area 54 square units; perimeter 30 units

Horizontal length 3 - (-3) = 6; vertical length 3 - (-6) = 9; area = 6 × 9.

6.G.A.3, 6.G.A.1 | Constructed response

Day 139 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Explain Data and Critique Mathematical Claims

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d, 6.EE.A.4, 6.EE.B.5, 6.EE.B.8

Learning target: I can explain Data and Critique Mathematical Claims, and explain and check my reasoning.

1. 6 observations of reading time in minutes. Mean 7/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

2. Yes; distribution gives 5x + 15

5(x + 3) = 5x + 15. This matches.

6.EE.A.4 | Constructed response

3. 9

Values below 10 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

4. Mean 23/2 min; MAD 5/2 min

Distances: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2; sum 15; divide by 6 = 5/2.

6.SP.B.5c | Constructed response

5. 6 observations of reading time in minutes. Mean 8 min, MAD 5/3 min; roughly symmetric, so mean/MAD are reasonable.

Calculate the center and spread before interpreting them in context. Other accurate, justified summaries are acceptable.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Constructed response

6. Yes; distribution gives 6x + 12

6(x + 2) = 6x + 12. This matches.

6.EE.A.4 | Constructed response

7. 10; no, infinitely many real values satisfy it

Values below 11 work; the endpoint does not work.

6.EE.B.5, 6.EE.B.8 | Constructed response

8. Mean 9 min; MAD 10/3 min

Mean = 54/6 = 9. Absolute deviations: 6, 2, 2, 2, 2, 6. Their sum divided by 6 is 10/3. This is the average distance from the mean.

6.SP.B.5c | Numeric entry and interpretation

9. 6 observations; minutes; median 14, IQR 6; mean 46/3, MAD 19/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 14 and half medians give IQR 6. Mean is 92/6; averaging distances gives MAD 19/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation

10. Yes; distribution gives 3x + 12

3(x + 4) = 3x + 12. This matches.

6.EE.A.4 | Constructed response

Day 140 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Course Final Test

Standards: 6.NS.B.3, 6.EE.A.1, 6.RP.A.1, 6.NS.C.5, 6.EE.A.2b, 6.SP.A.1, 6.NS.B.2, 6.EE.B.7, 6.RP.A.3a, 6.EE.A.2c, 6.G.A.1, 6.EE.A.2a, 6.EE.B.6, 6.NS.A.1, 6.RP.A.2, 6.RP.A.3b, 6.EE.A.3, 6.EE.A.4, 6.NS.B.4a, 6.SP.B.4, 6.SP.B.5c, 6.NS.C.7a, 6.NS.C.7b, 6.RP.A.3d, 6.NS.C.7c, 6.NS.C.7d, 6.RP.A.3c, 6.G.A.4, 6.EE.B.5, 6.NS.C.6b, 6.NS.C.6c, 6.EE.B.8, 6.G.A.2, 6.NS.C.8, 6.EE.C.9, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5d

Learning target: I can apply and explain Grade 6 mathematics across the full course.

1. 47.7

Align decimal points; 39.5 + 8.2 = 47.7.

6.NS.B.3 | Constructed response

2. 4 × 4 × 4 = 64

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

3. 1:1

Keep red first; divide 5:5 by 5.

6.RP.A.1 | Constructed response

4. -20 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. Terms 2x, 7y, 4; coefficients 2, 7; constant 4

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

6. B, D - How does mass vary among the dogs?; What is the mass in kilograms of each of the dogs?

Different dogs can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

7. 60

30 ÷ 5: write 6, subtract 30, remainder 0; bring down the next digit: 0 ÷ 5: write 0, subtract 0, remainder 0. Check: 5 × 60 = 300.

6.NS.B.2 | Constructed response

8. x = 4

Subtract 2 on both sides. Then x = 4; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

9. B - 97/100

Align decimal points; 2.6 - 1.63 = 0.97.

6.NS.B.3 | Multiple choice

10. C, D - (6, 4); (9, 6)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

11. 17

Parentheses: 5 - 3 = 2. Exponent: $3^2 = 9$. Multiply: 4 × 2 = 8. Add: 9 + 8 = 17. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

12. C - 336/125

Compute the whole-number product and restore decimal places: 1.92 × 1.4 = 2.688.

6.NS.B.3 | Multiple choice

13. B, C - 7 × 5; (14 × 5) ÷ 2

Triangle area is half the matching parallelogram: (14 × 5)/2 = 35 cm^2. Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

14. 5x + 2y - 2

Marker cost is 5x; folder cost is 2y. Add these subtotals, then subtract the one fixed coupon of 2. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

15. D - 11/2

Multiply both numbers by 10: 137.5 ÷ 25 = 5.5. Check: 2.5 × 5.5 = 13.75.

6.NS.B.3 | Multiple choice

16. Aligned pairs (0,0), (3,3), (6,6), (21,21); 21 m^2

Each aligned jump adds 3 liters and 3 m^2. Both quantities scale by 7; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

17. 39/2

Substitute: 7 × (3/2) + 3². The product is 21/2 and the square is 9. Add to get 39/2.

6.EE.A.2c | Numeric entry and reasoning

18. B - 2

Each small part is 1/4; 2 shaded parts divided into groups of 1 gives 2/1 = 2 groups.

6.NS.A.1 | Multiple choice

19. C, D - 3 notebooks cost $15.; Each notebook costs $5.

Cost per notebook = 25 ÷ 5 = 5. Multiply $5 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

20. B, C - 2x + 8; 8 + 2x

Distribute 2 to both terms to get 2x + 8. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

21. D - 35/12

7/4 ÷ 3/5 = 7/4 × 5/3 = 35/12.

6.NS.A.1 | Multiple choice

22. Removed 12 m²; remaining 88 m²

Outer area = 100. Triangle = 4 × 6/2 = 12. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

23. C, D - (5 + 6)x + 5; 11x + 5

Combine coefficients of like x terms: 5 + 6 = 11. The constant 5 stays separate.

6.EE.A.3 | Multiple select

24. A - 2

5 ÷ 5/2 = 5 × 2/5 = 2.

6.NS.A.1 | Multiple choice

25. A: $4 each; B: $5 each; A is the better buy.

Divide each cost by its own notebook count: 24/6 = 4; 35/7 = 5. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

26. B, D - 6x + 24; 24 + 6x

Distribute 6 to both terms to get 6x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

27. A, D - 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.; 6 divides both 12 and 18.

GCF(12, 18) = 6. Divide each count by 6: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

28. (5, 8, 19/2, 11, 14); IQR 3 min

Minimum and maximum anchor whiskers. Box ends are 8, 11; median line is 19/2. IQR = 11 - 8.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

29. A, B - 18/4; 9/2

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

30. B, D - In 2 hours it travels 40 km.; The speed is 20 km per hour.

Divide distance by time: 40 ÷ 2 = 20 km/h. Multiply this unit rate by 2 for the later distance.

6.RP.A.3b | Multiple select

31. B, C - -9 < -7; -7 lies to the right of -9 on a number line.

Among negatives, the number closer to zero is greater. Thus -7 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

32. A, D - 14/4; 7/2

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

33. A, C - 42 inches; 7/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 7/2 by 12, giving 42 inches. The physical length does not change.

6.RP.A.3d | Multiple select

34. A, D - -9 is farther from zero than 7.; |-9| = 9

Absolute value measures distance: 9 versus 7. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

35. A, C - 26/4; 13/2

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

36. 12 participants; n = (15/100) × 80; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 80 to obtain 12. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

37. 72 cm²; open box 60 cm²

Closed area = 2(12+18+6) = 72. Subtract exactly one top face, 12, for the open box.

6.G.A.4 | Two-part numeric entry

38. 16 observations; highest intervals 10–14

Use four equal-width touching bars with heights 3, 4, 5, 4. Frequencies sum to 16. Include every interval with frequency 5.

6.SP.B.4 | Data display and explanation

39. 1 bottles; 2/3 L left

2 ÷ 4/3 = 2 × 3/4 = 3/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

40. A, D - 5; 4

Subtract 2 conceptually: x must exceed 3. Equality at 3 fails because the inequality is strict.

6.EE.B.5 | Multiple select

41. A, C - (125/100)w = 75; w = 75 ÷ (125/100)

The part is percent rate times whole. Divide 75 by 5/4; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

42. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

43. A, C - n ≥ 4; 4 is an allowed membership count.

At least includes the boundary, so n ≥ 4. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

44. 65/4 m³; 130 cubes

Multiply edges: 13/2 × 5/2 × 1 = 65/4. The edge counts are 13, 5, 2, giving 130 cubes. Each cube is 1/8 m³; 130/8 = 65/4.

6.G.A.2 | Numeric entry and model reasoning

45. 20%; denominator 100

Percent = (20 ÷ 100) × 100 = 20%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

46. AB = 5 units; AC = 6 units

AB is horizontal: 2 + 3 = 5. AC is vertical: 3 + 3 = 6. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

47. (0,0), (1,4), (2,8); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

48. 6 observations; minutes; median 11, IQR 6; mean 37/3, MAD 19/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 11 and half medians give IQR 6. Mean is 74/6; averaging distances gives MAD 19/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation